

Answers (Even and Odd)

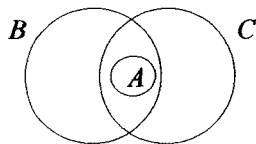
NOTE: We have not included "solutions" for Coding Exercises for several reasons. We believe each department (or instructor) should have very specific programming standards, and any code we presented would certainly violate someone's standards and set up unnecessary conflicts. More importantly, different programming languages support different constructions and what is a good choice for one can be a very bad one in another.

Chapter 1 Answers

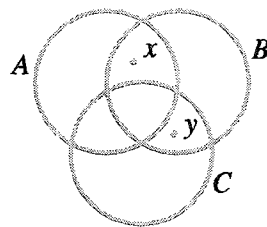
Exercise Set 1.1, page 4

1. (a) True (b) False (c) False (d) False (e) True (f) False
2. (a) True (b) False (c) False (d) True (e) True (f) False
3. (a) {A, R, D, V, K} (b) {B, O, K} (c) {M, I, S, P}
4. (a) {1, 2, 3, 4, 5, 6, 7, 8, 9} (b) {-3, -2, -1, 0, 1, 2, 3}
5. (a) False (b) True (c) False (d) True (e) False (f) False
6. $\{x \mid x \text{ is a positive even integer less than twelve}\}$
7. $\{x \mid x \text{ is a vowel}\}$
8. $\{x \mid x = y^3 \text{ and } y \in \{1, 2, 3, 4, 5\}\}$
9. $\{x \mid x \in \mathbb{Z} \text{ and } x^2 < 5\}$
10. (a) Yes (b) No (c) No (d) No (e) Yes (f) No
11. (b), (c), (e)
12. $\{\}, \{a\}, \{b\}, \{a, b\}$
13. $\{\}, \{\text{BASIC}\}, \{\text{PASCAL}\}, \{\text{ADA}\}, \{\text{BASIC, PASCAL}\}, \{\text{BASIC, ADA}\}, \{\text{PASCAL, ADA}\}, \{\text{BASIC, PASCAL, ADA}\}$
14. $\{\}$
15. (a) True (b) False (c) False (d) True (e) True (f) True
(g) True (h) True
16. (a) True (b) True (c) True (d) True (e) True
17. (a) $\subseteq$ (b) $\subseteq$ (c) $\not\subseteq$ (d) $\subseteq$ (e) $\not\subseteq$ (f) $\subseteq$
18. $\{a, b, c, d, e, f, g\}$
19. $\{1, 2, 3\}$
20. $\{2, 3, 4, 6, 9, 8, 10, 12, 14, 16, 18, 20, 21\}$
21. Yes, Yes, the complement of a set would not be defined unambiguously.
22. (a) False (b) True (c) Insufficient information
(d) True (e) True (f) False
23. (a) False (b) False (c) Insufficient information
(d) False (e) True (f) True
24. (a) two The regions are inside and outside the circle representing the set.
(b) four The regions are the part common to both sets, the part outside both sets, and the parts that represent what is left of each set when the common part is removed.
25. eight There are three parts that represent what is left of each set when common parts are removed, three regions that each represent the part shared by one of the three pairs of sets, a region that represents what all three sets have in common, and a region outside all three sets.
26. (a) $\{\{\}, \{3\}, \{7\}, \{3, 7\}\}$ (b) $|A| = 2$ (c) $|P(A)| = 4$
27. $B = \{m, n\}$
28. (a) $\{\{\}, \{3\}, \{7\}, \{2\}, \{3, 7\}, \{3, 2\}, \{7, 2\}, \{3, 7, 2\}\}$ (b) 3
(c) 8
29. $B = \{a, b, c\}$

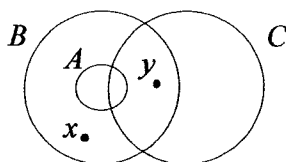
30.



31.



32. One solution

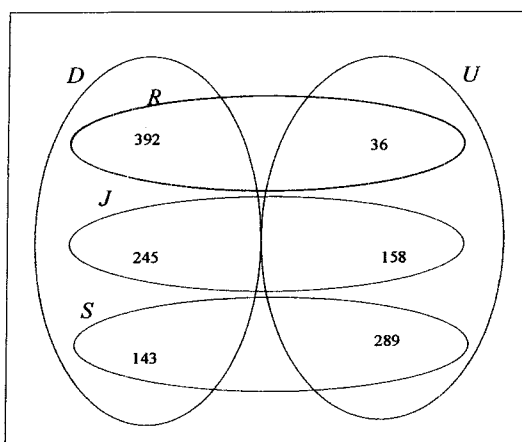
33. $\emptyset \subseteq \mathbb{Z}^+ \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$ 34. Choose any $x \in A$. Since $A \subseteq B$, $x \in B$. Also since $B \subseteq C$, $x \in C$. Hence any element of A belongs to C and $A \subseteq C$.35. $B; B$ 36. Let $b \in B$, $b \notin A$. Each subset of B either contains b or it does not. Let B_j be a subset of B . If it does not contain b , then B_j is one of the S_i . If it contains b , then the elements of B_j are all the elements of one of the S_i and b . In this case B_j is not one of the S_i , and B has exactly $2n$ subsets.

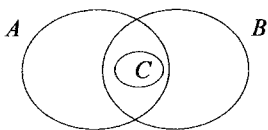
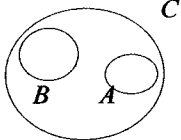
37. 4; 8

Exercise Set 1.2, page 11

1. (a) $\{a, b, c, d, e, f, g\}$ (b) $\{a, c, d, e, f, g\}$ (c) $\{a, c\}$ (d) $\{f\}$
 (e) $\{b, g, d, e\}$ (f) $\{a, b, c\}$ (g) $\{d, e, f, h, k\}$ (h) $\{a, b, c, d, e, f\}$
 (i) $\{b, g, f\}$ (j) $\{g\}$
2. (a) $\{a, b, c, g, f, h, k\}$ (b) $\{d, e, f, g, h, k\}$ (c) $\{f\}$ (d) $\{\}$ (e) $\{b\}$
 (f) $\{e, d, g\}$ (g) $\{a, b, c, h, k\}$ (h) $\{a, c\}$ (i) $\{a, c, h, k\}$ (j) $\{g\}$
3. (a) $\{a, b, c, d, e, f, g\}$ (b) $\{\}$ (c) $\{a, c, g\}$ (d) $\{a, c, f\}$ (e) $\{h, k\}$
 (f) $\{a, b, c, d, e, f, h, k\}$
4. (a) A (b) U (c) B (d) $\{\}$ (e) $\{b, d, e, g\}$
 (f) $\{a, b, c, d, e, g, h, k\}$
5. (a) $\{1, 2, 4, 5, 6, 8, 9\}$ (b) $\{1, 2, 3, 4, 6, 8\}$ (c) $\{1, 2, 4, 6, 7, 8\}$ (d) $\{1, 2, 3, 4, 5, 9\}$
 (e) $\{1, 2, 4\}$ (f) $\{8\}$ (g) $\{2, 4\}$ (h) $\{\}$
6. (a) $\{1, 6, 8\}$ (b) $\{5, 9\}$ (c) $\{1, 2, 3, 4\}$ (d) $\{5, 6, 7, 8, 9\}$
 (e) $\{3, 5, 7, 9\}$ (f) $\{1, 6, 8, 5, 9\}$ (g) $\{1, 2, 3, 4, 7, 8\}$ (h) $\{5, 9, 3, 1\}$
7. (a) $\{1, 2, 3, 4, 5, 6, 8, 9\}$ (b) $\{2, 4\}$ (c) $\{1, 2, 4\}$ (d) $\{8\}$
 (e) $\{3, 7\}$ (f) $\{1, 3, 5, 6, 7, 8, 9\}$
8. (a) $\{1, 2, 3, 4, 5, 7, 8, 9\}$ (b) $\{\}$ (c) A (d) $\{\}$
 (e) U (f) $\{6, 8\}$
9. (a) $\{b, d, e, h\}$ (b) $\{b, c, d, f, g, h\}$ (c) $\{b, d, h\}$ (d) $\{b, c, d, e, f, g, h\}$
 (e) $\{\}$ (f) $\{c, f, g\}$
10. (a) $\{b, d, h\}$ (b) $\{b, c, d, f, g, h, a, e\}$ (c) $\{b, d, e, h\}$ (d) $\{a, c, d, e, f, g\}$
 (e) $\{c, e, f, g\}$ (f) $\{a, e, h, b\}$
11. (a) all real numbers except -1 and 1 (b) all real numbers except -1 and 4
 (c) all real numbers except $-1, 1$, and 4 (d) all real numbers except -1
12. (a) False (b) True (c) False (d) True
13. (a) True (b) True (c) False (d) False
14. $C \cup (A \cap B)$ is one possible answer.
15. 1

16. (a) $|A \cup B| = 7, |A| = 4, |B| = 5, |A \cap B| = 2$. $|A \cup B| = |A| + |B| - |A \cap B|$ and Theorem 2 is verified.
 (b) $|A \cup B| = 9, |A| = 4, |B| = 5, |A \cap B| = 0$. Hence $|A \cup B| = |A| + |B| - |A \cap B|$.
17. (a) $|A \cup B| = 10, |A| = 6, |B| = 7, |A \cap B| = 3$. Hence $|A \cup B| = |A| + |B| - |A \cap B|$.
 (b) $|A \cup B| = 11, |A| = 5, |B| = 6, |A \cap B| = 0$. Hence $|A \cup B| = |A| + |B| - |A \cap B|$.
18. (a) $|A \cup B| = 7, |A| = 7, |B| = 4, |A \cap B| = 4$. Hence $|A \cup B| = |A| + |B| - |A \cap B|$.
 (b) $|A \cup B| = 9, |A| = 4, |B| = 5, |A \cap B| = 0$. Hence $|A \cup B| = |A| + |B| - |A \cap B|$.
19. B must be the empty set.
20. $|A| = 5, |B| = 7, |C| = 8, |A \cap B| = 2, |A \cap C| = 4, |B \cap C| = 3, |A \cap B \cap C| = 2$,
 $|A \cup B \cup C| = 13$. Hence $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$
 and Theorem 3 is verified.
21. $|A| = 6, |B| = 5, |C| = 6, |A \cap B| = 2, |A \cap C| = 3, |B \cap C| = 3, |A \cap B \cap C| = 2$,
 $|A \cup B \cup C| = 11$. Hence $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.
22. $|A| = 7, |B| = 3, |C| = 7, |A \cap B| = 3, |A \cap C| = 3, |B \cap C| = 2, |A \cap B \cap C| = 2$,
 $|A \cup B \cup C| = 11$. Hence $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.
23. (a) 106 (b) 60
24. (a) 20 (b) 325
25. 16; 23
26. 30; 28
27. (a) 162 (b) 118 (c) 236 (d) 290 (e) 264
28. 245 sophomores have declared their major; 388 sophomores and juniors have declared their major; 677 students have declared their major or are sophomores, but not seniors.
- 29.



30. $A \cup B = \{y | y \in A \text{ or } y \in B\}$
31. A and B
32. (a)  (b) C
 (c) Let $x \in C$. Then $x \in A$ since $C \subseteq A$ and hence $x \in A \cup B$.
33. (a)  (b) $A \cup B$
 (c) Let $x \in A \cup B$. Then $x \in A$ or $x \in B$. Since $A \subseteq C$ and $B \subseteq C$, $x \in A$ or $x \in B$ means $x \in C$. Hence $A \cup B \subseteq C$.

34. Let $x \in A - (A - B)$ so $x \in A$ and $x \notin (A - B)$. Using Exercise 36, $x \notin A \cap \overline{B}$ or $x \in \overline{A \cap \overline{B}}$. By De Morgan's law, $\overline{A \cap \overline{B}} = \overline{A} \cup B$. Since $x \in \overline{A \cap \overline{B}}$, $x \in \overline{A} \cup B$. But $x \notin \overline{A}$ since $x \in A$. Hence $x \in B$ and $A - (A - B) \subseteq B$.
35. Yes. Suppose $x \in B$. Either $x \in A$ or $x \notin A$. If $x \in A$, then $x \notin A \oplus B = A \oplus C$. But then x must be in C . If $x \notin A$, then $x \in A \oplus B = A \oplus C$, and again x must be in C . So $B \subseteq C$. A similar argument shows that $C \subseteq B$, so $B = C$.
36. Suppose $x \in A - B$. Then $x \in A$ and $x \notin B$. Thus $x \in A$ and $x \in \overline{B}$, so $x \in A \cap \overline{B}$. We have $A - B \subseteq A \cap \overline{B}$. Now choose any $y \in A \cap \overline{B}$. Then $y \in A$ and $y \in \overline{B}$ so $y \in A$ and $y \notin B$. This means $y \in A - B$ and $A \cap \overline{B} \subseteq A - B$. Hence $A - B = A \cap \overline{B}$.
37. No. Let $A = \{1, 2, 3\}$, $B = \{4\}$, and $C = \{3, 4\}$. Then $A \cup B = A \cup C$ and $B \neq C$.
38. No. Let $A = \{1, 2, 3\}$, $B = \{3, 4\}$, and $C = \{3, 5\}$. Then $A \cap B = A \cap C$ and $B \neq C$.
39. (a) Let $x \in A \cup C$. Then $x \in A$ or $x \in C$, so $x \in B$ or $x \in D$ and $x \in B \cup D$. Hence $A \cup C \subseteq B \cup D$.
(b) Let $x \in A \cap C$. Then $x \in A$ and $x \in C$, so $x \in B$ and $x \in D$. Hence $x \in B \cap D$. Thus $A \cap C \subseteq B \cap D$.
40. If $A = B$, then $A - B = \{\} = B - A$. If $A - B = B - A$, then $x \in A - B$ implies $x \in B - A$. But $x \in A - B$ means $x \in A$ and $x \notin B$, and $x \in B - A$ means $x \in B$ and $x \notin A$. This is impossible unless $A - B = \emptyset$ and $A \subseteq B$, and $B - A = \emptyset$ with $B \subseteq A$. Hence $A = B$.
41. We must subtract $|B \cap C|$, because each element in $B \cap C$ has been counted twice in $|A| + |B| + |C|$. But when we subtract both $|B \cap C|$ and $|A \cap C|$, we have "uncounted" all the elements of C that also belong to B and A . These (and the similar elements of A and B) are counted again by adding $|A \cap B \cap C|$.
42. $|A \cup B \cup C \cup D| = |A| + |B| + |C| + |D|$
 $- (|A \cap B| + |A \cap C| + |A \cap D| + |B \cap C| + |B \cap D| + |C \cap D|)$
 $+ (|A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D|)$
 $- |A \cap B \cap C \cap D|$
43. The cardinality of the union of n sets is the sum of the cardinalities of each of the n sets minus the sum of the cardinalities of the ${}_nC_2$ different intersections of two of the sets plus the sum of the ${}_nC_3$ different intersections of three of the sets and so on, alternating plus and minus the sum of the ${}_nC_k$ k -set intersections, $k = 4, \dots, n$.

Exercise Set 1.3, page 19

1. $\{1, 2\}$
2. $\{0, 2, 4, 6, 8, 10, \dots\}$
3. $\{a, b, c, \dots, z\}$
4. $\{a, b, d, c\}$
5. Possible answers include $xyzxyz\dots$, $xyzyzxyzxyz$, and $yzxyzx\dots$.
6. Possible answers include $1, 2, 3, 4, \dots$; $1, 1, 2, 2, 3, 3, 4, 4, \dots$; and $1, 2, 2, 2, 3, 2, 4, 2, 5, 2, \dots$.
7. 5, 25, 125, 625
8. -1, 10, 27, 50
9. 1, 2, 6, 24
10. $1, a + 1, a^2 + a + 1, a^3 + a^2 + a + 1$
11. 2.5, 4, 5.5, 7
12. -3, 7, -13, 27
13. 0, -2, -4, -6
14. 4, 8, 24, 96
15. $a_n = a_{n-1} + 2$, $a_1 = 1$, recursive; $a_n = 2n - 1$, explicit
16. $b_n = n^2 - 1$, explicit
17. $c_n = (-1)^{n+1}$, explicit
18. $d_n = 0$ if n is odd, $d_n = 2$ if n is even, explicit
19. $e_n = e_{n-1} + 3$, $e_1 = 1$, recursive
20. $f_n = \frac{1}{2}f_{n-1}$, $f_1 = 1$, recursive; $f_n = \frac{1}{2^{n-1}}$, explicit
21. $a_n = 2 + 3(n - 1)$

22. $b_n = b_{n-1} + b_{n-2}$, $b_1 = 2$, $b_2 = 5$
23. A , uncountable B , finite C , countable D , finite E , finite
24. countable; finite; countable; uncountable; uncountable
25. (a) Yes (b) No (c) Yes (d) Yes (e) No (f) No
26. (a) 01101110 (b) 01100000 (c) 00100000 (d) 11111011 (e) 00010100
27. (a) 1 (b) 0 (c) $f_B: 10000000$, $f_C: 01010011$, $f_D: 01000101$
(d) 11010011, 01010111, 01000001
28. $f_A(x) = 1$; $f_B(x) = 0$; $f_A(x)f_B(x) = 0$; $f_A(x) + f_B(x) - 2f_A(x)f_B(x) = 1$
 $f_A(x) = 0$; $f_B(x) = 1$; $f_A(x)f_B(x) = 0$; $f_A(x) + f_B(x) - 2f_A(x)f_B(x) = 1$
 $x \in A \cap B$ or $x \notin A \cup B$; $f_A(x) + f_B(x) - 2f_A(x)f_B(x) = 0$
29. $f_{(A \oplus B) \oplus C} = f_{A \oplus B} + f_C - 2f_{A \oplus B}f_C$ by Theorem 4
 $= (f_A + f_B - 2f_Af_B) + f_C - 2(f_A + f_B - 2f_Af_B)f_C$
 $= f_A + (f_B + f_C - 2f_Bf_C) - 2f_A(f_B + f_C - 2f_Bf_C)$
 $= f_A + f_{B \oplus C} - 2f_Af_{B \oplus C}$
 $= f_{A \oplus (B \oplus C)}$
- Since the characteristic functions are the same, the sets must be the same.
30. (a) a and b are regular by RE2. ab is then regular by RE3 and so by RE5, $(ab)^*$ is regular. Applying RE2 and RE3, we have $a + b(ab)^*$ and $a \times b$ are regular. By RE4, $(a \times b \vee a)$ is regular. Using RE3, $a + b(ab)^*(a \times b \vee a)$ is regular.
(b) By RE2, a , b , $+$, $\times$ are regular. By RE5, a^* is regular; $(a^* \vee b)$ is regular by RE4. Using RE3, we have $a + b \times (a^* \vee b)$ is regular.
(c) By RE2, a , b , $\vee$, $+$ are regular. Thus by RE5, a^* and b^* are regular. Using RE3 we have a^*b and $\times ab^*$ are regular. By RE4, $(a^*b \vee +)$ is regular. By RE5, $(a^*b \vee +)^*$ is regular. And finally by RE4, $((a^*b \vee +)^* \vee \times ab^*)$ is regular.
31. (a) Yes (b) Yes (c) Yes
32. (a) No (b) No (c) Yes
33. Possible answers include $\vee ac$ $a \vee$ $ab($
34. (a) $\{pr, prq, prqq, \dots; qr, qrq, qrqq, \dots\}$ (b) $\{pr, pqqr, pqqqqr, \dots\}$
35. (a) 01^*0 (b) $0(00)^* \vee (00)^*1$
36. T-numbers form the sequence 0, 3, 6, 9, The T-numbers are the non-negative multiples of 3.
37. By (1), 8 is an S -number. By (3), 1 is an S -number. By (2), all multiples of 1, that is, all integers are S -numbers.
38. 0, 1, 1, 3, 5, 11
39. 1, 2, 3, 7, 16

Exercise Set 1.4, page 30

1. $20 = 6 \cdot 3 + 2$
2. $64 = 1 \cdot 37 + 27$
3. $3 = 0 \cdot 22 + 3$
4. $48 = 4 \cdot 12 + 0$
5. (a) $828 = 2^2 \cdot 3^2 \cdot 23$ (b) $1666 = 2 \cdot 7^2 \cdot 17$ (c) $1781 = 13 \cdot 137$
(d) $1125 = 3^2 \cdot 5^3$ (e) 107
6. $d = 20$; $20 = 2 \cdot 60 - 100$
7. $d = 3$; $3 = 3 \cdot 45 - 4 \cdot 33$
8. $d = 2$; $2 = 12 \cdot 34 - 7 \cdot 58$
9. $d = 1$; $1 = 5 \cdot 77 - 3 \cdot 128$
10. 216
11. 1050
12. 1225
13. 864
14. (a) 3 (b) 6 (c) 3 (d) 4 (e) 2 (f) 1
15. (a) 6 (b) 1 (c) 0 (d) 1 (e) 20 (f) 14
16. (a) 5 (b) 5 (c) 6 (d) 6

17. (a) 10 (b) 22 (c) 2 (d) 14
18. $k \cdot f(a)$ does not always equal $f(k \cdot a)$.
19. $f(a) + f(b)$ may be greater than n .
20. Only if $f(a) + f(b)$ is less than n will $f(a + b) = f(a) + f(b)$.
21. (a) $\{2, 7, 12, 17, \dots\}$ (b) $\{1, 6, 11, 16, \dots\}$
22. (a) $\{3, 9, 15, 21, \dots\}$ (b) $\{1, 7, 13, 19, \dots\}$
23. The only divisors of p are $\pm p$ and ± 1 , but p does not divide a . (Multiply both sides by b)
 $p \mid sab$ and $p \mid tpb$. If p divides the right side of the equation, then it must divide the left side also.
24. If $\text{GCD}(a, c) = 1$, there are integers s, t such that $1 = sa + tc$. Thus $b = sab + tcb$. Since $c \mid ab$ and $c \mid c$, we have $c \mid (sab + tcb)$. That is, $c \mid b$.
25. Because $a \mid m$, $ac \mid mc$ and because $c \mid m$, $ac \mid am$. If $\text{GCD}(a, c) = 1$, there are integers s, t such that $1 = sa + tc$. Thus $m = sam + tcm$. But ac divides each term on the left so $ac \mid m$.
26. Since $a \mid b$ and $c \mid b$, $ac \mid ab$ and $ac \mid bb$. We can write d as $sa + tb$ and $bd = sab + tbb$. Because ac divides both sab and tbb , ac also divides bd .
27. Let $d = \text{GCD}(a, b)$. Then $cd \mid ca$ and $cd \mid cb$; that is, cd is a common divisor of ca and cb . Let $e = \text{GCD}(ca, cd)$. Then $cd \mid e$ and $e = cdk$ is a divisor of ca and cb . But then $dk \mid a$ and $dk \mid b$. Because d is the greatest common divisor of a and b , k must be 1 and $e = cd$.
28. Clearly, $a \mid ab$ and $ab \mid ab$, so ab is a common multiple of a and ab . No smaller multiple of ab exists so $ab = \text{LCM}(a, ab)$.
29. By Theorem 6, $\text{GCD}(a, b) \cdot \text{LCM}(a, b) = ab$. Since $\text{GCD}(a, b) = 1$, we have $\text{LCM}(a, b) = ab$.
30. Let $a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}$, $b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$, and $k = p_1^{k_1} p_2^{k_2} \dots p_n^{k_n}$ where some of the a_i, b_i , and k_i may be 0. Since $a \mid k$, $k_i \geq a_i, i = 1, 2, \dots, n$ and since $b \mid k$, $k_i \geq b_i, i = 1, 2, \dots, n$. If $c = \text{LCM}(a, b)$, then $c = p_1^{c_1} p_2^{c_2} \dots p_n^{c_n}$ where $c_i = \max(a_i, b_i)$. Then $k_i \geq c_i, i = 1, 2, \dots, n$ and $c \mid k$.
31. $a \mid b$ means $b = am$. $b \mid a$ means $a = bn$. Thus $b = am = bnm$. Hence $nm = 1$ and $n = m = 1$, because a and b are positive.
32. $p \mid a$ and $p \mid p$ so p is a common divisor of a and p . Hence $p \mid \text{GCD}(a, p)$. But $\text{GCD}(a, p) \mid p$ so $\text{GCD}(a, p) = p$. (See Exercise 31.)
33. No; consider $a = 6, b = 4, c = 3$.
34. Consider $m = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ and $n = q_1^{b_1} q_2^{b_2} \dots q_l^{b_l}$, the prime factorizations of m and n . Since $\text{GCD}(m, n) = 1$, $\{p_1, p_2, \dots, p_k\} \cap \{q_1, q_2, \dots, q_l\} = \{\}$. Thus for mn to be a perfect square, each a_i and b_j must be even. But this means m and n are each perfect squares.
35. Yes. Using the same reasoning as in Exercise 34, m and n share no prime factors, so for mn to be a perfect t^{th} power, each prime in the factorizations of m and n must appear a multiple of t times. But this means each of m and n are also perfect t^{th} powers.
36. (a) 563 (b) 337 (c) 153
37. (a) 112 (b) 10 (c) 30
38. (a) 788 (b) 870
39. (a) i. $(104)_5$ ii. $(243)_5$ iii. $(1330)_5$ iv. $(10412)_5$
 (b) i. 49 ii. 85 iii. 197 iv. 816
40. (a) i. $(41)_7$ ii. $(133)_7$ iii. $(425)_7$ iv. $(2264)_7$
 (b) i. 51 ii. 92 iii. 238 iv. 647
41. (a) $(11101)_2$ $(1001001)_2$ $(11010111)_2$ $(1011011100)_2$
 (b) $(131)_4$ $(1021)_4$ $(3113)_4$ $(23130)_4$
 (c) $(1D)_{16}$ $(49)_{16}$ $(D7)_{16}$ $(2DC)_{16}$
42. (a) $4 = 2^2, 16 = 2^4 = 2^4$
 (b) i. From right to left, group the digits in pairs. Convert each pair to its base 4 equivalent.
 ii. From left to right, convert each digit in the base 16 expansion to its base 2 equivalent;
 $(D)_{16}$ is $(1101)_2$.
 iii. From right to left, group the digits in pairs. Convert each pair to its base 16 equivalent.
43. (a) Answers will vary, but the pattern of italicized and non-italicized letters should match
 AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
 (b) STUDY WELL
44. (a) The base 2 expansion of 25 is 11001, so five digits are needed to write the position of an

arbitrary letter of the alphabet.

(b) There are now $26 + 1 + 10$ or 37 symbols to encode. This will require 6 digits for the largest binary expansion, so the dummy message should be six times as long as the original. An ordering for all 37 symbols must be agreed upon.

45. BIJS

46. THE EARTH IS ROUND

Exercise Set 1.5, page 39

1. (a) $-2, 1, 2$ (b) $3, 4$ (c) $4, -1, 8$ (d) $2, 6, 8$

2. **B**, **C**, and **E** are diagonal matrices.

3. a is 3, b is 1, c is 8, and d is -2 .

4. a is 0, b is 2, c is 1, and d is 2.

5. (a) $\begin{bmatrix} 4 & 0 & 2 \\ 9 & 6 & 2 \\ 3 & 2 & 4 \end{bmatrix}$ (b) $\mathbf{AB} = \begin{bmatrix} 7 & 13 \\ -3 & 0 \end{bmatrix}$

(c) not possible (d) $\begin{bmatrix} 21 & 14 \\ -7 & 17 \end{bmatrix}$

6. (a) $\mathbf{A(BD)} = \begin{bmatrix} 31 & 27 \\ 9 & -6 \end{bmatrix} = (\mathbf{AB})\mathbf{D}$

(b) $\mathbf{A(C + E)} = \begin{bmatrix} 26 & 12 & 18 \\ 19 & 2 & 2 \end{bmatrix} = \mathbf{AC + AE}$

(c) $\mathbf{FD + AB} = \begin{bmatrix} 25 & 12 \\ 5 & 13 \end{bmatrix}$

7. (a) $\mathbf{EB}$ is 3×2 and $\mathbf{FA}$ is 2×3 ; the sum is undefined.

(b) $\mathbf{B + D}$ does not exist.

(c) $\begin{bmatrix} 10 & 0 & -25 \\ 40 & 14 & 12 \end{bmatrix}$

(d) $\mathbf{DE}$ does not exist.

8. (a) $\mathbf{A}^T = \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 3 & -2 \end{bmatrix}$, $(\mathbf{A}^T)^T = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 1 & -2 \end{bmatrix}$

(b) $(\mathbf{C + E})^T = \begin{bmatrix} 4 & 9 & 3 \\ 0 & 6 & 2 \\ 2 & 2 & 4 \end{bmatrix} = \mathbf{C}^T + \mathbf{E}^T$

(c) $(\mathbf{AB})^T = \begin{bmatrix} 7 & -3 \\ 13 & 0 \end{bmatrix} = \mathbf{B}^T \mathbf{A}^T$ (d) $\begin{bmatrix} 12 & 5 & 12 \\ 22 & 6 & 17 \end{bmatrix}$

9. (a) $\begin{bmatrix} 22 & 34 \\ 3 & 11 \\ -31 & 3 \end{bmatrix}$ (b) $\mathbf{BC}$ is not defined. (c) $\begin{bmatrix} 25 & 5 & 26 \\ 20 & -3 & 32 \end{bmatrix}$

(d) $\mathbf{D}^T + \mathbf{E}$ is not defined.

10. $\begin{bmatrix} -67 & 30 \\ 60 & -7 \end{bmatrix}$

11. Let $\mathbf{B} = [b_{jk}] = \mathbf{I}_m \mathbf{A}$. Then $b_{jk} = \sum_{l=1}^m i_{jl} a_{lk}$, for $1 \leq j \leq m$ and $1 \leq k \leq n$. But $i_{jj} = 1$ and

$i_{jl} = 0$ if $j \neq l$. Hence $b_{jk} = i_{jj} a_{jk}$, $1 \leq j \leq m$, $1 \leq k \leq n$. This means $\mathbf{B} = \mathbf{I}_m \mathbf{A} = \mathbf{A}$.

Similarly, if $\mathbf{C} = \mathbf{A} \mathbf{I}_n = [c_{jk}]$, $c_{jk} = \sum_{l=1}^n a_{jl} i_{lk} = a_{jk} i_{kk} = a_{jk}$ for $1 \leq j \leq m$, $1 \leq k \leq n$.

12. $\mathbf{AB} = \begin{bmatrix} 1 & 8 \\ -9 & -2 \end{bmatrix}$ and $\mathbf{BA} = \begin{bmatrix} 4 & -5 \\ 18 & -5 \end{bmatrix}$.

13. $\mathbf{A}^3 = \begin{bmatrix} 27 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & 64 \end{bmatrix}$ or $\begin{bmatrix} 3^3 & 0 & 0 \\ 0 & (-2)^3 & 0 \\ 0 & 0 & 4^3 \end{bmatrix}$ $\mathbf{A}^k = \begin{bmatrix} 3^k & 0 & 0 \\ 0 & (-2)^k & 0 \\ 0 & 0 & 4^k \end{bmatrix}$
14. Let $\mathbf{B} = [b_{ij}] = \mathbf{A}\mathbf{O}$. Then $b_{ij} = \sum_{k=1}^n a_{ik} \cdot 0 = 0$ for all i, j .
15. The entries of $\mathbf{I}_n^T$ satisfy $i'_{kj} = i_{jk}$. But $i_{jk} = 1$ if $j = k$, and is 0 otherwise. Thus, $i'_{kj} = 1$ if $k = j$ and is 0 if $k \neq j$ for $1 \leq j \leq n, 1 \leq k \leq n$.
16. (a) Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times t$. Suppose $a_{ij} = 0, 1 \leq j \leq n$. Consider $\mathbf{C} = [c_{ij}] = \mathbf{A}\mathbf{B}$, then $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = 0$ for $1 \leq k \leq t$. Hence the i^{th} row of $\mathbf{A}\mathbf{B}$ is all zeros.
 (b) As in part (a), if $b_{jk} = 0$ for $1 \leq k \leq n, c_{ik} = \sum_{j=1}^n a_{ij}b_{jk} = 0$ for $1 \leq i \leq m$. Hence the k^{th} column of $\mathbf{A}\mathbf{B}$ is all zeros.
17. The j^{th} column of $\mathbf{A}\mathbf{B}$ has entries $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$. Let $\mathbf{D} = [d_{ij}] = \mathbf{A}\mathbf{B}_j$, where $\mathbf{B}_j$ is the j^{th} column of $\mathbf{B}$. Then $d_{ij} = \sum_{m=1}^n a_{im}b_{mj} = c_{ij}$.
18. One such pair is $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & -2 \\ 0 & 1 \end{bmatrix}$.
19. $\mathbf{A}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}_2$.
20. $a = 1/b; b$, any real number; $c = 0$.
21. (a) $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$ by Theorem 3. Since $\mathbf{A}$ and $\mathbf{B}$ are symmetric, $\mathbf{A}^T + \mathbf{B}^T = \mathbf{A} + \mathbf{B}$ and $\mathbf{A} + \mathbf{B}$ is also symmetric.
 (b) $(\mathbf{A}\mathbf{B})^T = \mathbf{B}^T\mathbf{A}^T = \mathbf{B}\mathbf{A}$, but this may not be $\mathbf{A}\mathbf{B}$ so $\mathbf{A}\mathbf{B}$ may not be symmetric. Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$. Then $\mathbf{A}\mathbf{B}$ is not symmetric.
22. (a) Both $\mathbf{A}\mathbf{A}^T$ and $\mathbf{A}^T\mathbf{A}$ are defined because $\mathbf{A}$ is $n \times n$. $(\mathbf{A}\mathbf{A}^T)^T = (\mathbf{A}^T)^T\mathbf{A}^T$ (Theorem 3) $= \mathbf{A}\mathbf{A}^T$. Hence, $\mathbf{A}\mathbf{A}^T$ is symmetric. Similarly, we have $(\mathbf{A}^T\mathbf{A})^T = \mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{A}$ is symmetric.
 (b) By Theorem 3, $(\mathbf{A} + \mathbf{A}^T)^T = \mathbf{A}^T + (\mathbf{A}^T)^T = \mathbf{A}^T + \mathbf{A} = \mathbf{A} + \mathbf{A}^T$ and $\mathbf{A} + \mathbf{A}^T$ is symmetric.
23. (a) The ij -element of $(\mathbf{A}^T)^T$ is the ji -element of $\mathbf{A}^T$. But the ji -element of $\mathbf{A}^T$ is the ij -element of $\mathbf{A}$. Thus $(\mathbf{A}^T)^T = \mathbf{A}$.
 (b) The ij -element of $(\mathbf{A} + \mathbf{B})^T$ is the ji -element of $\mathbf{A} + \mathbf{B}$, $a_{ji} + b_{ji}$. But this is the sum of the ji -entry of $\mathbf{A}$ and the ji -entry of $\mathbf{B}$. It is also the sum of the ij -entry of $\mathbf{A}^T$ and the ij -entry of $\mathbf{B}^T$. Thus $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$.
 (c) Let $\mathbf{C} = [c_{ij}] = (\mathbf{A}\mathbf{B})^T$. Then $c_{ij} = \sum_{k=1}^n a_{jk}b_{ki}$, the ji -entry of $\mathbf{A}\mathbf{B}$. Let $\mathbf{D} = [d_{ij}] = \mathbf{B}^T\mathbf{A}^T$, then $d_{ij} = \sum_{k=1}^n b'_{ik}a'_{kj} = \sum_{k=1}^n b_{ki}a_{jk} = \sum_{k=1}^n a_{jk}b_{ki} = c_{ij}$. Hence $(\mathbf{A}\mathbf{B})^T = \mathbf{B}^T\mathbf{A}^T$.
24. Yes, if $\mathbf{A}$ is symmetric and $\mathbf{A}^{-1}$ exists, then $\mathbf{A} = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $\mathbf{A}^{-1} = \begin{bmatrix} \frac{a}{a^2-b^2} & \frac{-b}{a^2-b^2} \\ \frac{-b}{a^2-b^2} & \frac{a}{a^2-b^2} \end{bmatrix}$.
25. (a). $\begin{bmatrix} \frac{7}{9} & \frac{-1}{9} \\ \frac{-5}{9} & \frac{2}{9} \end{bmatrix}$ (b). $\begin{bmatrix} \frac{-9}{27} & \frac{4}{27} \\ \frac{0}{27} & \frac{3}{27} \end{bmatrix}$ (c). $\begin{bmatrix} \frac{2}{32} & \frac{5}{32} \\ \frac{4}{32} & \frac{-6}{32} \end{bmatrix}$
26. (a). $\begin{bmatrix} \frac{2}{11} & \frac{3}{11} \\ \frac{9}{11} & \frac{8}{11} \end{bmatrix}$ (b). $\begin{bmatrix} \frac{10}{34} & \frac{-4}{34} \\ \frac{1}{34} & \frac{3}{34} \end{bmatrix}$ (c). $\begin{bmatrix} \frac{4}{46} & \frac{-9}{46} \\ \frac{-6}{46} & \frac{2}{46} \end{bmatrix}$
27. (a). $\mathbf{C} = \mathbf{A}^{-1}$ since $\mathbf{CA} = \mathbf{I}_3$. (b). $\mathbf{D} = \mathbf{B}^{-1}$ since $\mathbf{DB} = \mathbf{I}_3$.

28. $CD = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 2 & 0 \\ 1 & 0 & -1 \end{bmatrix}, AB = \begin{bmatrix} 0.4 & -0.2 & 0.4 \\ 0.2 & -0.6 & 0.2 \\ 0 & 0 & 1 \end{bmatrix}$, but $(CD)(AB) \neq I_3$.

29. Since $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_n B = B^{-1}B = I_n, (B^{-1}A^{-1}) = (AB)^{-1}$.

30. $A \vee B$ $A \wedge B$ $A \odot B$

(a) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

31. (a) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

32. (a) 0, because $a_{ij} = 1$ would force $b_{ij} = 1$.
 (b) 1, because $a_{ij} = 0$ would force $b_{ij} = 0$.

33. Let $C = [c_{ij}] = A \vee B$ and $D = [d_{ij}] = B \vee A$. $c_{ij} = \begin{cases} 1 & a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & a_{ij} = 0 \text{ or } b_{ij} = 0 \end{cases} = d_{ij}$

Hence $C = D$.

34. An argument similar to that for Exercise 27 shows that $A \wedge B = B \wedge A$.

35. Let $[d_{ij}] = B \vee C$, $[e_{ij}] = A \vee (B \vee C)$, $[f_{ij}] = A \vee B$, and $[g_{ij}] = (A \vee B) \vee C$.

Then $d_{ij} = \begin{cases} 1 & \text{if } b_{ij} = 1 \text{ or } c_{ij} = 1 \\ 0 & \text{otherwise} \end{cases}$ $e_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } d_{ij} = 1 \\ 0 & \text{otherwise} \end{cases}$

But this means $d_{ij} = 1$ if $a_{ij} = 1$ or $b_{ij} = 1$ or $c_{ij} = 1$ and $d_{ij} = 0$ otherwise.

$f_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & \text{otherwise} \end{cases}$ $g_{ij} = \begin{cases} 1 & \text{if } f_{ij} = 1 \text{ or } c_{ij} = 1 \\ 0 & \text{otherwise} \end{cases}$

But this means $g_{ij} = 1$ if $a_{ij} = 1$ or $b_{ij} = 1$ or $c_{ij} = 1$ and $g_{ij} = 0$ otherwise. Hence

$A \vee (B \vee C) = (A \vee B) \vee C$.

36. A similar argument to that in Exercise 35 using "and" in place of "or" shows

$A \wedge (B \wedge C) = (A \wedge B) \wedge C$.

37. Let $[d_{ij}] = B \odot C$, $[e_{ij}] = A \odot (B \odot C)$, $[f_{ij}] = A \odot B$, and $[g_{ij}] = (A \odot B) \odot C$.

Then $d_{ij} = \begin{cases} 1 & \text{if } b_{ik} = c_{kj} \text{ for some } k \\ 0 & \text{otherwise} \end{cases}$ and $e_{ij} = \begin{cases} 1 & \text{if } a_{il} = 1 = d_{lj} \text{ for some } l \\ 0 & \text{otherwise} \end{cases}$

But this means $e_{ij} = \begin{cases} 1 & a_{il} = 1 = b_{lk} = c_{kj} \text{ for some } k, l \\ 0 & \text{otherwise} \end{cases}$.

$f_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 = b_{kj} \text{ for some } k \\ 0 & \text{otherwise} \end{cases}$ and $g_{ij} = \begin{cases} 1 & \text{if } f_{il} = 1 = c_{lj} \text{ for some } l \\ 0 & \text{otherwise} \end{cases}$

But then $g_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 = b_{kl} = c_{lj} \text{ for some } k, l \\ 0 & \text{otherwise} \end{cases}$

and $A \odot (B \odot C) = (A \odot B) \odot C$.

38. Let $[d_{ij}] = \mathbf{B} \vee \mathbf{C}$, $[e_{ij}] = \mathbf{A} \wedge (\mathbf{B} \vee \mathbf{C})$, $[f_{ij}] = \mathbf{A} \wedge \mathbf{B}$, $[g_{ij}] = \mathbf{A} \wedge \mathbf{C}$, and $[h_{ij}] = (\mathbf{A} \wedge \mathbf{B}) \vee (\mathbf{A} \wedge \mathbf{C})$. Then $e_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 = d_{ij} \\ 0 & \text{otherwise} \end{cases}$ or
- $$e_{ij} = \begin{cases} 1 & \text{if } (a_{ij} = 1 = b_{ij}) \text{ or } (a_{ij} = 1 = c_{ij}) \\ 0 & \text{otherwise} \end{cases}$$
- $$h_{ij} = \begin{cases} 1 & \text{if } f_{ij} = 1 \text{ or } g_{ij} = 1 \\ 0 & \text{otherwise} \end{cases} \text{ or } h_{ij} = \begin{cases} 1 & \text{if } (a_{ij} = 1 = b_{ij}) \text{ or } (a_{ij} = 1 = c_{ij}) \\ 0 & \text{otherwise} \end{cases}$$
- But then $e_{ij} = h_{ij}$ for all i, j and $\mathbf{A} \wedge (\mathbf{B} \vee \mathbf{C}) = (\mathbf{A} \wedge \mathbf{B}) \vee (\mathbf{A} \wedge \mathbf{C})$.
39. An argument similar to that in Exercise 38 shows that $\mathbf{A} \vee (\mathbf{B} \wedge \mathbf{C}) = (\mathbf{A} \vee \mathbf{B}) \wedge (\mathbf{A} \vee \mathbf{C})$.
40. Example 8 shows that matrix multiplication is not commutative.
41. Since $c_{ij} = \sum_{t=1}^n a_{it}b_{tj}$ and $k \mid a_{it}$ for any i and t , k divides each term in c_{ij} , and thus $k \mid c_{ij}$ for all i and j .
42. Let $[c_{ij}] = \mathbf{A} + \mathbf{B}$ and $[d_{ij}] = \mathbf{A} - \mathbf{B}$. We know that $p \mid c_{ij}$ for all i, j and $p \mid d_{ij}$ for all i, j . Consider $[c_{ij}] + [d_{ij}]$. This matrix is $2\mathbf{A}$. Since $p \mid (c_{ij} + d_{ij})$, $p \mid 2a_{ij}$, but p does not divide 2, so $p \mid a_{ij}$ all i, j . Similarly, p divides all entries of $[c_{ij}] - [d_{ij}] = 2\mathbf{B}$ and $p \mid b_{ij}$ for all i, j .
43. (a) $\begin{bmatrix} 6 & -9 & -3 \\ 0 & 15 & 6 \\ 12 & -12 & 18 \end{bmatrix}$ (b) $\begin{bmatrix} 10 & 20 & -30 \\ 20 & 0 & 45 \\ 35 & -5 & 15 \end{bmatrix}$ (c) $\begin{bmatrix} -4 & 0 \\ -3 & -1 \\ 2 & -5 \end{bmatrix}$
44. (a) $\begin{bmatrix} 12 & 3 & -21 \\ 12 & 15 & 33 \\ 33 & -15 & 27 \end{bmatrix}$ (b) $\begin{bmatrix} 12 & 3 & -21 \\ 12 & 15 & 33 \\ 33 & -15 & 27 \end{bmatrix}$
- (c) $\begin{bmatrix} -2 & 16 \\ -22 & -30 \\ 16 & -52 \end{bmatrix}$ (d) $\begin{bmatrix} -2 & 16 \\ -22 & -30 \\ 16 & -52 \end{bmatrix}$
45. Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be two $m \times n$ matrices. Then $k(\mathbf{A} + \mathbf{B}) = k[a_{ij} + b_{ij}] = [k(a_{ij} + b_{ij})] = [ka_{ij} + kb_{ij}] = [ka_{ij}] + [kb_{ij}] = k\mathbf{A} + k\mathbf{B}$.
46. Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix and $\mathbf{B} = [b_{ij}]$ be an $n \times p$ matrix. Then $k(\mathbf{AB}) = k[a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}] = [ka_{i1}b_{1j} + ka_{i2}b_{2j} + \cdots + ka_{in}b_{nj}] = [ka_{ij}] \cdot [b_{ij}] = (k\mathbf{A})\mathbf{B}$. But also $[ka_{i1}b_{1j} + ka_{i2}b_{2j} + \cdots + ka_{in}b_{nj}] = [a_{i1}kb_{1j} + a_{i2}kb_{2j} + \cdots + a_{in}kb_{nj}] = [a_{ij}] \cdot [kb_{ij}] = \mathbf{A}(k\mathbf{B})$.
47. Let $\mathbf{K}$ be the $m \times m$ diagonal matrix with each diagonal entry equal to k . Then $\mathbf{KA} = k\mathbf{A}$.

Exercise Set 1.6, page 44

1. (a) yes (b) yes
2. (a) yes (b) no
3. (a) no (b) yes
4. (a) yes (b) no
5. $A \oplus B = \{x \mid (x \in A \cup B) \text{ and } x \notin A \cap B\} = \{x \mid (x \in B \cup A) \text{ and } x \notin B \cap A\} = B \oplus A$
6. (a) $0 \square (0 \square 0) = 0 \square 0 = 0 = (0 \square 0) \square 0$
 $0 \square (0 \square 1) = 0 \square 1 = 1 = (0 \square 0) \square 1$
 $0 \square (1 \square 0) = 0 \square 1 = 1 = (0 \square 1) \square 0$
 $0 \square (1 \square 1) = 0 \square 0 = 0 = (0 \square 1) \square 1$
 $1 \square (0 \square 0) = 1 \square 0 = 1 = (1 \square 0) \square 0$
 $1 \square (0 \square 1) = 1 \square 1 = 0 = (1 \square 0) \square 1$
 $1 \square (1 \square 0) = 1 \square 1 = 0 = (1 \square 1) \square 0$
 $1 \square (1 \square 1) = 1 \square 0 = 1 = (1 \square 1) \square 1$
- (b) A case-by-case check as in part (a) shows that ∇ is associative.

7.	x	y	z	$y \sqcap z$	$x \nabla (y \sqcap z)$	$x \nabla y$	$x \nabla z$	$(x \nabla y) \sqcap (x \nabla z)$
	0	0	0	0	0	0	0	0
	0	0	1	1	0	0	0	0
	0	1	0	1	0	0	0	0
	0	1	1	0	0	0	0	0
	1	0	0	0	0	0	0	0
	1	0	1	1	0	1	1	1
	1	1	0	1	1	1	0	1
	1	1	1	0	0	1	1	0
	(A)				(B)			

Since columns (A) and (B) are identical, the distributive property

$$x \nabla (y \sqcap z) = (x \nabla y) \sqcap (x \nabla z) \text{ holds.}$$

8. (a) 0 for $+$; 1 for $*$ (b) $\{ \}$ for $\cup$; U for $\cap$
(c) 0 for $\sqcap$; 1 for ∇ (d) $\{ \}$ for $\oplus$; none for $-$
9. 5×5 zero matrix for $\vee$; 5×5 matrix of 1's for $\wedge$; I_5 for $\odot$
10. Let $\mathbf{A}$ and $\mathbf{B}$ be two $n \times n$ diagonal matrices. Let $[e_{ij}] = \mathbf{A} + \mathbf{B}$. Then $e_{ij} = a_{ij} + b_{ij}$, but $a_{ij} = 0$ if $i \neq j$ and $b_{ij} = 0$ if $i \neq j$. Hence $e_{ij} = 0$ if $i \neq j$ and $\mathbf{A} + \mathbf{B}$ is an $n \times n$ diagonal matrix.
11. Let $\mathbf{A}, \mathbf{B}$ be $n \times n$ diagonal matrices. Let $[c_{ij}] = \mathbf{AB}$. Then $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$, but $a_{ik} = 0$ if $i \neq k$. Hence $c_{ij} = a_{ii}b_{ij}$. But $b_{ij} = 0$ if $i \neq j$. Thus $c_{ij} = 0$ if $i \neq j$ and $\mathbf{AB}$ is an $n \times n$ diagonal matrix.
12. Let $\mathbf{A}$ be an $n \times n$ diagonal matrix. Then $\mathbf{A}^T$ is $n \times n$ and the ij -entry of $\mathbf{A}^T$ is the ji -entry of $\mathbf{A}$. If $j \neq i$, this entry is 0, so $\mathbf{A}^T$ is also a diagonal matrix.
13. Yes, the $n \times n$ zero matrix, which is a diagonal matrix.
14. Yes, $\mathbf{I}_n$ is an $n \times n$ diagonal matrix.
15. $-\mathbf{A}$ is the diagonal matrix with ii -entry $-a_{ii}$.
16. $\mathbf{A}^{-1}$ is the diagonal matrix with ii -entries $1/a_{ii}$. If any a_{ii} is 0, then $\mathbf{A}^{-1}$ does not exist.
17. $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix}$ or $\begin{bmatrix} a+b & 0 \\ 0 & 0 \end{bmatrix}$ belongs to M .
18. $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix}$ or $\begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix}$ belongs to M .
19. $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}^T$ or $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$ belongs to M .
20. Yes, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ since 0 is a real number.
21. Yes, $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. By Exercise 18, we see this is the identity and $1 \in \mathbb{R}$.
22. If $\mathbf{A} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$, then $-\mathbf{A} = \begin{bmatrix} -a & 0 \\ 0 & 0 \end{bmatrix}$. $-a$ is also in $\mathbb{R}$.
23. If $a \neq 0$, then $\mathbf{A}^{-1} = \begin{bmatrix} a^{-1} & 0 \\ 0 & 0 \end{bmatrix}$.
24. Yes
25. Yes
26. No
27. No
28. No
29. (a) Yes (b) Yes (c) Yes
30. (a) Yes, $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$ (b) Yes, $\begin{bmatrix} -x \\ -y-2 \end{bmatrix}$
31. (a) Yes (b) Yes (c) No
32. (a) No (b) No

33. (a) $k\left(\begin{bmatrix} x \\ y \end{bmatrix} \nabla \begin{bmatrix} w \\ z \end{bmatrix}\right) = \left(k\begin{bmatrix} x \\ y \end{bmatrix} \nabla k\begin{bmatrix} w \\ z \end{bmatrix}\right)$ (b) No
34. (a) $k([x \ y] \square [w \ z]) = k[x \ y] \square k[w \ z]$ (b) Yes
35. Let $\mathbf{C} = [c_{ij}] = \text{comp}(\mathbf{A} \vee \mathbf{B})$ and $\mathbf{D} = [d_{ij}] = \text{comp}(\mathbf{A}) \wedge \text{comp}(\mathbf{B})$.
 Then $c_{ij} = \begin{cases} 0 & a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 1 & a_{ij} = 0 = b_{ij} \end{cases}$ and $d_{ij} = \begin{cases} 0 & a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 1 & a_{ij} = 0 = b_{ij} \end{cases}$. Hence, $\mathbf{C} = \mathbf{D}$. Similarly, we can show that $\text{comp}(\mathbf{A} \wedge \mathbf{B}) = \text{comp}(\mathbf{A}) \vee \text{comp}(\mathbf{B})$
36. A
37. $A \cap \overline{B}$
38. $\overline{A \cap B}$
39. $\{\}$

Review Questions, page 47

- $P(A)$ is a set of sets.
- $|P(A)|$ is a counting number or infinity.
- $\text{LCM}(a, b)$ is a positive integer.
- $k\mathbf{A}$ is a matrix of the same size as $\mathbf{A}$.
- A mathematical structure consists of a set of objects, operations on those objects, and the properties of those operations.

Chapter 2 Answers

Exercise Set 2.1, page 55

- (b), (d), and (e) are statements.
- (a) $2 + 7 > 11$ (b) 2 is not an even integer or 8 is not an odd integer.
- (a) It will not rain tomorrow and it will not snow tomorrow.
(b) It is not the case that if you drive, I will walk.
- (a) $3 + 1 < 5$ and $7 = 3 \cdot 6$ $3 + 1 < 5$ or $7 = 3 \cdot 6$
(b) I am rich and I am happy. I am rich or I am happy.
- (a) I will drive my car and I will be late. I will drive my car or I will be late.
(b) $10 < \text{NUM} \leq 15$ $\text{NUM} > 10$ or $\text{NUM} \leq 15$
- (a) True (b) False (c) False (d) False
- (a) True (b) True (c) True (d) False
- (a) False (b) False (c) True (d) False
- (a) False (b) True (c) True (d) False
- (d) is the negation.
- (d) is the negation.
- (a) $p \wedge \sim r$ (b) $q \vee p$ (c) $\sim p \wedge \sim q$ (d) $r \wedge q$
- (a) The dish did not run away with the spoon and the grass is wet.
(b) The grass is dry or the dish ran away with the spoon.
(c) It is not true that the today is Monday or the grass is wet.
(d) Today is Monday or the dish did not run away with the spoon.
- (a) For all x , x is even.
(b) There exists an x such that x is a prime number.
- (a) For all x there exists a y such that $x + y$ is even.
(b) There exists an x such that for all y , $x + y$ is even.
- (a) For all x , x is not a prime number.
(b) There exists a y such that y is odd.
- (a) It is not true that there is an x such that x is even.
(b) It is not true that for all x , x is a prime number.
- (a) $\forall x (\sim P(x))$ (b) $\forall x \forall y R(x, y)$.
(c) $\sim (\exists x (P(x) \wedge Q(x)))$ (d) $\forall x (P(x) \vee Q(x))$

19. 14 (a) false (b) true 15 (a) true (b) false 16 (a) false (b) true
 17 (a) false (b) true 18 (a) false (b) false (c) false (d) false
 20. $\forall a \forall b a + b = b + a$
 21. $\forall A \forall B (\overline{A \cup B}) = \overline{A} \cap \overline{B}$; $\forall A \forall B (\overline{A \cap B}) = \overline{A} \cup \overline{B}$
 22. $\forall x \neq 0 \exists y (xy = 1)$

23.

p	q	$(\sim p \wedge q) \vee p$
T	T	F
T	F	F
F	T	T
F	F	T

(1) (2) ↑

24.

p	q	$(p \vee q) \vee \sim q$
T	T	T
T	F	T
F	T	T
F	F	F

(1) ↑ (2)

25.

p	q	r	$(p \vee q) \wedge r$
T	T	T	T
T	T	F	F
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	F
F	F	F	F

(1) ↑

26.

p	q	r	$(\sim p \vee q) \wedge \sim r$
T	T	T	F
T	T	F	T
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	T
F	F	T	F
F	F	F	T

(1) ↑

27.

p	q	r	$(p \downarrow q) \downarrow r$
T	T	T	F
T	T	F	T
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

(1) ↑

28.

p	q	r	$(p \downarrow q) \wedge (p \downarrow r)$
T	T	T	F
T	F	T	F
T	F	F	F
F	T	T	F
F	T	F	F
F	F	T	T
F	F	F	T

(1) ↑ (2)

29.

p	q	r	$(p \downarrow q) \downarrow (p \downarrow r)$
T	T	T	F
T	T	F	F
T	F	T	F
T	F	F	F
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

(1) ↑ (2)

30. (a)

p	q	$p \triangle q$
T	T	F
T	F	T
F	T	T
F	F	F

(b)

p	$p \triangle \sim p$
T	F
F	T

↑

31.

p	q	$(p \wedge q) \triangle p$
T	T	F
T	F	T
F	T	T
F	F	F
	(1)	↑

32.

p	q	r	$(p \triangle q) \triangle (q \triangle r)$
T	T	T	F
T	T	F	F
T	F	T	T
T	F	F	T
F	T	T	T
F	T	F	T
F	F	T	F
F	F	F	F
	(1)	↑	(2)

33. $(x = \text{max}) \text{ or } (y \leq 4)$
 34. $(\text{key} \neq \text{"open"}) \text{ and } (t \geq \text{limit})$
 35. WHILE (item = sought or index $\geq$ 101) take action
 36. IF (cell $\leq$ 0 and not found) THEN take action

Exercise Set 2.2, page 60

1. (a) $p \Rightarrow q$ (b) $r \Rightarrow p$ (c) $q \Rightarrow p$ (d) $\sim r \Rightarrow p$
2. (a) $\sim p \Leftrightarrow r$ (b) $r \Rightarrow (p \wedge q)$ (c) $\sim q \Rightarrow (p \wedge \sim r)$ (d) $(\sim p \wedge r) \Rightarrow q$
3. (a) If I am not the Queen of England, then $2 + 2 = 4$.
 (b) If I walk to work, then I am not the President of the United States.
 (c) If I did not take the train to work, then I am late.
 (d) If I go to the store, then I have time and I am not too tired.
 (e) If I buy a car and I buy a house, then I have enough money.
4. (a) If I am the Queen of England, then $2 + 2 \neq 4$.
 (b) If I don't walk to work, then I am the President of the United States.
 (c) If I take the train to work, then I am not late.
 (d) If I don't go to the store, then I will not have time or I am too tired.
 (e) If I don't buy a car or a house, then I will not have enough money.
5. (a) True (b) False (c) True (d) True
6. (a) $\sim r \Rightarrow q$ (b) $\sim q \wedge p$ (c) $q \Rightarrow \sim p$ (d) $\sim p \Rightarrow \sim r$
7. (a) If I don't study discrete structures and I go to a movie, then I am in a good mood.
 (b) If I am in a good mood, then I will study discrete structures or I will go to a movie.
 (c) If I am not in a good mood, then I will not go to a movie or I will study discrete structures.
 (d) I will go to a movie and I will not study discrete structures if and only if I am in a good mood.
8. (a) $p \vee q$ (b) $r \rightarrow s$ (c) $\sim (s \vee p)$
9. (a) If $4 > 1$ and $2 > 2$, then $4 < 5$. (b) It is not true that $3 \leq 3$ and $4 < 5$.
 (c) If $3 > 3$, then $4 > 1$.
10. (a)

p	$p \wedge \sim p$
T	F
F	F

 absurdity
 (b)

p	q	$q \vee (\sim q \wedge p)$
T	T	F
T	F	T
F	T	F
F	F	F

 contingency
 ↑
11. (a)

p	q	$p \Rightarrow (q \Rightarrow p)$
T	T	T
T	F	T
F	T	F
F	F	T

 ↑ tautology
 (b)

p	q	$q \Rightarrow (q \Rightarrow p)$
T	T	T
T	F	T
F	T	F
F	F	T

 ↑ contingency

12. (a) $p \quad q \quad (q \wedge p) \vee (q \wedge \sim p)$
- | | | | | |
|---|---|---|---|---|
| T | T | T | T | F |
| T | F | F | F | F |
| F | T | F | T | T |
| F | F | F | F | F |
- (1) ↑ (2)
- contingency
- (b) $p \quad q \quad (p \wedge q) \Rightarrow p$ tautology
- | | | | |
|---|---|---|---|
| T | T | T | T |
| T | F | F | T |
| F | T | F | T |
| F | F | F | T |
- (1) ↑

- (c) $p \quad q \quad p \Rightarrow (q \wedge p)$
- | | | | |
|---|---|---|---|
| T | T | T | T |
| T | F | F | F |
| F | T | T | F |
| F | F | T | F |
- ↑
- contingency

13. Yes. If $p \Rightarrow q$ is false, then p is true and q is false. Hence $p \wedge q$ is false, $\sim(p \wedge q)$ is true, and $(\sim(p \wedge q)) \Rightarrow q$ is false.
14. Yes. If $p \Rightarrow q$ is false, then p is true and q is false. Hence $\sim p$ is false and $p \Leftrightarrow q$ is false. Thus, $\sim p \vee p \Leftrightarrow q$ is false.
15. No, because if $p \Rightarrow q$ is true, it may be that both p and q are true and so $(p \wedge q) \Rightarrow \sim q$ is false. But it could also be that p and q are both false, and then $(p \wedge q) \Rightarrow \sim q$ is true.
16. Yes, because if $p \Rightarrow q$ is true, then $\sim(p \Rightarrow q)$ is false and $\sim(p \Rightarrow q) \wedge \sim p$ is false.
17. (a) False (b) True (c) False (d) True
18. (a) True (b) True (c) True (d) True
19. $p \quad q \quad p \wedge q \quad (p \downarrow p) \downarrow (q \downarrow q)$ Since columns (A) and (B) are the same, the statements are equivalent.
- | | | | | | |
|---|---|---|---|---|---|
| T | T | T | F | T | F |
| T | F | F | F | F | T |
| F | T | F | T | F | F |
| F | F | F | T | F | T |
- (A) (B)

20. (a) The weather is good or I will go to work.
 (b) Carol is not sick and goes to the picnic and doesn't have a good time.
 (c) I will win the game and enter the contest.
21. (a) Jack did eat fat or he did not eat broccoli.
 (b) Mary did not lose her lamb and the wolf did not eat the lamb.
 (c) Tom stole a pie and ran away and the three pigs have some supper.
22. (a) (i) (b) (iv)

23. $p \quad q \quad r \quad p \wedge (q \vee r) \quad (p \wedge q) \vee (p \wedge r)$
- | | | | | | | |
|---|---|---|---|---|---|---|
| T | T | T | T | T | T | T |
| T | T | F | T | T | T | F |
| T | F | T | T | T | F | T |
| T | F | F | F | F | F | F |
| F | T | T | F | T | F | F |
| F | T | F | F | T | F | F |
| F | F | T | F | T | F | F |
| F | F | F | F | F | F | F |
- (A) (B)

Because (A) and (B) are the same the statements are equivalent.

24. $p \quad q \quad \sim(p \wedge q) \quad \sim p \vee \sim q$ Because columns (A) and (B) are the same, the statements are equivalent.
- | | | | | | | |
|---|---|---|---|---|---|---|
| T | T | F | T | F | F | F |
| T | F | T | F | F | T | T |
| F | T | T | F | T | T | F |
| F | F | T | F | T | T | T |
- (A) (B)

25.	p	q	$\sim (p \Leftrightarrow q)$	$(p \wedge \sim q) \vee (q \wedge \sim p)$
	T	T	F	T
	T	F	T	F
	F	T	T	F
	F	F	F	T
			(A)	(B)

Because columns (A) and (B) are the same, the statements are equivalent.

26. The statement $\exists x (P(x) \vee Q(x))$ is true if and only if $\exists x$ such that either $P(x)$ or $Q(x)$ is true, but this means either $\exists x P(x)$ or $\exists x Q(x)$ is true. This holds if and only if $\exists x P(x) \vee \exists x Q(x)$ is true.
27. The statement $\forall x (P(x) \wedge Q(x))$ is true if and only if $\forall x$ both $P(x)$ and $Q(x)$ are true, but this means $\forall x P(x)$ is true and $\forall x Q(x)$ is true. This holds if and only if $\forall x P(x) \wedge \forall x Q(x)$ is true.

28.	p	q	$(p \wedge q) \Rightarrow p$
	T	T	T
	T	F	F
	F	T	T
	F	F	T
			$\uparrow$
29.	p	q	$q \Rightarrow (p \vee q)$
	T	T	T
	T	F	T
	F	T	T
	F	F	T
			$\uparrow$

30.	p	q	$(p \wedge (p \Rightarrow q)) \Rightarrow q$
	T	T	T
	T	F	F
	F	T	T
	F	F	T
			$\uparrow$

31.	p	q	r	$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$
	T	T	T	T
	T	T	F	F
	T	F	T	F
	T	F	F	F
	F	T	T	T
	F	T	F	T
	F	F	T	T
	F	F	F	T
			(1)	(3)
			(2)	$\uparrow$
			(4)	

32. The statements in parts (e) and (f) of Theorem 4 are the contrapositives of each other so if one of them is true so is the other.
33. Because $(p \wedge q) \equiv (q \wedge p)$, parts (a) and (b) of Theorem 4 say the same thing; that is, either p or q can be considered the first statement.

Exercise Set 2.3, page 66

- Valid $((d \Rightarrow t) \wedge \sim t) \Rightarrow \sim d$
- Invalid
- Invalid
- Valid $((d \Rightarrow t) \wedge d) \Rightarrow t$
- Valid $((f \vee \sim w) \wedge w) \Rightarrow f$
- Valid $((f \vee w) \wedge \sim w) \Rightarrow f$
- Valid $[(ht \Rightarrow m) \wedge (m \Rightarrow hp)] \Rightarrow [\sim hp \Rightarrow \sim ht]$
- Valid $((g \rightarrow p) \wedge (\sim s \rightarrow \sim p) \wedge (s \rightarrow \sim v)) \rightarrow (v \rightarrow \sim g)$
- Invalid
- (a) Valid $((p \vee q) \wedge \sim q) \rightarrow p$ (b) Invalid
- (a) $((p \vee q) \wedge \sim q) \Rightarrow p$ (b) $((p \Rightarrow q) \wedge \sim p) \Rightarrow \sim q$
- (a) Invalid (b) Valid $(\sim (p \rightarrow q) \wedge p) \rightarrow \sim q$
- (a) $((p \Rightarrow q) \wedge (q \Rightarrow r)) \wedge ((\sim q) \wedge r) \Rightarrow p$ (b) $(\sim (p \Rightarrow q) \wedge p) \Rightarrow \sim q$
- Suppose m and n are even numbers. Then there exist integers j and k such that $m = 2j$ and

- $n = 2k$. $m + n = 2j + 2k = 2(j + k)$. Since $j + k$ is an integer, $m + n$ is even.
15. Suppose m and n are odd numbers. Then there exist integers j and k such that $m = 2j + 1$ and $n = 2k + 1$. $m + n = 2j + 1 + 2k + 1 = 2j + 2k + 2 = 2(j + k + 1)$. Since $j + k + 1$ is an integer, $m + n$ is even.
 16. Suppose m and n are even. Then there exist integers j and k such that $m = 2j$ and $n = 2k$. $mn = 2j \cdot 2k = 2(2jk)$. Since $2jk$ is an integer, mn is even and the system is closed with respect to multiplication.
 17. Suppose m and n are odd. Then there exist integers j and k such that $m = 2j + 1$ and $n = 2k + 1$. $mn = 2j \cdot 2k + 2j + 2k + 1 = 2(2jk + j + k) + 1$. Since $2jk + j + k$ is an integer, mn is odd and the system is closed with respect to multiplication.
 18. If n is even, Exercise 16 shows that $n^2 = n \cdot n$ is even. To show that n^2 is even implies that n is even, we show the contrapositive. Assume n is odd (not even), then Exercise 17 shows n^2 is odd.
 19. If $A = B$, then clearly, $A \subseteq B$ and $B \subseteq A$.
If $A \subseteq B$ and $B \subseteq A$, then $A \subseteq B \subseteq A$ and B must be the same as A .
 20. Assume $A \subseteq B$. Let $x \in \bar{B}$. Then $x \notin B$ and hence x could not belong to A . That is, $x \in \bar{A}$ and $\bar{B} \subseteq \bar{A}$.
Assume $\bar{B} \subseteq \bar{A}$. Let $x \in A$. Then $x \notin \bar{A}$ and so $x \notin \bar{B}$. But this means $x \in B$ so $A \subseteq B$.
 21. (a) If $A \subseteq B$, then $A \cup B \subseteq B$. But $B \subseteq A \cup B$. Hence $A \cup B = B$.
If $A \cup B = B$, then since $A \subseteq A \cup B$, we have $A \subseteq B$.
(b) If $A \subseteq B$, then $A \subseteq A \cap B$. But $A \cap B \subseteq A$. Hence $A \cap B = A$. If $A \cap B = A$, then since $A \cap B \subseteq B$, we have $A \subseteq B$.
 22. If k is odd, then $k = 2j + 1$ for an integer j , and $k^3 = (2j + 1)^3 = 8j^3 + 12j^2 + 6j + 1 = 2(j^3 + 6j^2 + 3j) + 1$ which is clearly an odd integer.
The converse is shown by using the contrapositive. If k is even, then $k = 2j$ for some integer j and $k^3 = (2j)^3 = 8j^3 = 2(4j^3)$ which is clearly an even integer.
 23. For $n = 41$, we have a counterexample. $41^2 + 41 \cdot 41 + 41$ is $41(41 + 41 + 1)$ or $41 \cdot 83$.
 24. Any five consecutive integers can be represented by $n, n + 1, n + 2, n + 3, n + 4$. Their sum is $5n + 10$ or $5(n + 2)$. This is clearly divisible by 5.
 25. $n^3 - n = n(n - 1)(n + 1)$, the product of three consecutive integers. One of these must be a multiple of 3, so $3 \mid n^3 - n$.
 26. For $n = 1$, $1 + 2^n = 3$ and $3^n = 3$. The statement is false.
 27. Invalid Multiplying by $x - 1$ may or may not preserve the order of the inequality.
 28. Invalid $A \neq 0$ does not guarantee A^{-1} exists.
 29. Valid
 30. Valid
 31. Let x and y be prime numbers, each larger than 2. Then x and y are odd and their sum is even (Exercise 15). The only even prime is two so $x + y$ is not a prime.
 32. We give a proof by contradiction. Let $l_2 \perp l_1$ and $l_3 \perp l_1$. If l_2 and l_3 intersect, then there is a triangle formed with two right angles. This is not possible in Euclidean geometry. Thus l_2 and l_3 cannot intersect.
 33. Suppose $x + y$ is rational. Then there are integers a and b such that $x + y = \frac{a}{b}$. Since x is rational, we can write $x = \frac{c}{d}$ with integers c and d . But now $y = x + y - x = \frac{a}{b} - \frac{c}{d}$. $y = \frac{ad - bc}{bd}$. Both $ad - bc$ and bd are integers. This is a contradiction since y is an irrational number and cannot be expressed as the quotient of two integers.
 34. Suppose that y is rational. Then there are integers m and n such that $y = \frac{m}{n}$. But then we would have $2y = \frac{2m}{n}$. This is a contradiction.

Exercise Set 2.4, page 72

Note: Only the outlines of the induction proofs are given. These are not complete proofs.

1. Basis Step: $n = 1$ $P(1)$: $2(1) = 1(1 + 1)$ is true.
Induction Step: $P(k)$: $2 + 4 + \cdots + 2k = k(k + 1)$
 $P(k + 1)$: $2 + 4 + \cdots + 2(k + 1) = (k + 1)(k + 2)$.
LHS of $P(k + 1)$: $2 + 4 + \cdots + 2k + 2(k + 1) = k(k + 1) + 2(k + 1) = (k + 1)(k + 2)$ RHS $P(k + 1)$

2. Basis Step: $n = 1$ $P(1)$: $(2 \cdot 1 - 1)^2 = \frac{1(2+1)(2-1)}{3}$ is true.
 Induction Step: LHS of $P(k+1)$: $1^2 + 3^2 + \dots + (2k-1)^2 + (2(k+1)-1)^2$

$$= \frac{k(2k+1)(2k-1)}{3} + (2(k+1)-1)^2 = \frac{k(2k+1)(2k-1)}{3} + \frac{3(2k+1)^2}{3}$$

$$= \frac{(2k+1)(k(2k-1)+3(2k+1))}{3} = \frac{(2k+1)(2k^2+5k+3)}{3}$$

$$= \frac{(k+1)(2k+1)(2k+3)}{3} = \frac{(k+1)(2(k+1)-1)(2(k+1)+1)}{3} \text{ RHS of } P(k+1)$$
3. Basis Step: $n = 0$ $P(0)$: $2^0 = 2^{0+1} - 1$ is true.
 Induction Step: LHS of $P(k+1)$: $1 + 2^1 + 2^2 + \dots + 2^k + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1}$

$$= 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1 \text{ RHS of } P(k+1)$$
4. Basis Step: $n = 1$ $P(1)$: $5 = \frac{5(1)(1+1)}{2}$ is true.
 Induction Step: LHS of $P(k+1)$: $5 + 10 + \dots + 5k + 5(k+1) = \frac{5k(k+1)}{2} + 5(k+1)$

$$= 5(k+1)\left(\frac{k}{2} + 1\right) = 5(k+1)\frac{(k+2)}{2} \text{ RHS of } P(k+1)$$
5. Basis Step: $n = 1$ $P(1)$: $1^2 = \frac{1(1+1)(2+1)}{6}$ is true.
 Induction Step: LHS of $P(k+1)$: $1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

$$= (k+1)\left(\frac{k(2k+1)}{6} + (k+1)\right) = \frac{k+1}{6}(2k^2 + k + 6(k+1)) = \frac{(k+1)}{6}(2k^2 + 7k + 6)$$

$$= \frac{(k+1)(k+2)(2k+3)}{6} = \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6} \text{ RHS of } P(k+1)$$
6. Basis Step: $n = 1$ $P(1)$: $1 = \frac{a^1-1}{a-1}$ is true.
 Induction Step: LHS of $P(k+1)$: $1 + a + \dots + a^{k-1} + a^k = \frac{a^k-1}{a-1} + a^k = \frac{a^k-1+a^{k+1}-a^k}{a-1}$

$$= \frac{a^{k+1}-1}{a-1} \text{ RHS of } P(k+1)$$
7. Basis Step: $n = 1$ $P(1)$: $a = \frac{a(1-r^1)}{1-r}$ is true.
 Induction Step: LHS of $P(k+1)$: $a + ar + \dots + ar^{k-1} + ar^k = \frac{a(1-r^k)}{1-r} + ar^k$

$$= \frac{a-ar^k+ar^k-ar^{k+1}}{1-r} = \frac{a(1-r^{k+1})}{1-r} \text{ RHS of } P(k+1)$$
8. (a) LHS of $P(k+1)$: $1^3 + 2^3 + 3^3 + \dots + (k+1)^3 = \frac{k^2(k+1)^2+4}{4} + (k+1)^3 = \frac{k^2(k+1)^2+4+4(k+1)^3}{4}$

$$= \frac{(k+1)^2(k^2+4(k+1))+4}{4} = \frac{(k+1)^2(k+2)^2+4}{4} \text{ RHS of } P(k+1)$$

 (b) No; $P(1)$: $1^3 = \frac{1^2(2)^2+4}{4}$ is false.
9. (a) LHS of $P(k+1)$: $1 + 5 + 9 + \dots + (4(k+1) - 3) = (2k+1)(k-1) + 4(k+1) - 3$

$$= 2k^2 + 3k = (2k+3)(k) = (2(k+1)+1)((k+1)-1) \text{ RHS of } P(k+1)$$

 (b) No; $P(1)$: $1 = (2 \cdot 1 + 1)(1 - 1)$ is false.
10. Basis Step: $n = 2$ $P(2)$: $1 + 2^2 < 3^2$ is true.
 Induction Step: LHS of $P(k+1)$: $1 + 2^{k+1} = 1 + 2 \cdot 2^k = 1 + 2^k + 2^k < 3^k + 2^k < 3^k + 2 \cdot 3^k$

$$= 3 \cdot 3^k = 3^{k+1} \text{ RHS of } P(k+1)$$
11. Basis Step: $n = 2$ $P(2)$: $2 < 2^2$ is true.
 Induction Step: LHS of $P(k+1)$: $k + 1 < 2^k + 1 < 2^k + 2^k = 2 \cdot 2^k = 2^{k+1} \text{ RHS of } P(k+1)$
12. Basis Step: $n = 1$ $P(1)$: $1 < \frac{(2+1)^2}{8}$ is true.
 Induction Step: LHS of $P(k+1)$: $1 + 2 + \dots + k + (k+1) < \frac{(2k+1)^2}{8} + (k+1)$

$$= \frac{4k^2+4k+1+8k+8}{8} = \frac{4k^2+12k+9}{8}$$

$$= \frac{(2k+3)^2}{8} = \frac{(2(k+1)+1)^2}{8} \text{ RHS of } P(k+1)$$

13. Basis Step: $n = 5$. $P(5)$: $1 + 5^2 < 2^5$ is true.
Induction Step: LHS of $P(k+1)$: $1 + (k+1)^2 = k^2 + 1 + 2k + 1 < 2^k + 2k + 1 < 2^k + k^2 + 1 < 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$ RHS of $P(k+1)$
14. Basis Step: $n = 4$. $P(4)$: $10 \cdot 4 < 3^4$ is true.
Induction Step: LHS of $P(k+1)$: $10(k+1) = 10k + 10 < 3^k + 10 < 3^k + 3^k + 3^k = 3^{k+1}$ RHS of $P(k+1)$
15. Basis Step: $n = 0$. $A = \{\}$ and $P(A) = \{\{\}\}$ so $|P(A)| = 2^0$ and $P(0)$ is true.
Induction Step: Use $P(k)$: If $|A| = k$, then $|P(A)| = 2^k$ to show $P(k+1)$: If $|A| = k+1$, then $|P(A)| = 2^{k+1}$. Suppose $|A| = k+1$. Set aside one element x of A . Then $|A - \{x\}| = k$ and $A - \{x\}$ has 2^k subsets. These subsets are also subsets of A . We can form another 2^k subsets of A by forming the union of $\{x\}$ with each subset of $A - \{x\}$. None of these subsets are duplicates. Now A has $2^k + 2^k$, or 2^{k+1} , subsets.
16. Basis Step: $n = 1$. $P(1)$: $3 \mid (1^3 - 1)$ is true.
Induction Step: Use $P(k)$: $3 \mid (k^3 - k)$ to show $P(k+1)$: $3 \mid ((k+1)^3 - (k+1))$.
 $((k+1)^3 - (k+1)) = k^3 + 3k^2 + 3k + 1 - (k+1) = k^3 - k + 3k^2 + 3k$
Since $3 \mid 3k^2 + 3k$, we have $3 \mid ((k+1)^3 - (k+1))$.
17. Basis Step: $n = 1$. $P(1)$: $\overline{A_1} = \overline{A_1}$ is true.
Induction Step: LHS of $P(k+1)$: $\overline{\bigcap_{i=1}^{k+1} A_i} = \overline{\left(\bigcap_{i=1}^k A_i\right) \cap A_{k+1}}$
 $= \overline{\bigcap_{i=1}^k A_i \cup \overline{A_{k+1}}}$ (De Morgan's laws)
 $= \left(\bigcup_{i=1}^k \overline{A_i}\right) \cup \overline{A_{k+1}} = \bigcup_{i=1}^{k+1} \overline{A_i}$ RHS of $P(k+1)$
18. Basis Step: $n = 1$. $P(1)$: $A_1 \cap B = A_1 \cap B$ is true.
Induction Step: LHS of $P(k+1)$: $\left(\bigcup_{i=1}^{k+1} A_i\right) \cap B = \left(\left(\bigcup_{i=1}^k A_i\right) \cup A_{k+1}\right) \cap B$
 $= \left(\left(\bigcup_{i=1}^k A_i\right) \cap B\right) \cup (A_{k+1} \cap B)$ (distributive property)
 $= \left(\bigcup_{i=1}^k (A_i \cap B)\right) \cup (A_{k+1} \cap B) = \bigcup_{i=1}^{k+1} (A_i \cap B)$ RHS of $P(k+1)$
19. Basis Step: $n = 1$. $P(1)$: $A_1 \cup B = A_1 \cup B$ is true.
Induction Step: LHS of $P(k+1)$: $\left(\bigcap_{i=1}^{k+1} A_i\right) \cup B = \left(\left(\bigcap_{i=1}^k A_i\right) \cap A_{k+1}\right) \cup B$
 $= \left(\left(\bigcap_{i=1}^k A_i\right) \cup B\right) \cap (A_{k+1} \cup B)$ (distributive property)
 $= \left(\bigcap_{i=1}^k (A_i \cup B)\right) \cap (A_{k+1} \cup B) = \bigcap_{i=1}^{k+1} (A_i \cup B)$ RHS of $P(k+1)$
20. (a) The only case to check is to show that if $2 \mid (2k-1)$, then $2 \mid (2(k+1)-1)$. Since $2(k+1)-1 = 2k-1+2$, $2 \mid (2k-1)$ implies $2 \mid (2k-1)+2$.
(b) $2n-1$ is never divisible by 2 since 2 does not divide 1.
(c) No, mathematical induction requires both the basis and the induction steps.
21. (a) $(k+1)^2 + (k+1) = k^2 + 2k + 1 + k + 1 = k^2 + k + 2(k+1)$. Using $P(k)$, $k^2 + k$ is odd; $2(k+1)$ is clearly even. Hence their sum is odd.
(b) No, $P(1)$ is false.
22. In the induction step if $k = 0$, we have $z^1 = \frac{z^0}{z^{-1}} \cdot z^0 = z$ which need not be 1.
23. The flaw is that to carry out the procedure in the induction step, you must have at least three trucks, but the basis step was done for one truck.

24. Basis Step: $n = 1$ $P(1)$: $A_1^T = A_1^T$ is true.
 Induction Step: LHS of $P(k+1)$: $(A_1 + A_2 + \cdots + A_k + A_{k+1})^T$
 $= (A_1 + A_2 + \cdots + A_k)^T + A_{k+1}^T = A_1^T + A_2^T + \cdots + A_k^T + A_{k+1}^T$ RHS of $P(k+1)$
25. Basis Step: $n = 1$ $P(1)$: $A^2 \cdot A = A^{2+1}$ is true.
 Induction Step: LHS of $P(k+1)$: $A^2 \cdot A^{k+1} = A^2(A^k \cdot A) = (A^2 \cdot A^k) \cdot A$
 $= A^{2+k} \cdot A = A^{2+k+1}$ RHS of $P(k+1)$
26. Basis Step: $n = 1$ $P(1)$: $(AB)^1 = A^1 \cdot B^1$ is true.
 Induction Step: LHS of $P(k+1)$: $(AB)^{k+1} = (AB)(AB)^k = AB \cdot A^k B^k = BA \cdot A^k B^k$
 $= A \cdot A^k B \cdot B^k = A^{k+1} \cdot B^{k+1}$ RHS of $P(k+1)$
27. Basis Step: $n = 5$ $P(5)$: A restaurant bill of \$5 can be paid exactly with a \$5 bill.
 Induction Step: We use $P(j)$: A restaurant bill of \$j can be paid exactly using \$2 and \$5 bills for $j = 5, 6, \dots, k$ to show $P(k+1)$. Write $k+1$ as $2m+r$, $0 \leq r < 2$. If $r = 0$, then the bill can be paid with m \$2 bills. If $r = 1$, then $2m+r = 2m+1 = 2(m-2)+5$. Since $k+1 \geq 7$, $m \geq 2$, and $m-2 \geq 0$. Thus a bill of $k+1$ dollars can be paid exactly with $(m-2)$ \$2 bills and a \$5 bill.
28. Basis Step: $n = 28$ $P(28)$: 28 can be written as $5 \cdot 4 + 8 \cdot 1$.
 Induction Step: We use $P(j)$ for $j = 28, 29, \dots, k$ to show $P(k+1)$. Consider $k+1 = (k+1-5) + 5 = (k-4) + 5$. $P(k-4)$ guarantees $k-4 = 5a' + 8b$, a', b in Z^+ . Hence, $k+1 = 5a + 8b$, a, b in Z^+ .
29. Basis Step: $n = 1$ $P(1)$: If p is prime and $p \mid a^1$, then $p \mid a$ is true.
 Induction Step: If $p \mid a^{k+1}$, then $p \mid a^k \cdot a$ and either $p \mid a^k$ or $p \mid a$. If $p \mid a^k$, then (using $P(k)$), $p \mid a$.
30. Basis Step: $n = 1$ $P(1)$: If $\text{GCD}(a, b) = 1$, then $\text{GCD}(a^1, b^1) = 1$ is true.
 Induction Step: Suppose $\text{GCD}(a, b) = 1$. Let $d = \text{GCD}(a^{k+1}, b^{k+1})$. If $d \neq 1$, then let p be a prime factor of d . Then $p \mid a^{k+1}$ and $p \mid b^{k+1}$. Thus $p \mid a$ and $p \mid b$. But this is a contradiction. Hence d must be 1.
31. (a) 5
 (b) Basis Step: $n = 5$ $P(5)$: $2^5 > 5^2$ is true.
 Induction Step: LHS of $P(k+1)$: $2^{k+1} = 2 \cdot 2^k = 2^k + 2^k > k^2 + k^2 > k^2 + 3k$
 $> k^2 + 2k + 1$ (since $k \geq 5$) But $k^2 + 2k + 1 = (k+1)^2$ RHS of $P(k+1)$
32. For $n = 3$, we have $2 + 8 + 18 = 9 + 3$ which is false.
33. Basis Step: $n = 1$ $P(1)$: $x - y$ divides $x - y$ is true.
 Induction Step: $x^{k+1} - y^{k+1} = x \cdot x^k - y \cdot y^k = x \cdot x^k - x \cdot y^k + x \cdot y^k - y \cdot y^k$
 $= x(x^k - y^k) + y^k(x - y)$. This rewriting gives an expression where each term is divisible by $x - y$ and so the sum is as well.
34. Loop invariant check:
 Basis Step: $n = 0$ $P(0)$: $Y \times W_0 + Z_0 = X + Y^2$ is true because $W_0 = Y$ and $Z_0 = X$.
 Induction Step: LHS of $P(k+1)$: $Y \times W_{k+1} + Z_{k+1} = Y \times (W_k - 1) + (Z_k + Y)$
 $= Y \times W_k + Z_k = X + Y^2$ RHS of $P(k+1)$
 Exit condition check: When $W = 0$, $Y \times W + Z = X + Y^2$ yields $Z = X + Y^2$
35. Loop invariant check:
 Basis Step: $n = 0$ $P(0)$: $X - Z_0 + W_0 = Y$ is true, because $Z_0 = X$ and $W_0 = Y$.
 Induction Step: LHS of $P(k+1)$: $X - Z_{k+1} + W_{k+1} = X - (Z_k - 1) + (W_k - 1)$
 $= X - Z_k + W_k = Y$ RHS of $P(k+1)$
 Exit condition check: $W = 0$ $X - Z + W = Y$ yields $X - Z = Y$ or $Z = X - Y$.
36. Loop invariant check:
 Basis Step: $n = 0$ $P(0)$: $R_0 \times N^{K_0} = N^{2M}$ is true, because $R_0 = 1$ and $K_0 = 2M$.
 Induction Step: LHS of $P(k+1)$: $R_{k+1} \times N^{K_{k+1}} = (R_k \times N) \times N^{K_k-1} = R_k \times N^{K_k}$
 $= N^{2M}$ RHS of $P(k+1)$
 Exit condition check: When $K = 0$, $R \times N^K = N^{2M}$ yields $R \times N^0 = N^{2M}$ or $R = N^{2M}$.
37. Loop invariant check:
 Basis Step: $n = 0$ $P(0)$: $Z_0 + (X \times W_0) = X \times Y$ is true, because $Z_0 = 0$ and $W_0 = Y$.
 Induction Step: LHS of $P(k+1)$: $Z_{k+1} + (X \times W_{k+1}) = Z_k + X + (X \times (W_k - 1))$
 $= Z_k + (X \times W_k) = X \times Y$ RHS of $P(k+1)$
 Exit condition check: $W = 0$ $Z + (X \times W) = X \times Y$ yields $Z = X \times Y$.
 Loop invariant check:
 Basis Step: $n = 0$ $P(0)$: $Z_0 + (X \times Y \times W_0) = X + Y^2$ is true, because $Z_0 = X \times Y$ and $W_0 = Y - 1$.

Induction Step: LHS of $P(k+1)$: $Z_{k+1} + (X \times Y \times W_{k+1}) = Z_k + X \times Y + (X \times Y \times (W_k - 1))$
 $= Z_k + X \times Y \times W_k = X + Y^2$ RHS of $P(k+1)$
Exit condition check: $W = 0$ $Z + X \times Y \times W = X + Y^2$ yields $Z = X + Y^2$.

38. Loop invariant check:

Basis Step: $n = 0$ $P(0)$: $R_0 + K_0 \times Y = X$ is true, because $R_0 = X$ and $K_0 = 0$.

Induction Step: LHS of $P(k+1)$: $R_{k+1} + K_{k+1} \times Y = (R_k - Y) + (K_k + 1) \times Y$
 $= R_k + K_k \times Y$
 $= X$ RHS of $P(k+1)$

Exit condition check: $R < Y$. $R + K \times Y = X$ and $R = 0$ if and only if $Y \mid X$.

39. Loop invariant check:

Basis Step: $n = 0$ $P(0)$: $Z_0 + (X \times W_0) = Y + X^2$ is true, because $Z_0 = Y$ and $W_0 = X$.

Induction Step: LHS of $P(k+1)$: $Z_{k+1} + (X \times W_{k+1}) = (Z_k + X) + (X \times (W_k - 1))$
 $= Z_k + X \times W_k = Y + X^2$ RHS of $P(k+1)$

Exit condition check: $W = 0$ $Z + (X \times W) = Y + X^2$ yields $Z = Y + X^2$

Loop invariant check:

Basis Step: $n = 0$ $P(0)$: $Z_0 + (Y \times W_0) = X^2 + Y^2$ is true, because $Z_0 = Y + X^2$ and $W_0 = Y - 1$.

Induction Step: LHS of $P(k+1)$: $Z_{k+1} + (Y \times W_{k+1}) = Z_k + Y + Y \times (W_k - 1)$
 $= Z_k + Y \times W_k = X^2 + Y^2$ RHS of $P(k+1)$

Exit condition check: $W = 0$ $Z + Y \times W = X^2 + Y^2$ yields $Z = X^2 + Y^2$

Review Questions, page 75

1. The converse is not equivalent to the original statement, but the contrapositive is. In some cases, the contrapositive may be easier to prove than the original statement.
2. In the strong form of induction all the statements $P(n_0)$, $P(n_0 + 1)$, ..., $P(k)$ may be used to show $P(k + 1)$, not just $P(k)$.
3. The mathematical structure of sets, union, intersection, and complement has the same properties as (logical statements, $\vee$, $\wedge$, $\sim$)
4. An indirect proof proves the contrapositive of the statement or assumes the statement is false and derives a contradiction.
5. A proof by contradiction proceeds by assuming the negation of the conclusion of the statement to be proved. Then definitions, previous theorems, and commonly known facts are used to derive a contradiction.

Chapter 3 Answers

Exercise Set 3.1, page 82

- | | | |
|---------------------|--------------------|------------------------|
| 1. 67,600 | 2. 120 | 3. 16 |
| 4. 120 | 5. 1296 | |
| 6. (a) 6 | (b) 2^7 or 128 | |
| 7. (a) 0 | (b) 1 | |
| 8. (a) 24 | (b) 720 | (c) 42 |
| 9. (a) $n!$ | (b) $\frac{n!}{2}$ | (c) $\frac{(n+1)!}{2}$ |
| 10. 24 | 11. 120 | |
| 12. 720 | | |
| 13. $4!$ or 24 | | |
| 14. 210 | | |
| 15. 30 | | |
| 16. 840 | | |
| 17. (a) 479,001,600 | (b) 1,036,800 | |

18. 120
19. 240
20. 2520
21. 360
22. 4,989,600
23. 39,916,800
24. 720
25. $(n-1)!$
26. Let $P(n)$: There are $(n-1)!$ ways to seat n people around a circular table. For $n=1$ and $n=2$, it is clear that this can be done in only one way. Since $(1-1)! = 1 = (2-1)!$, $P(1)$ and $P(2)$ are true. We now use $P(k)$: There are $(k-1)!$ ways to seat k people around a circular table to show $P(k+1)$: There are $k!$ ways to seat $k+1$ people around a circular table. Select one of the $k+1$ people to wait as the remaining k people are seated around the table. Now the last person can be seated to the right of any one of the k seated people. So there are $k \cdot (k-1)! = k!$ ways to seat the $k+1$ people.
27. 67,200
28. 216
29. $n \cdot {}_{n-1}P_{n-1} = n \cdot (n-1)(n-2) \cdots 2 \cdot 1 = n! = {}_nP_n$
30. 2,037,468,266,496
31. 190
32. Assuming that any digit may be used in any of the 9 positions, there could be 10^9 or 1,000,000,000. In fact, the first three digits are a geographic code.
33. 2; 6; 12
34. <no solution given>
35. (a) 14 (b) 11
36. (a) 196 (b) 98
37. (a) 16 (b) 12

Exercise Set 3.2, page 86

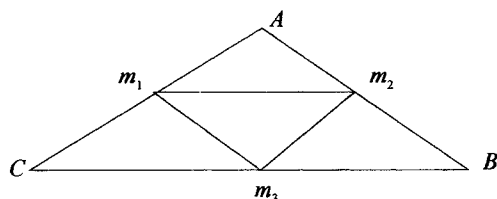
1. (a) 1 (b) 35 (c) 4368
2. (a) n (b) $\frac{n(n-1)}{2}$ (c) $\frac{n(n+1)}{2}$
3. ${}_nC_r = \frac{n!}{r!(n-r)!} = \frac{n!}{(n-(n-r))!(n-r)!} = {}_nC_{n-r}$
4. 980
5. 20,358,520
6. 5,096,520
7. (a) 1 (b) 360
8. 171,028,000
9. (a) One of size 0, four of size 1, six of size 2, four of size 3, and one of size 4.
(b) For each r , $0 \leq r \leq n$, there are ${}_nC_r$ subsets of size r .
10. (a) 56 (b) 21
11. (a) 980 (b) 1176
12. 2702 13. 2702
14. Since it does not matter where in the string the four ones are, the number of strings of the required type is the number of ways to choose 4 places out of 6 for the ones.
15. Because three people can be arranged in only one way from youngest to oldest, the problem is to count the number of ways to choose three people from seven.
16. 1260
17. 177,100
18. 35
19. ${}_nC_{r-1} + {}_nC_r = \frac{n!}{(r-1)!(n-(r-1))!} + \frac{n!}{r!(n-r)!} = \frac{n!r+n!(n-r+1)}{r!(n-r+1)!} = \frac{n!(n+1)}{r!(n+1-r)!}$

$$= \frac{(n+1)!}{r!(n+1-r)!} = {}_{n+1}C_r$$

20. (a) 45 (b) 21
 21. (a) 32 (b) 5 (c) 10
 22. (a) 216 (b) 75 (c) 15
 23. (a) 2^n (b) ${}_nC_3$ (c) ${}_nC_k$
 24. (a) 6^n (b) $n \cdot 5^{n-1}$ (c) ${}_nC_2 \cdot 5^{n-2}$
 25. 525
 26. 19,997,331,200
 27. (a) 1 5 10 10 5 1
 1 6 15 20 15 6 1
 1 7 21 35 35 21 7 1
 (b) Begin the row with a 1; write the sum of each consecutive pair of numbers in the previous row, moving left to right; end the row with a 1.
 28. The k^{th} -entry in the n^{th} row is ${}_nC_k$, $k = 0, 1, 2, \dots, n$.
 29. Exercise 19 shows another way to express the results of Exercise 27(b) and Exercise 28.
 30. Since ${}_nC_r = {}_nC_{n-r}$, each row is a palindrome.
 31. (a) 2 (b) 4 (c) 8
 32. It appears that the sum of the entries in the n^{th} row of Pascal's triangle is 2^{n-1} . One proof of this fact is that the entries in the n^{th} row of Pascal's triangle are the numbers of subsets of each possible size for a set with cardinality $n - 1$. (See Exercises 9 and 28.) But we know the sum of these numbers is the cardinality of the power set, or 2^{n-1} .
 33. 15

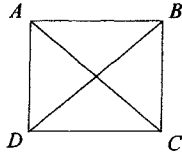
Exercise Set 3.3, page 90

1. Let the birth months play the role of the pigeons and the calendar months, the pigeonholes. Then there are 13 pigeons and 12 pigeonholes. By the pigeonhole principle, at least two people were born in the same month.
 2. Let $A_1 = \{1, 12\}$, $A_2 = \{2, 11\}$, $A_3 = \{3, 10\}$, $A_4 = \{4, 9\}$, $A_5 = \{5, 8\}$, and $A_6 = \{6, 7\}$ play the role of the pigeonholes. Assign each of the selected numbers (pigeons) to the set to which it belongs. At least two numbers will be assigned to the same set and their sum is 13.
 3.



- m_1, m_2, m_3 are the midpoints of sides AC, AB, and BC, respectively. Let the four small triangles created be the pigeonholes. For any five points in or on triangle ABC at least two must be in or on the same small triangle and are thus no more than $\frac{1}{2}$ unit apart.
 4. The possible remainders when a positive integer is divided by 7 are 0, 1, 2, 3, 4, 5, 6. The remainders are the pigeonholes. Assign each number (pigeon) to the pigeonhole labeled by the remainder when the number is divided by 7. Since there are 8 pigeons at least two of them must have the same remainder.
 5. By the extended pigeonhole principle, at least $\lfloor (50 - 1)/7 \rfloor + 1$ or 8 will be the same color.
 6. There are ${}_{10}C_3$ or 120 three-person committees (pigeons) and 10 hats (pigeonholes). By the extended pigeonhole principle, at least one hat has 12 or more slips.
 7. Let 2161 cents be the pigeons and the six friends, the pigeonholes. Then at least one friend has $\lfloor (2161 - 1)/6 \rfloor + 1$ or 361 cents.
 8. Let the pigeonholes be labeled 1 through 175, the possible charges. There are ${}_{12}C_5$ or 792 different choices that a customer can make. By the extended pigeonhole principle, there are at least $\left\lfloor \frac{(792-1)}{175} \right\rfloor + 1$ or 5 choices that cost the same.
 9. If repetitions are allowed, there are ${}_{16}C_5$ or 4368 choices. At least $\lfloor 4367/175 \rfloor + 1$, or 25, choices have the same cost.

10. There are ${}_{15}C_6$ or 5005 ways to choose the six numbers. The smallest possible sum is 21; the largest is 75. Hence there are 55 possible sums. At least $\left\lfloor \frac{5004}{55} \right\rfloor + 1$ or 91 choices must have the same sum.
11. You must have at least 49 friends.
12. If $AB = BC = 1$ inch, then $BD = AC = 2\sqrt{2}$ inches. For any five points in $\square ABCD$, at least two lie in the same triangle and can be at most 1 inch from each other, but $1 < \sqrt{2}$.



13. Consider the first eight rows; one row must have at least 7 ones since there are 51 ones in all. Similarly, there is a column with at least 7 ones. The sum of the entries in this row and this column is at least 14.
14. < no solution is given >
15. Label the pigeonholes with 1, 3, 5, ..., 25, the odd numbers between 1 and 25 inclusive. Assign each of the selected 14 numbers to the pigeonhole labeled with its odd part. There are only 13 pigeonholes so two numbers must have the same odd part. One is a multiple of the other.
16. If we label a set of pigeonholes with 1, 3, 5, ..., 49, then there are 25 pigeonholes. Assign a number to a pigeonhole labeled by its odd part. To guarantee that two numbers are assigned to the same pigeonhole we must have at least 26 numbers. Then as shown in Example 3, one of these numbers is a multiple of the other.
17. Using an argument similar to that for Exercise 16, the subset must contain at least $\left\lfloor \frac{n-1}{2} \right\rfloor + 1$ elements.
18. There are ten pairs whose sum is 21. It is possible to choose 10 disks without producing one of these pairs.
19. No. At least one pair of the 12 disks must add up to 21.
20. 10, because there are ten places to begin a triple. 3; $10 \cdot 15$ or 150; 3; 3 times 55 or 165
21. Consider the six sums $c_1, c_1 + c_2, c_1 + c_2 + c_3, \dots, c_1 + c_2 + c_3 + c_4 + c_5 + c_6$. If one of these has remainder 0 when divided by 6, then we are done. If none have remainder 0 when divided by 6, then two of them must give the same remainder. The positive difference of these two is the sum of a subsequence that is divisible by 6.
22. There are n sums $c_1, c_1 + c_2, c_1 + c_2 + c_3, \dots, c_1 + c_2 + c_3 + \dots + c_n$ and n possible remainders from division by n . If $\sum_{i=1}^k c_i$ has remainder 0 when divided by n , then we are done. Otherwise, two of the sums have the same remainder when divided by n and their positive difference is a subsequence whose sum is divisible by n .
23. Consider the cases: 3 or more ones, 2 ones, 1 one, and no ones. If there are at least 3 ones, we are done. If there are 2 ones and no two, then the sum is at least $1 + 1 + 4 \cdot 3$, but this is not possible. So if there are 2 ones, there is at least 1 two and we are done. If there is 1 one and no twos, then the sum is at least $1 + 5 \cdot 3$. This is impossible. If there are no ones and no three, then the sum is $6 \cdot 2$ or at least $5 \cdot 2 + 5$. But again, neither of these is possible.
24. Consider how division by n proceeds. At each step there is a remainder of 0, 1, ..., or $n - 1$. If the remainder is zero, the division ends and we have a finite decimal part. Let 1, ..., $n - 1$ label the pigeonholes. After at most n steps a remainder must appear a second time. After this the decimal part repeats.
25. There need to be at least 40 connections. If one printer has seven or fewer connections to PCs, then there can be a set of 5 PCs requesting printer access, but only four printers are available.

Exercise Set 3.4, page 98

1. {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}
2. {ab, ac, ad, bc, bd, cd}
3. {sb, sr, sg, cb, cr, cg}
4. {4, 5, 6, 7, 8, 9, 10, 11, 12}
5. (a) {}, {1}, {2}, {3}, {2, 3}, {1, 2}, {1, 3}, {1, 2, 3} (b) 2^n
6. (a) The card is black or is an ace. (b) The card is a black ace.
7. (a) The card is a red ace. (b) The card is black or a diamond or an ace.

8. (a) The card is black or not a diamond or an ace. (b) The card is black or it is a diamond that is not the ace.
9. (a) $\{(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (1, 5), (2, 5), (3, 5), (4, 5), (6, 5)\}$
(b) $\{\}$
10. (a) $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (5, 1)\}$
(b) $\{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}$
11. (a) No, 3 satisfies both descriptions. (b) No, 2 satisfies both descriptions.
(c) Yes, $E \cup F = \{3, 4, 5, 1, 2, 3\}$ (d) No, $E \cap F = \{3\}$
12. (a) $\frac{12}{52}$ (b) $\frac{5}{52}$ (c) $\frac{37}{52}$
13. (a) No (b) No (c) No (d) Yes
14. (a) Let S be the sample space. Then $E \cup \bar{E} = S$ and hence $E \cup \bar{E}$ is a certain event.
(b) $E \cap \bar{E} = \{\}$ so $E \cap \bar{E}$ is an impossible event.
15. (a) $\{dls, dln, dms, dmn, dus, dun, als, aln\}$
(b) $\{als, aln\}$ (c) $\{dls, dln, als\}$
16. $E \cup F = \{1, 2, 3, 4, 5\}$ $E \cap F = \{3\}$ $\bar{F} = \{1, 4, 5, 6\}$
17. $\bar{E} \cup F = \{2, 6, 3\}$ $\bar{F} \cap G = \{4\}$
18. $\{a\}, \{e\}, \{i\}, \{o\}, \{u\}$
19. $\{\text{club}\}, \{\text{spade}\}, \{\text{diamond}\}, \{\text{heart}\}$
20. (a) $\frac{1}{10,000}$ (b) $\frac{1}{24}$
21. (a) $\frac{10}{18}$ (b) $\frac{11}{18}$ (c) $\frac{12}{18}$ (d) $\frac{9}{18}$
22. (a) $\frac{3}{6}$ (b) $\frac{3}{6}$ (c) $\frac{4}{6}$ (d) $\frac{3}{6}$
23. (a) 0.7 (b) 0 (c) 0.7 (d) 1
24. (a) $\frac{5}{7}$ (b) $\frac{4}{7}$
25. $p(A) = \frac{6}{11}$ $p(B) = \frac{3}{11}$ $p(C) = \frac{1}{11}$ $p(D) = \frac{1}{11}$
26. $p(1) = \frac{1}{4}$ $p(2) = \frac{1}{4}$ $p(3) = \frac{1}{4}$ $p(4) = \frac{1}{8}$ $p(5) = \frac{1}{8}$
27. $\frac{10}{32}$
28. (a) $\frac{5}{6}$ (b) $\frac{1}{6}$
29. (a) $\frac{1}{6}$ (b) 1
30. (a) $\frac{5}{6}$ (b) 0
31. (a) $\frac{11}{36}$ (b) $\frac{27}{36}$ (c) $\frac{6}{36}$ (d) $\frac{21}{36}$
32. $\frac{20}{221}$
33. (a) $\frac{35}{220}$ (b) $\frac{80}{220}$ (c) $\frac{210}{220}$ (d) $\frac{210}{220}$
34. (a) $\frac{15}{216}$ (b) $\frac{20}{216}$ (c) $\frac{91}{216}$ (d) $\frac{200}{216}$ (e) $\frac{125}{216}$
35. (a) $\frac{4}{24}$ (b) $\frac{20}{24}$
36. (a) $\frac{1}{120}$ (b) $\frac{1}{120}$ (c) $\frac{76}{120}$
37. $\frac{n+1}{2}$
38. There is an $(n+1)$ -st outcome, not found. $P(\text{finding the key of determining the key is not in the array}) = \frac{n+2}{2}$.
39. $-\frac{11}{9}$ dollars
40. As close to $\$1\frac{7}{9}$ as possible. $\frac{1}{36} \left(-\frac{16}{9} \cdot 30 + 10 \cdot 3 + 11 \cdot 2 + 12 \right) = 0$.
41. (a) $\frac{8}{52} \cdot \frac{7}{51}$ (b) $\frac{8}{52} \cdot \frac{8}{52}$

Exercise Set 3.5, page 104

1. 4, 10, 25, 62.5 Yes, degree 1.
2. -2, 4, -8, 16 Yes, degree 2
3. 3, 12, 24, 48 No
4. 2, 4, 12, 48 No
5. 1, 8, 43, 216 No
6. 1, 3, 2, $\sqrt{2}$ No
7. $s_1 = 2, s_2 = 3, s_n = s_{n-1} + 1$
8. $s_1 = 2, s_2 = 4, s_3 = 7, s_n = s_{n-1} + s_{n-2} + s_{n-3}$
9. $A_1 = 100\left(1 + \frac{0.06}{12}\right), A_n = \left(1 + \frac{0.06}{12}\right)(A_{n-1} + 100)$
10. $A_1 = 10,000\left(1 + \frac{0.08}{12}\right) - 250, A_n = \left(1 + \frac{0.08}{12}\right)A_{n-1} - 250$
11. $c_n = c_{n-2} + c_{n-3}, c_1 = 0, c_2 = 1, c_3 = 1$
12. $a_n = 4(2.5)^{n-1}$ 13. $b_n = 3 \cdot 5^{n-1} + \frac{3}{4}(5^{n-1} - 1)$
14. $c_n = 3 + \frac{n(n+1)}{2}$ 15. $d_n = 5(-1.1)^{n-1}$
16. $e_n = -2(n-1)$ 17. $g_n = n! \cdot 6$
18. $a_n = \frac{8}{30} \cdot 5^n - \frac{20}{30}(-1)^n$ 19. $b_n = (-2)^n$
20. $c_n = -\frac{19.7}{9}(-3)^n + \frac{12.2}{9}n(-3)^n$
21. $d_n = -\frac{3}{4} \cdot 2^n + \frac{5}{4} \cdot n \cdot 2^n$
22. $e_n = 2(\sqrt{2})^n + (-\sqrt{2})^n$ 23. $g_n = \frac{-1-2i}{2}(1+i)^n + \frac{-1+2i}{2}(1-i)^n$
24. $a_n = r^{n-1}a_1 + s\left(\frac{r^{n-1}-1}{r-1}\right)$
25. <no solution given>
26. (a) $r_n = r_{n-1} + n, r_1 = 2$ (b) $r_n = 1 + \frac{n(n+1)}{2}$
27. (a) $a_n = 2a_{n-1} + 1, a_1 = 0$ (b) $a_n = 2^{n-1} - 1$
28. For $x^2 - r_1x - r_2 = 0$ to have a unique solution, s , we must have $r_1^2 = -4r_2$. Then $s = \frac{r_1}{2}$. We now show that $a_n = us^n + vns^n$ defines the same sequence as $a_n = r_1a_{n-1} + r_2a_{n-2}$ when u and v are chosen so that $a_1 = us + vs$ and $a_2 = us^2 + 2vs^2$. Consider

$$\begin{aligned}
 a_n &= us^n + vns^n = (us^{n-2} + v(n-1)s^{n-2})s^2 + vs^n \\
 &= (us^{n-2} + v(n-1)s^{n-2})(r_1s + r_2) + vs^n \\
 &= r_1(us^{n-1} + v(n-1)s^{n-1}) + r_2(us^{n-2} + v(n-2)s^{n-2}) + r_2vs^{n-2} + vs^n \\
 &= r_1a_{n-1} + r_2a_{n-2} + vs^{n-2}(r_2 + s^2) \\
 &= r_1a_{n-1} + r_2a_{n-2} \quad \text{since } r_2 + s^2 = 0.
 \end{aligned}$$
29. $c_1 = 2, c_2 = 3, c_n = u\left(\frac{1+\sqrt{5}}{2}\right)^n + v\left(\frac{1-\sqrt{5}}{2}\right)^n, u = \frac{5+3\sqrt{5}}{10}, v = \left(\frac{5-3\sqrt{5}}{10}\right)$
30. There are r places to insert an element in a sequence of length $r-1$, but there are $n-(r-1)$ different elements that could be used for the insertion.
31. For $n \geq 2, f_{n+1}^2 - f_n^2 = (f_{n+1} - f_n)(f_{n+1} + f_n) = f_{n-1}f_{n+2}$, by the definition of f_k .
32. $s_n = n + 1$
33. $A_1 = 100.5, A_n = 1.005^{n-1}(100.5) - 20, 100(1.005^{n-1} - 1)$
34. (a) 3 (b) $a_n = -\frac{1}{2}(-1)^n + \frac{1}{2}(-2)^n - \frac{1}{2}(3)^n$
35. $a_n = -2(-2)^n + \left(\frac{5}{2} - \sqrt{2}\right)(\sqrt{2})^n + \left(\frac{5}{2} + \sqrt{2}\right)(-\sqrt{2})^n$

36. Basis Step: $n = 1$ $P(1)$: $b_1 < \frac{5}{2}$ is true, because b_1 is 1.
 Induction Step: LHS of $P(k+1)$: $b_{k+1} = b_k + 2b_{k-1} < \left(\frac{5}{2}\right)^k + 2\left(\frac{5}{2}\right)^{k-1}$

$$= \left(\frac{5}{2}\right)^{k-1} \left(\frac{5}{2} + 2\right) = \left(\frac{5}{2}\right)^{k-1} \left(\frac{9}{2}\right) < \left(\frac{5}{2}\right)^{k-1} \left(\frac{25}{4}\right) = \left(\frac{5}{2}\right)^{k+1} \text{ RHS of } P(k+1)$$
37. Basis Step: $n = 0$ $P(0)$: $5 \mid a_1$ is clearly true.
 Induction Step: We use $P(k)$: $5 \mid a_{3k+1}$ to show $P(k+1)$: $5 \mid a_{3(k+1)+1}$. Consider $a_{3(k+1)+1}$

$$= 2a_{3(k+1)} + a_{3k+2} = 2(2a_{3k+2} + a_{3k+1}) + a_{3k+2}$$

$$= 5a_{3k+2} + 2a_{3k+1}. \text{ Clearly, } 5 \mid 5a_{3k+2} \text{ and } P(k) \text{ guarantees } 5 \mid a_{3k+1}.$$
38. $C_1 = 1, C_2 = 1, C_3 = 2, C_4 = 5, C_5 = 14$
 39. $C_n = C_1C_{n-1} + C_2C_{n-2} + \dots + C_{n-1}C_1, C_1 = 1$
 40. <no solution given>

Review Questions, page 106

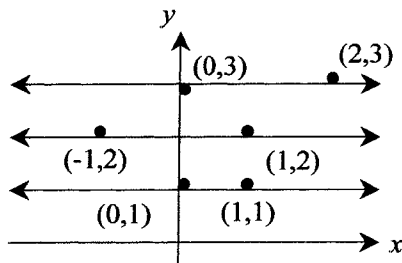
1. If all items to be chosen are not from the same set, then the problem is not a simple combination or permutation problem. If the items to be chosen are from the same set, and if the order in which they are chosen matters, permutations should be counted; otherwise, combinations.
2. Pigeons can often be identified by the phrases "at least k items have the property $P(x)$ " and the corresponding pigeonholes are the possible values for $P(x)$.
3. The recursive form may be easier to find or to justify, but this form may be difficult to evaluate for large n .
4. The solution form may be more efficient to evaluate for large n , but this form may be difficult to find.
5. The counting rules from Sections 3.1 and 3.2 are the primary tools needed to answer the probability questions presented in this chapter.

Chapter 4 Answers

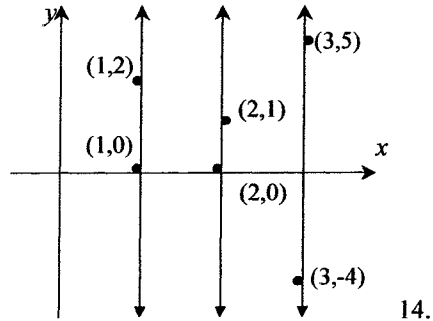
Exercise Set 4.1, page 114

1. (a) x is 4. (b) y is 3.
2. (a) x is 2. (b) y is C^{++} ; x is PASCAL.
3. (a) x is 4; y is 6. (b) x is 4; y is 2.
4. (a) x is 7 or -7 ; y is 25. (b) x is 0 or 1; y is 0 or 1.
5. (a) $\{(a, 4), (a, 5), (a, 6), (b, 4), (b, 5), (b, 6)\}$
 (b) $\{(4, a), (5, a), (6, a), (4, b), (5, b), (6, b)\}$
6. (a) $\{(a, a), (a, b), (b, a), (b, b)\}$ (b) $\{(4, 4), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$
7. (a) $\{(\text{Fine}, \text{president}), (\text{Fine}, \text{vice-president}), (\text{Fine}, \text{secretary}), (\text{Fine}, \text{treasurer}),$
 $(\text{Yang}, \text{president}), (\text{Yang}, \text{vice-president}), (\text{Yang}, \text{secretary}), (\text{Yang}, \text{treasurer})\}$
 (b) $\{(\text{president}, \text{Fine}), (\text{vice-president}, \text{Fine}), (\text{secretary}, \text{Fine}), (\text{treasurer}, \text{Fine}),$
 $(\text{president}, \text{Yang}), (\text{vice-president}, \text{Yang}), (\text{secretary}, \text{Yang}), (\text{treasurer}, \text{Yang})\}$
 (c) $\{(\text{Fine}, \text{Fine}), (\text{Fine}, \text{Yang}), (\text{Yang}, \text{Fine}), (\text{Yang}, \text{Yang})\}$
8. (a) 6 (b) ms, mm, ml, fs, fm, fl
9. gs, ds, gc, dc, gv, dv
10. $(a, 1, \square), (a, 1, *), (a, 2, \square), (a, 2, *), (b, 1, \square), (b, 1, *), (b, 2, \square), (b, 2, *), (c, 1, \square), (c, 1, *),$
 $(c, 2, \square), (c, 2, *)$
11. (outline) Basis Step: $n = 1$ $P(1)$: If $|A| = 3$ and $|B| = 1$, then $|A \times B| = 3$. Since $A \times B = \{(a_1, *), (a_2, *), (a_3, *)\}$, $|A \times B| = 3$.
 Induction Step: Suppose $|B| = k > 1$. Let $x \in B$ and $C = B - \{x\}$. Then $|C| = k - 1 \geq 1$ and using $P(k)$, we have $|A \times C| = 3(k - 1)$. $|A \times \{x\}| = 3$. Since $(A \times C) \cap (A \times \{x\}) = \{\}$ and $(A \times C) \cup (A \times \{x\}) = A \times B$, $|A \times B| = 3(k - 1) + 3$ or $3k$.

12.

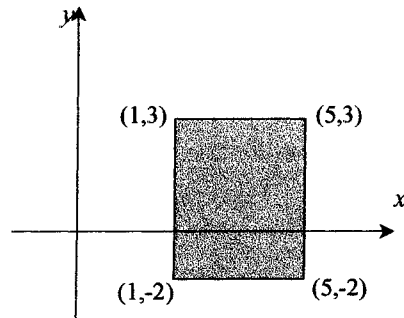
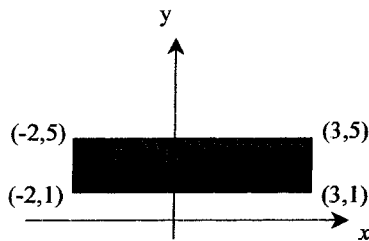


13.



14.

15.



16. By Theorem 2, Section 3.1, the number of ways to form an ordered triple in $A_1 \times A_2 \times A_3$ is $n_1 \cdot n_2 \cdot n_3$.
17. Let $(x, y) \in A \times B$, then $x \in A$ and $y \in B$. Since $A \subseteq C$ and $B \subseteq D$, $x \in C$ and $y \in D$ and $(x, y) \in C \times D$.
18. One possible answer is *project(select Employees[Department = Public Relations or Department = Research])[Employee ID, Last Name]*
19. One possible answer is *project(select Employees[Department = Human Resources])[Last Name]*
20. One possible answer is *| select Employees[Department = Sales] |*
21. One possible answer is *project(select Employees[Department = Research])[Years with Company]*
22. (a) Yes (b) No
23. (a) Yes (b) No
24. No, $A_1 \cup A_2 \neq Z$.
25. Answers will vary.
26. $\{\{a, e, i, o, u\}, \{b, c, d, f, g, q\}, \{h, j, k, l, m, n, p, r, s, t, v, w, x, y, z\}\}$ is a possible answer.
27. No, $|A| = 26$.
28. (a) $\{\{0, 3, 9, 15, 21, \dots\}, \{6, 12, 18, \dots\}\}$
 (b) $\{x \mid x = 0 \text{ or } 3 \cdot 5^k, k \geq 0\}, \{x \mid x = 3 \cdot 2^k, k \geq 1\},$
 $\{x \mid x = 3t, t \neq 2^k \text{ and } t \neq 5^k \text{ for any integer } k\}$
29. $\{\{1\}, \{2\}, \{3\}\}, \{\{1\}, \{2, 3\}\}, \{\{2\}, \{1, 3\}\}, \{\{3\}, \{1, 2\}\}, \{\{1, 2, 3\}\}$
30. $\{\{a\}, \{b\}, \{c\}, \{d\}\}, \{\{a\}, \{b, c, d\}\}, \{\{b\}, \{a, c, d\}\}, \{\{c\}, \{a, b, d\}\}, \{\{d\}, \{a, b, c\}\},$
 $\{\{a, b\}, \{c, d\}\}, \{\{a, c\}, \{b, d\}\}, \{\{a, d\}, \{b, c\}\}, \{\{a, b, c, d\}\}, \{\{a\}, \{b\}, \{c, d\}\},$
 $\{\{a\}, \{c\}, \{b, d\}\}, \{\{a\}, \{d\}, \{b, c\}\}, \{\{b\}, \{c\}, \{a, d\}\}, \{\{b\}, \{d\}, \{a, c\}\}, \{\{c\}, \{d\}, \{a, b\}\}$
31. 3 There are three 2-element partitions listed in the solution to Exercise 29.
32. $S(4, 2) = 7$
33. 6
34. 15
35. Let $(x, y) \in A \times (B \cup C)$. Then $x \in A, y \in B \cup C$. Hence $(x, y) \in A \times B$ or $(x, y) \in A \times C$. Thus $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$. Let $(x, y) \in (A \times B) \cup (A \times C)$. Then $x \in A, y \in B$ or $y \in C$. Hence $y \in B \cup C$ and $(x, y) \in A \times (B \cup C)$. So, $(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$.
36. $A \times (B \cap C) = \{(1, 7), (2, 7), (4, 7)\}$. $A \times B = \{(1, 2), (1, 5), (1, 7), (2, 2), (2, 5), (2, 7), (4, 2), (4, 5), (4, 7)\}$, $A \times C = \{(1, 1), (1, 3), (1, 7), (2, 1), (2, 3), (2, 7), (4, 1), (4, 3), (4, 7)\}$. Hence $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

37. Let $\{B_1, B_2, \dots, B_m\}$ be a partition of B . Form the set $\{B_1 \cap A, B_2 \cap A, \dots, B_m \cap A\}$. Delete any empty intersections from this set. The resulting set is a partition of A . If $a \in A$, then $a \in B$. We know that as an element of B , a is in exactly one of the B_i and hence in exactly one of the $B_i \cap A$.
38. Let (a, b) be any element of $A \times B$. Then a is an element of exactly one of the A_i 's and b is an element of exactly one of the B_j 's. Thus (a, b) must belong to exactly one of the $A_i \times B_j$'s.

39. (a)

# of elements	1	2	3	4	5	6
1	1					
2	1	1				
3	1	3	1			
4	1	7	6	1		
5	1	15	25	10	1	
6	1	31	90	65	10	1

(b) It appears that each entry in the $S(n, 2)$ -column can be calculated as $2^{n-1} - 1$.

Exercise Set 4.2, page 122

1. (a) No (b) No (c) Yes (d) Yes (e) Yes (f) Yes
 2. (a) Yes (b) No (c) Yes (d) No (e) Yes (f) No
 3. (a) No (b) No (c) Yes (d) Yes (e) No (f) only if $n = 1$

4. Domain $\{a, b, c, d\}$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Range $\{1, 2\}$

5. Domain $\{\text{IBM, Dell, COMPAQ, Gateway}\}$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

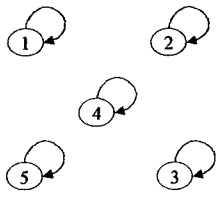
Range $\{750C, 466V, 450SV, \text{PS60}\}$

6. Domain $\{1, 2, 3\}$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Range $\{1, 4, 9\}$

7. Domain $\{1, 2, 3, 4, 8\}$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$


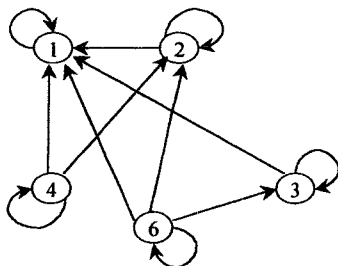
Range $\{1, 2, 3, 4, 8\}$

8. Domain $\{1, 2, 3, 4\}$ Range $\{1, 4, 6, 9\}$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

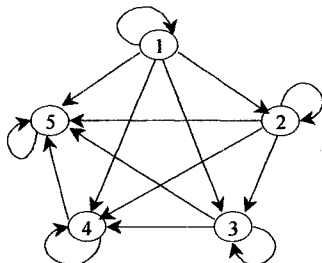
9. Domain $\{1, 2, 3, 4, 6\}$ Range $\{1, 2, 3, 4, 6\}$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$



10. Domain $= \{1, 2, 3, 4, 5\} = \text{Range}$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

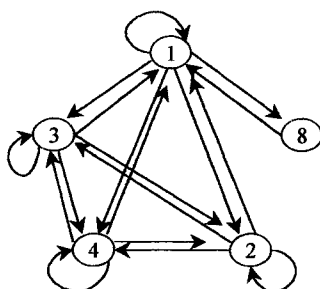


11. Domain $\{3, 5, 7, 9\}$ Range $\{2, 4, 6, 8\}$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

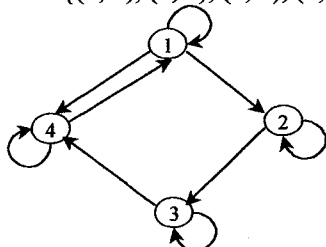
12. Domain $= \{1, 2, 3, 4, 8\} = \text{Range}$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

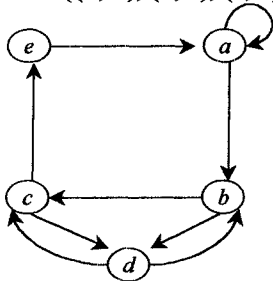


13. (a) no (b) no (c) yes (d) yes (e) no (f) no
 14. $\text{Dom}(R) = \mathbb{R}$ $\text{Ran}(R) = \mathbb{R}$
 15. $\text{Dom}(R) = [-5, 5]$ $\text{Ran}(R) = [-5, 5]$
 16. (a) $\left\{ \frac{-3\sqrt{3}}{2}, \frac{3\sqrt{3}}{2} \right\}$ (b) $\{ \}$ (c) $\{ \}$
 17. (a) $\{1, 3\}$ (b) $\{1, 2, 3, 6\}$ (c) $\{1, 2, 4, 3, 6\}$

18. (a) $\{2, 4, 6\}$ (b) $\{2, 4, 6, 8\}$ (c) $\{2\}$
 19. (a) $\{3\}$ (b) $\{2, 4\}$ (c) $\{3, 2, 4\}$ (d) $\{2, 4\}$
 20. Suppose $y \in R(a) \cap R(b)$, $a \neq b$. Then $y \in R(\{a\} \cap \{b\}) = R(\{\}) = \{\}$. Thus there can be no such y and $R(a) \cap R(b) = \{\}$, if $a \neq b$.
 Let $y \in R(A_1) \cap R(A_2)$, then there exist $a_1 \in A_1$ and $a_2 \in A_2$ such that $y \in R(a_1) \cap R(a_2)$. This implies $a_1 = a_2$. Thus $y \in R(A_1 \cap A_2)$.
 Let $y \in R(A_1 \cap A_2)$. Then aRy , $a \in A_1 \cap A_2$. This means $a \in A_1$ and $a \in A_2$ so $y \in R(A_1) \cap R(A_2)$.
 21. aRb if and only if $0 \leq a \leq 3$ and $0 \leq b \leq 2$.
 22. 2^{mn}
 23. $R = \{(1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 3), (3, 4), (4, 1)\}$



24. $R = \{(a, a), (a, b), (b, c), (b, d), (c, d), (c, e), (d, b), (d, c), (e, a)\}$



25. $R = \{(1, 2), (2, 2), (2, 3), (3, 4), (4, 4), (5, 1), (5, 4)\}$

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

26. $R = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (4, 4), (4, 1), (4, 5)\}$

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

27.

vertex	1	2	3	4	5
in-degree	1	2	1	3	0
out-degree	1	2	1	1	2

28.

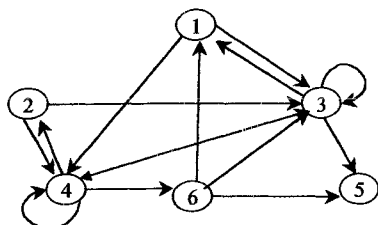
vertex	1	2	3	4	5
in-degree	1	2	2	2	1
out-degree	3	2	0	3	0

29. The in-degree of a vertex is the number of ones in the column labeled by that vertex.
 The out-degree of a vertex is the number of ones in the row labeled by that vertex.
 30. $\{(1, 2), (1, 4), (2, 5)\}$
 31. $\{(2, 3), (3, 6)\}$
 32. Delete from M_R every row and column labeled by an element of $A - B$.
 33. Delete any vertex labeled by an element of $A - B$. Then delete any edges that do not point to a vertex.

34. (a) The elements of $R(a_k)$ are those elements of A that label columns with a one in row k .
 (b) The elements of $R(\{a_1, a_j, a_n\})$ are the elements of A that label columns with a one in row 1, j , or n .
35. (a) The elements of $R(a_k)$ are those elements of A that can be reached from a_k in one step.
 (b) The elements of $R(\{a_1, a_j, a_n\})$ are those elements of A that reached from a_1, a_j , or a_n in one step.
36. $|S| = 6, |S \times S| = 36, |P(S \times S)| = 2^{36}$.
37. 2^{mn}

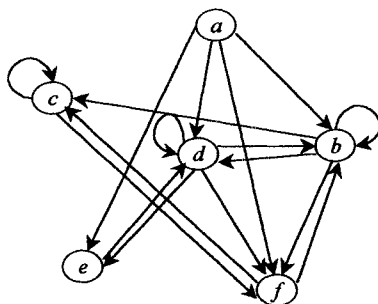
Exercise Set 4.3, page 128

1. 1,2 1,6 2,3 3,3 3,4 4,3 4,5 4,1 6,4
2. (a) 2, 3, 3 2, 3, 4
 (b) 2, 3, 3 2, 3, 4 1, 6, 4 1, 2, 3
 3, 3, 3 3, 3, 4 3, 4, 5 3, 4, 3
 4, 3, 3 6, 4, 3 4, 1, 6 6, 4, 5
 4, 1, 2 3, 4, 1 4, 3, 4 6, 4, 1
3. (a) 3, 3, 3, 3 3, 3, 4, 3 3, 3, 4, 5 3, 4, 1, 6 3, 4, 1, 2 3, 4, 3, 3
 3, 4, 3, 4 3, 3, 4, 1 3, 3, 3, 4
 (b) In addition to those in part (a), 1, 2, 3, 4 1, 6, 4, 1
 1, 6, 4, 5 2, 3, 3, 3 2, 3, 3, 4 2, 3, 4, 3
 2, 3, 4, 5 4, 1, 2, 3 4, 1, 6, 4 6, 4, 3, 3
 6, 4, 3, 4 6, 4, 1, 2 6, 4, 1, 6 1, 6, 4, 3
 2, 3, 4, 1 4, 3, 3, 3 4, 3, 4, 3 4, 3, 4, 1
 4, 3, 4, 5 4, 3, 3, 4 1, 2, 3, 3
4. One is 2, 3, 4, 1, 2.
5. One is 6, 4, 1, 6.
- 6.



7.
$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$
8. (a) $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$
 (b)
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$
9. a, c a, b b, b b, f c, d c, e d, c d, b e, f f, d
10. (a) c, e, f c, d, c c, d, b
 (b) a, c, d a, c, e a, b, b a, b, f b, b, f
 b, f, d b, b, b c, d, c c, d, b c, e, f
 d, b, b d, b, f d, c, d d, c, e e, f, d
 f, d, b f, d, c

11. (a) a, c, d, c a, c, d, b a, c, e, f a, b, b, b a, b, b, f a, b, f, d
 (b) In addition to those in part (a), b, b, b, b b, b, b, f b, b, f, d b, f, d, b
 b, f, d, c c, d, c, d c, d, c, e c, d, b, b d, b, b, b
 d, b, b, f d, c, d, c c, d, b, f c, e, f, d e, f, d, b
 e, f, d, c d, c, d, b f, d, c, d f, d, c, e f, d, b, b
 f, d, b, f d, b, f, d d, c, e, f
12. One is c, d, c .
 13. One is d, b, f, d .
 14. No cycle can begin at a , because the in-degree of a is 0.
 15.



16.
$$\begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$
 Yes

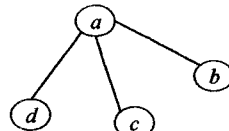
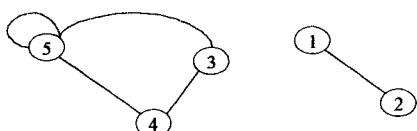
17. (a)
$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

- (b) $\{(a, c), (a, d), (a, b), (a, e), (a, f), (b, b), (b, c), (b, d), (b, e), (b, f), (c, b), (c, c), (c, d), (c, e), (c, f), (d, b), (d, c), (d, d), (d, e), (d, f), (e, b), (e, c), (e, d), (e, e), (e, f), (f, b), (f, c), (f, d), (f, e), (f, f)\}$
18. Let $\mathbf{M}_{R \cup S} = [m_{ij}]$. $m_{ij} = 1$ if and only if iRj or iSj . That is, either the ij -entry of $\mathbf{M}_R = 1$ or the ij -entry of $\mathbf{M}_S = 1$. Hence $m_{ij} = 1$ if and only if the ij -entry of $\mathbf{M}_R \vee \mathbf{M}_S$ is 1.
19. $x_i R^* x_j$ if and only if $x_i = x_j$ or $x_i R^n x_j$ for some n . The ij -entry of $\mathbf{M}_{R^*}$ is 1 if and only if $i = j$ or the ij -entry of $\mathbf{M}_{R^n}$ is 1 for some n . Since $R^\infty = \bigcup_{k=1}^{\infty} R^k$, the ij -entry of $\mathbf{M}_{R^*}$ is 1 if and only if $i = j$ or the ij -entry of $\mathbf{M}_{R^\infty}$ is 1. Hence, $\mathbf{M}_{R^*} = \mathbf{I}_n \vee \mathbf{M}_{R^\infty}$.
20. 1, 2, 4, 3, 5, 6, 4
 21. 1, 7, 5, 6, 7, 4, 3
 22. 3, 4, 5, 6, 7, 4, 3, 5
 23. 2, 3, 5, 6, 7, 5, 6, 4
 24. 7, 4, 3, 5, 6, 7; 5, 6, 7, 5 are possible answers.
 25. 7, 4, 3, 5, 6, 7 is a possible answer.
 26. (a) $R^2 = \{(1, 3), (1, 4), (1, 5), (2, 4), (2, 5), (3, 5)\}$ $R^3 = \{(1, 4), (1, 5), (2, 5)\}$
 (b) $b - a \geq 2$ (c) $b - a \geq 3$
 27. The ij -entry of $\mathbf{M}_R \cdot \mathbf{M}_R$ is the number of paths from i to j of length two, because it is also the number of k 's such that $a_{ik} = b_{kj} = 1$.
 28. Yes, the ij -entry of $(\mathbf{M}_R)^n$ is the number of paths of length n from i to j .
 29. direct; Boolean multiplication

30. (a) The statement is of the form $P(n)$, $n \geq 2$.
 (b) A path of length $k + 1$ is the composition of a path of length k and a path of length 1.
31. Suppose each vertex has out-degree at least one. Choose a vertex, say v_i . Construct a path in R $v_i, v_{i+1}, v_{i+2}, \dots$. This is possible since each vertex has an edge leaving it. But there are only a finite number of vertices so for some k and j , $v_j = v_k$ and a cycle is created.
32. A possible digraph is
- 
33. The essentials of the digraph are the connections made by the arrows. Compare the arrows leaving each vertex in turn to pairs in R with that vertex as first element.

Exercise Set 4.4, page 134

1. reflexive, symmetric, transitive
2. irreflexive, asymmetric, antisymmetric, transitive
3. none
4. reflexive, symmetric, transitive, antisymmetric
5. irreflexive, symmetric, asymmetric, antisymmetric, transitive
6. reflexive, symmetric, transitive
7. transitive
8. symmetric
9. antisymmetric, transitive
10. irreflexive, asymmetric, antisymmetric, transitive
11. irreflexive, symmetric
12. reflexive, symmetric, transitive
13. reflexive
14. reflexive, symmetric
15. reflexive, antisymmetric, transitive
16. reflexive, symmetric, transitive
17. irreflexive, symmetric
18. symmetric
19. symmetric
20. reflexive, symmetric, transitive
21. reflexive, symmetric, transitive
22. reflexive, symmetric, transitive
23. reflexive, symmetric, transitive
24. irreflexive, symmetric, transitive
- 25.



27. $\{(1, 5), (5, 1), (1, 6), (6, 1), (5, 6), (6, 5), (1, 2), (2, 1), (2, 7), (7, 2), (2, 3), (3, 2)\}$
28. $\{(a, b), (b, a), (a, c), (c, a), (b, c), (c, b), (d, d), (d, e), (e, d)\}$
29. Let $a_1, a_2, \dots, a_n$ be the elements of the base set. The graph of R is connected if for each a_i there is a 1 in the i^{th} column of $\mathbf{M}_R \odot^k$ for some k .
30. All of the properties will be inherited.
31. Let R be transitive and irreflexive. Suppose aRb and bRa . Then aRa since R is transitive. But this contradicts the fact that R is irreflexive. Hence R is asymmetric.
32. Suppose aR^2b and bR^2c . Then aRd, dRb, bRe, eRc for some $d, e \in A$. Then aRb and bRc since R is transitive. Thus, aR^2c and R^2 is transitive.
33. Let $R \neq \{ \}$ be symmetric and transitive. There exists $(x, y) \in R$, and $(y, x) \in R$. Since R is transitive, we have $(x, x) \in R$, and R is not irreflexive.
34. Let R be symmetric. Suppose aR^2b . Then there exists $c \in A$ with aRc, cRb . Since R is symmetric, cRa, bRc and bR^2a .

35. (outline) Basis Step: $n = 1$ $P(1)$: If R is symmetric, then R^1 is symmetric is true.
 Induction Step: Use $P(k)$: If R is symmetric, then R^k is symmetric to show $P(k+1)$. Suppose $aR^{k+1}b$. Then there is a $c \in A$ such that $aR^k c$ and cRb . We have bRc and $cR^k a$. Hence $bR^{k+1}a$.
36. < no solution is given >
37. (a) One answer is $\{(a, a), (b, b), (c, c), (d, d)\}$.
 (b) One answer is $\{(a, a), (b, b), (c, c), (d, d), (a, b)\}$.
38. (a) One answer is $\{(a, b), (b, c), (a, c)\}$.
 (b) One answer is $\{(a, a), (b, b), (c, c), (d, d)\}$.
39. (a) One answer is $\{(a, a), (b, b), (c, c), (d, d)\}$.
 (b) One answer is $\{(a, b), (b, c), (a, c)\}$.
40. First, suppose R is transitive. Let $aR^n b$, $n \geq 1$. Then there is a path $a, x_1, x_2, \dots, x_{n-1}, b$ in R . Since R is transitive, we have $aRx_1 \wedge x_1Rx_2 \Rightarrow aRx_2$; $aRx_2 \wedge x_2Rx_3 \Rightarrow aRx_3$; ...; $aRx_{n-1} \wedge x_{n-1}Rb \Rightarrow aRb$. Hence $R^n \subseteq R$, $n \geq 1$. Conversely, if $R^n \subseteq R$, $n \geq 1$, then $R^2 \subseteq R$ and R is transitive.

Exercise Set 4.5, 139

1. Yes
2. No
3. Yes
4. No
5. No
6. Yes
7. No
8. Yes
9. Yes
10. Yes
11. Yes
12. Yes
13. $\{(a, a), (a, c), (a, e), (c, a), (c, c), (c, e), (e, a), (e, c), (e, e), (b, b), (b, f), (b, d), (d, b), (d, d), (d, f), (f, b), (f, d), (f, f)\}$
14. $\{(1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3), (5, 5), (2, 2), (2, 4), (4, 2), (4, 4)\}$
15. $\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (8, 8), (9, 9), (10, 10), (1, 3), (3, 1), (1, 5), (5, 1), (1, 7), (7, 1), (1, 9), (9, 1), (3, 5), (5, 3), (3, 7), (7, 3), (3, 9), (9, 3), (5, 7), (7, 5), (5, 9), (9, 5), (7, 9), (9, 7), (2, 4), (4, 2), (2, 6), (6, 2), (2, 8), (8, 2), (2, 10), (10, 2), (4, 6), (6, 4), (4, 8), (8, 4), (4, 10), (10, 4), (6, 8), (8, 6), (6, 10), (10, 6), (8, 10), (10, 8)\}$
16. $\{(a, a), (e, e), (i, i), (o, o), (u, u), (a, i), (i, a), (e, o), (o, e)\}$
17. $\{\dots, -3, -1, 1, 3, 5, \dots\}, \{\dots, -4, -2, 0, 2, 4, \dots\}$
18. $\{1\}, \{2, 3, 4\}$
19. (a) R is reflexive because $a^2 + b^2 = a^2 + b^2$. R is clearly symmetric. R is transitive because if $a^2 + b^2 = c^2 + d^2$ and $c^2 + d^2 = e^2 + f^2$, certainly $a^2 + b^2 = e^2 + f^2$.
 (b) The equivalence classes of A/R are circles with center at $(0, 0)$ including the circle with radius 0.
20. $\{a, b, c, e\}, \{d\}$
21. (a) $(a, b)R(a, b)$ because $ab = ba$. Hence R is reflexive. If $(a, b)R(a', b')$, then $ab' = ba'$. Then $a'b = b'a$ and $(a', b')R(a, b)$. Hence R is symmetric. Now suppose that $(a, b)R(a', b')$ and $(a', b')R(a'', b'')$. Then $ab' = ba'$ and $a'b'' = b'a''$. $ab'' = a \frac{b'a''}{a'} = ab' \frac{a''}{a'} = ba' \frac{a''}{a'} = ba''$. Hence $(a, b)R(a'', b'')$ and R is transitive.
 (b) $\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}, \{(1, 2), (2, 4)\}, \{(1, 3)\}, \{(1, 4)\}, \{(1, 5)\}, \{(2, 1), (4, 2)\}, \{(2, 3)\}, \{(2, 5)\}, \{(3, 1)\}, \{(3, 2)\}, \{(3, 4)\}, \{(3, 5)\}, \{(4, 1)\}, \{(4, 3)\}, \{(4, 5)\}, \{(5, 1)\}, \{(5, 2)\}, \{(5, 3)\}, \{(5, 4)\}$
22. (a) $(a, b)R(a, b)$ since $a + b = b + a$. That is, R is reflexive.
 If $(a, b)R(a', b')$, then $a + b = a' + b'$. Then $(a', b')R(a, b)$. Hence R is symmetric.
 Now suppose that $(a, b)R(a', b')$ and $(a', b')R(a'', b'')$. Then $a + b = a' + b'$ and $a' + b' = a'' + b''$. Hence $a + b = a'' + b''$ and $(a, b)R(a'', b'')$. R is transitive.
 (b) $\{(1, 1)\}, \{(1, 2), (2, 1)\}, \{(1, 3), (3, 1), (2, 2)\}, \{(1, 4), (4, 1), (2, 3), (3, 2)\}, \{(2, 4), (4, 2), (3, 3)\}, \{(4, 3), (3, 4)\}, \{(4, 4)\}$
23. Let R be reflexive and circular. If aRb , then aRb and bRb , so bRa . Hence R is symmetric. If aRb and bRc , then cRa . But R is symmetric so aRc , and R is transitive.
 Let R be an equivalence relation. Then R is reflexive. If aRb and bRc , then aRc (transitivity) and cRa (symmetry), so R is also circular.

24. aR_1a and aR_2a , $\forall a \in A$. Hence $a(R_1 \cap R_2)a$; $R_1 \cap R_2$ is reflexive. If $a(R_1 \cap R_2)b$, then aR_1b and aR_2b . Hence bR_1a and bR_2a ; $b(R_1 \cap R_2)a$ and $R_1 \cap R_2$ is symmetric. If $a(R_1 \cap R_2)b$, $b(R_1 \cap R_2)c$, then aR_1b , bR_1c , aR_2b , bR_2c , and aR_1c , aR_2c . $a(R_1 \cap R_2)c$ and $R_1 \cap R_2$ is transitive.
25. aRb if and only if $ab > 0$.
26. aRb if and only if $a \pmod{3} = b \pmod{3}$.
27. If z is even (or odd), then $R(z)$ is the set of even (or odd) integers. Thus, if a and b are both even (or odd), then $R(a) + R(b) = \{x \mid x = s + t, s \in R(a), t \in R(b)\} = \{x \mid x \text{ is even}\} = R(a + b)$. If a and b have opposite parity, then $R(a) + R(b) = \{x \mid x = s + t, s \in R(a), t \in R(b)\} = \{x \mid x \text{ is odd}\} = R(a + b)$.
28. $R((a, b)) + R((m, n)) = \{x \mid x = (y, b) + (z, n), y, z \in \mathbb{Z}^+\}$
 $= \{x \mid x = (w, b + n), w \in \mathbb{Z}^+\} = R((a, b) + (m, n)).$
29. $(1, 2)R(2, 4)$ and $(1, 3)R(1, 3)$, but $((1, 2) + (1, 3)) \not R ((2, 4) + (1, 3))$ so the set $R((a, b)) + R((a', b'))$ is not an equivalence class.

Exercise Set 4.6, page 146

1.

VERT[1] = 9 (1,6)	NEXT[9] = 10 (1, 3)	NEXT[10] = 1 (1, 2)	NEXT[1] = 0
VERT[2] = 3 (2, 1)	NEXT[3] = 2 (2, 3)	NEXT[2] = 0	
VERT[3] = 6 (3, 4)	NEXT[6] = 4 (3, 5)	NEXT[4] = 7 (3, 6)	NEXT[7] = 0
VERT[4] = 0			
VERT[5] = 5 (5, 4)	NEXT[5] = 0		
VERT[6] = 8 (6, 1)	NEXT[8] = 0		
2. FUNCTION EDGE(I, J)
 1. $V \leftarrow F$
 2. $K \leftarrow \text{VERT}[I]$
 3. WHILE ($K \neq 0$)
 - a. IF ($\text{TAIL}[K] = J$) THEN
 1. $V \leftarrow T$
 2. $K \leftarrow 0$
 - b. ELSE
 1. $K \leftarrow \text{NEXT}[K]$
 4. RETURN (V)
 END OF EDGE
3. On average, EDGE must look at the average number of edges from any vertex. If R has P edges and N vertices, then EDGE examines $\frac{\sum P_{ij}}{N} = \frac{P}{N}$ edges on average.
4. FUNCTION LOOK (NUM, NEXT, START, N, K)
 1. $V \leftarrow 0$
 2. $C \leftarrow \text{START}$
 3. IF ($\text{NUM}[C] = K$) THEN
 - a. $V \leftarrow C$
 - b. $C \leftarrow 0$
 4. ELSE
 - a. $I \leftarrow 1$
 - b. WHILE ($C \neq 0$)
 1. $C \leftarrow \text{NEXT}[I]$
 2. IF ($\text{NUM}[C] = K$) THEN
 - a. $V \leftarrow C$
 - b. $C \leftarrow 0$
 3. ELSE
 - a. $I \leftarrow I + 1$
 5. IF ($V \neq 0$) THEN
 - a. PRINT V
 6. ELSE
 - a. PRINT 'NOT FOUND'
 END OF LOOK

5.

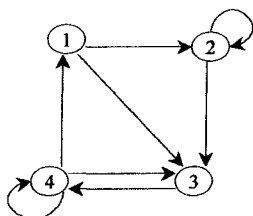
VERT	TAIL	HEAD	NEXT
1	1	1	2
4	1	2	3
6	1	3	0
8	2	3	5
	2	4	0
	3	1	7
	3	4	0
	4	2	0

5. (con't)

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

7.

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$



9.

VERT	TAIL	HEAD	NEXT
1	1	1	2
3	1	4	0
5	2	2	4
6	2	3	0
9	3	4	0
	4	1	7
	4	3	8
	4	5	0
	5	2	10
	5	5	0

11.

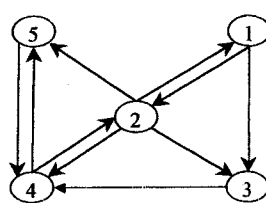
VERT	TAIL	HEAD	NEXT
1	a	a	2
4	a	b	3
6	a	d	0
9	b	b	5
	b	c	0
	c	b	7
	c	c	8
	c	d	0
	d	a	10
	d	b	11
	d	c	12
	d	d	0

6.

VERT	TAIL	HEAD	NEXT
1	1	1	2
4	1	2	3
7	1	5	0
9	2	1	5
	2	4	6
	2	3	0
	3	4	8
	3	3	0
	4	1	10
	4	3	0

8.

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$



10.

VERT	TAIL	HEAD	NEXT
1	a	a	2
3	a	d	0
5	b	c	4
8	b	d	0
10	c	a	6
	c	b	7
	c	e	0
	d	b	9
	d	d	0
	e	a	11
	e	e	0

12.

VERT	TAIL	HEAD	NEXT
1	F	F	2
4	F	M	3
7	F	W	0
11	M	M	5
	M	F	6
	M	W	0
	R	M	8
	R	R	9
	R	W	10
	R	R	0
	W	M	12
	W	W	13
	W	F	0

Exercise Set 4.7, page 155

1. (a) $\{(1, 3), (2, 1), (2, 2), (3, 2), (3, 3)\}$ (b) $\{(3, 1)\}$
(c) $\{(1, 1), (1, 2), (2, 1), (2, 3), (3, 1), (3, 2), (3, 3)\}$ (d) $\{(1, 2), (1, 3), (2, 3), (3, 3)\}$
2. (a) $\{(a, 2), (a, 3), (b, 2), (b, 3), (c, 1)\}$ (b) $\{(a, 1), (b, 1)\}$
(c) $\{(a, 1), (a, 2), (b, 1), (b, 2), (c, 2), (c, 3)\}$ (d) $\{(1, a), (2, a), (1, b), (2, b)\}$
3. $\{(a, b) \mid a, b \text{ are sisters or } a, b \text{ are brothers}\}$
4. $a(R \cap S)b$ if and only if a is an older brother of b .
5. $a(R \cup S)b$ if and only if a is a parent of b .
6. (a) $\{(2, 3), (3, 2), (3, 6), (6, 3), (3, 12), (12, 3)\}$
(b) $\{(2, 2), (3, 3), (6, 6), (12, 12), (6, 12), (12, 6)\}$
(c) $\{(2, 2), (3, 3), (6, 6), (12, 12), (2, 6), (2, 12), (12, 2), (6, 2), (6, 12), (12, 6), (3, 6), (6, 3), (3, 12), (12, 3)\}$
(d) $\{(2, 2), (3, 3), (6, 6), (12, 12), (3, 6), (6, 3), (3, 12), (12, 3), (6, 12), (12, 6)\}$
7. (a) $\{(2, 1), (3, 1), (3, 2), (3, 3), (4, 2), (4, 3), (4, 4), (1, 4)\}$
(b) $\{(1, 1), (1, 2), (2, 2), (2, 3), (2, 4), (4, 1), (3, 4)\}$
(c) $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 4)\}$
(d) $\{(1, 1), (2, 1), (2, 2), (1, 4), (4, 1), (2, 3), (3, 2), (1, 3), (4, 2), (3, 4), (4, 4)\}$
8. (a) $\{(a, c), (a, e), (b, a), (b, c), (b, d), (c, c), (c, a), (c, d), (c, e), (d, a), (d, b), (d, e), (e, c), (e, d), (e, e)\}$
(b) $\{(a, a), (a, d), (c, b), (e, b), (e, a)\}$
(c) $\{(a, a), (a, b), (a, d), (b, b), (c, b), (b, e), (d, c), (d, d), (e, a), (e, b), (b, a), (a, e), (e, e), (e, d), (c, d)\}$
(d) $\{(a, a), (a, b), (d, a), (b, c), (d, c), (a, e), (e, a), (e, e), (d, e), (b, e)\}$
9. (a) $\{(1, 1), (1, 4), (2, 2), (2, 3), (3, 3), (3, 4)\}$
(b) $\{(1, 2), (2, 4), (3, 1), (3, 2)\}$
(c) $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 4), (3, 1), (3, 2), (3, 3)\}$
(d) $\{(1, 1), (2, 1), (4, 1), (4, 2), (1, 3), (2, 3), (3, 3)\}$
10. (a) $\{(2, 1), (2, 2), (2, 3), (3, 1), (3, 3)\}$ (b) $\{(1, 1), (1, 3), (2, 4), (3, 2)\}$
(c) $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4)\}$
(d) $\{(1, 1), (3, 1), (4, 2), (1, 3), (2, 3), (3, 3)\}$
11. (a) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
12. (a) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$
13. (a) $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$
14. $R \cap S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$ $\{\{1, 2\}, \{3\}, \{4\}, \{5\}, \{6\}\}$
15. $R \cap S = \{(a, a), (b, b), (b, c), (c, b), (c, c), (d, d), (e, e)\}$ $\{\{a\}, \{b, c\}, \{d\}, \{e\}\}$
16. (a) $\{(2, 1), (2, 3), (3, 2), (3, 3), (2, 2), (4, 2), (1, 1), (4, 4)\}$
(b) $\{(2, 1), (2, 3), (3, 2), (3, 3), (2, 2), (2, 4), (1, 2), (4, 2)\}$

17. (a) $\{(a, a), (a, d), (a, e), (b, b), (b, c), (b, e), (c, a), (c, b), (c, c), (d, b), (d, c), (d, d), (e, c), (e, e)\}$
 (b) $\{(a, a), (a, d), (d, a), (a, e), (e, a), (b, c), (c, b), (b, e), (e, b), (c, a), (a, c), (c, c), (d, b), (b, d), (d, c), (c, d), (e, c), (c, e), (e, e)\}$
18. (a) Add to the digraph of R a loop at each vertex that does not already have one.
 (b) For every arrow from v_i to v_j in the digraph, add an arrow from v_j to v_i (if one does not exist).
19. The definitions of irreflexive, asymmetric, and antisymmetric each require that a certain pair does not belong to R . We can not "fix" this by including more pairs in R .
20. (a) Yes (b) No
21. (a) Yes (b) Yes (c) $x(S \circ R)y$ if and only if $x \leq 6y$.
22. (a) Yes (b) Yes (c) Yes
 (d) $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 4), (2, 1), (2, 2), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), (4, 4)\}$
 (e) $\{(1, 1), (1, 4), (1, 3), (2, 1), (2, 4), (3, 4), (4, 1), (4, 4), (4, 3)\}$
 (f) $\{(1, 1), (1, 2), (2, 4), (2, 1), (2, 2), (3, 1), (3, 2), (4, 1), (4, 2)\}$
 (g) $\{(1, 1), (1, 4), (2, 1), (2, 4), (3, 1), (3, 4), (4, 4)\}$
23. (a) reflexive $aRa \wedge aSa \Rightarrow a(S \circ R)a$ irreflexive no $1R2 \wedge 2S1 \Rightarrow 1(S \circ R)1$
 symmetric no $1R3, 3R1, 3S2, 2S3 \Rightarrow 1(S \circ R)2$, but $2(S \circ R)1$
 asymmetric no $R = \{(1, 2), (3, 4)\}$ and $S = \{(2, 3), (4, 1)\}$ provide a counterexample.
 antisymmetric no $R = \{(a, b), (c, d)\}$ and $S = \{(b, c), (d, a)\}$ provide a counterexample.
 transitive no $R = \{(a, d), (b, e)\}$ and $S = \{(d, b), (e, c)\}$ provide a counterexample.
 (b) No, the symmetric and transitive properties are not preserved.

24. (a)
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

(b)
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

(c)
$$\begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

(d)
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

25. (a)
$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(b)
$$\begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

(c)
$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(d)
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

26. $R \cap S$ is asymmetric. If $a(R \cap S)b$ and $b(R \cap S)a$, then aRb and bRa . This contradicts the fact that R is asymmetric. $R \cup S$ may not be asymmetric. Let $R = \{(1, 2)\}$, $S = \{(2, 1)\}$.
27. $R \cap S$ is antisymmetric. If $a(R \cap S)b$ and $b(R \cap S)a$, then aRb and bRa . Hence $a = b$ because R is antisymmetric. $R \cup S$ may not be antisymmetric. Let $R = \{(1, 2)\}$, $S = \{(2, 1)\}$.
28. Let $(x, y) \in (S \cup T) \circ R$. Then xRz , $z(S \cup T)y$ for some $z \in B$. Either zSy or zTy and $x(S \circ R)y$ or $x(T \circ R)y$. Hence $(x, y) \in (S \circ R) \cup (T \circ R)$. Now let $(x, y) \in (S \circ R) \cup (T \circ R)$. Say $(x, y) \in T \circ R$. Then xRz , zTy for some $z \in B$. Thus $z(S \cup T)y$ and $(x, y) \in (S \cup T) \circ R$.
29. Let $R = \{(x, z), (x, m)\}$, $S = \{(z, y)\}$, and $T = \{(m, y)\}$. Then $(x, y) \in ((S \circ R) \cap (T \circ R))$, but $(S \cap T) \circ R = \{\}$.

30. Let $(x, y) \in T \circ R$. Then $xRz, zTy, z \in B$. Since $R \subseteq S, xSz$ and $(x, y) \in T \circ S$.
31. (a) Let $\mathbf{M}_{R \cap S} = [m_{ij}], \mathbf{M}_R = [r_{ij}], \mathbf{M}_S = [s_{ij}]$. $m_{ij} = 1$ if and only if $(i, j) \in R \cap S$. $(i, j) \in R$ if and only if $r_{ij} = 1$ and $(i, j) \in S$ if and only if $s_{ij} = 1$. But this happens if and only if ij -entry of $\mathbf{M}_R \wedge \mathbf{M}_S$ is 1.
- (b) Let $\mathbf{M}_{R \cup S} = [m_{ij}], \mathbf{M}_R = [r_{ij}], \mathbf{M}_S = [s_{ij}]$. $m_{ij} = 1$ if and only if $(i, j) \in R \cup S$. $(i, j) \in R$ if and only if $r_{ij} = 1$ or $(i, j) \in S$ if and only if $s_{ij} = 1$. But this happens if and only if ij -entry of $\mathbf{M}_R \vee \mathbf{M}_S$ is 1.
- (c) The ij -entry of $\mathbf{M}_{R^{-1}}$ is 1 if and only if $(i, j) \in R^{-1}$ if and only if $(j, i) \in R$ if and only if the ji -entry of $\mathbf{M}_R$ is 1 if and only if the ij -entry of $\mathbf{M}_R^T$ is 1.
- (d) The ij -entry of $\mathbf{M}_{\bar{R}}$ is 1 if and only if $(i, j) \in \bar{R}$ if and only if $(i, j) \notin R$ if and only if the ij -entry of $\mathbf{M}_R$ is 0 if and only if the ij -entry of $\bar{\mathbf{M}}_R$ is 1.
32. (outline) Basis Step: $n = 1$ $R \cap S \subseteq R \cap S$ is true.
Induction Step: Let $(x, y) \in (R \cap S)^{k+1}$. Then $(x, z) \in (R \cap S)^k$ and $(z, y) \in R \cap S$, for some $z \in A$. $(x, z) \in (R \cap S)^k = R^k \cap S^k$. Hence $(x, y) \in R^{k+1}$ and $(x, y) \in S^{k+1}$. Thus $(R \cap S)^{k+1} \subseteq R^{k+1} \cap S^{k+1}$.
33. To form the digraph of R^{-1} , reverse the arrows in the digraph of R .
34. The edges of the digraph of $R \cap S$ are the edges that appear on both the digraph of R and the digraph of S .
35. The edges of the digraph of $R \cup S$ are the edges that appear on either the digraph of R or the digraph of S .
36. Let $(a, b) \in R - S$. Then $(a, b) \in R$ and $(a, b) \notin S$. Since R is symmetric, $(b, a) \in R$. But $(b, a) \notin S$ (or (a, b) would be also). Hence $(b, a) \in R - S$ and $R - S$ is symmetric.
37. (a) R is symmetric if and only if $xRy \Rightarrow yRx$ if and only if $R \subseteq R^{-1} \subseteq R$.
- (b) Suppose R is antisymmetric. Let $(x, y) \in R \cap R^{-1}$, then $x = y$ and $(x, y) \in \Delta$. Suppose $R \cap R^{-1} \subseteq \Delta$. If xRy and yRx , then xRy and $xR^{-1}y$. Thus $(x, y) \in \Delta$, so $x = y$.
- (c) Suppose R is asymmetric. Let $(x, y) \in R \cap R^{-1}$. This contradicts the fact that R is asymmetric. Hence $R \cap R^{-1} = \emptyset$. Let xRy and yRx . Then $(x, y) \in R \cap R^{-1} = \emptyset$. Hence R is asymmetric.

Exercise Set 4.8, page 162

1. (a) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (b) $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
2. $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)\}$
3. $W_1 = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$ $W_2 = W_1 = W_3$
4. $\{(1, 1), (2, 2), (3, 4), (1, 4), (3, 5), (3, 2), (4, 1), (4, 4), (5, 2), (5, 5)\}$
5. $aR^\infty b$ if and only if $b > a$.
6. $aR^\infty b$ if and only if b is a female ancestor of a .
7. Let R be reflexive and transitive. Suppose $xR^n y$. Then $x, a_1, a_2, \dots, a_{n-1}, y$ is a path of length n from x to y . $xRa_1 \wedge a_1Ra_2 \Rightarrow xRa_2$. Similarly, we have $xRa_k \wedge a_kRa_{k+1} \Rightarrow xRa_{k+1}$ and finally $xRa_{n-1} \wedge a_{n-1}Ry \Rightarrow xRy$. Hence $R^n \subseteq R$.
If xRy , then since R is reflexive we can build a path of length n , $x, x, x, \dots, x, y$ from x to y and $xR^n y$.
8. Let $aS^\infty b$. $(a, b) \in S^\infty$ if and only if there is a path $a, c_1, c_2, \dots, c_{k-1}, b$ in S from a to b if and only if $a, c_1, c_2, \dots, c_{k-1}, b$ is a path of length k in R^2 if and only if $a, d_1, c_1, d_2, c_2, \dots, d_{k-1}, c_{k-1}, b$ is a path of length $2k$ in R .
9. $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
10. $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$$11. \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$$12. \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

13. The sets are the same. This can be shown by computing each set.
 14. The sets are not the same. This can be shown by computing each set.
 15. $A \times A$
 16. $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5)\}$
 17. $A/R = \{\{1, 2, 3\}, \{4, 5\}\}$, $A/S = \{\{1\}, \{2, 3, 4\}, \{5\}\}$, $A/(R \cup S)^\infty = \{A\}$
 18. $A/R = \{\{1\}, \{2, 3\}, \{4, 5\}\}$, $A/S = \{\{1, 2\}, \{3\}, \{4\}, \{5\}\}$,
 $A/(R \cup S)^\infty = \{\{1, 2, 3\}, \{4, 5\}\}$
 19. The collection of elements in A/R and A/S can be separated into sub-collections of non-disjoint sets. Each element of $A/(R \cup S)^\infty$ is the union of the sets in one of these sub-collections.
 20. Warshall's algorithm finds the transitive closure for any R .
 21. $\{(1, 1), (1, 2), (1, 4), (2, 2), (3, 2), (3, 3), (4, 2), (4, 3), (4, 4), (1, 3)\}$
 22. $A \times A$
 23. We first show R^∞ is transitive. Then we show it is the smallest transitive relation that contains R . It is a direct proof.
 24. The proof is an analysis of any path in R between a and b , generic elements of A . It is a direct proof.
 25. $\mathbb{R} \times \mathbb{R}$

Review Questions, page 165

- $A \times B$ is a set of ordered pairs.
- A partition of A is a set of non-empty subsets of A .
- A relation may be given by a verbal description, a set of ordered pairs, a digraph (in some cases) and as a matrix (in some cases).
- Unless $|R|$ is very small, using the matrix representation will be the most efficient way to determine if R is transitive.
- The transitive closure of R is the connectivity relation R^∞ ; it is also the smallest transitive relation containing R .

Chapter 5 Answers

Exercise Set 5.1, page 176

- (a) yes $\text{Ran}(R) = \{1, 2\}$ (b) no
- (a) yes $\text{Ran}(R) = \{1, 2, 3\}$ (b) yes $\text{Ran}(R) = \{1\}$
- yes
- yes
- Each integer has a unique square that is also an integer.
- Each power of e is a unique real number.
- Each $r \in \mathbb{R}$ is either an integer or it is not.
- The description determines a unique integer.
- (a) 3 (b) 1 (c) $(x - 1)^2$ (d) $x^2 - 1$ (e) $y - 2$ (f) y^4
- (a) 3 (b) 7 (c) $(x + 1)^2 + 2$ (d) $x^2 + 3$ (e) $y + 2$ (f) $(y^2 + 2)^2 + 2$
- (a) both (b) neither
- (a) one to one (b) onto
- (a) both (b) onto
- (a) onto (b) neither
- (a) both (b) onto
- onto
- neither
- The given statement is the contrapositive of Theorem 1(a).

19. (a) $(g \circ f)(a) = g\left(\frac{a+1}{2}\right) = 2\left(\frac{a+1}{2}\right) - 1 = a$
 (b) $(g \circ f)(a) = g(a^2 - 1) = \sqrt{a^2 + 1} - 1 = |a| = a$, since $a \geq 0$.
20. (a) $(g \circ f)(X) = g(\overline{X}) = \overline{X} = X$ (properties of the complement)
 (b) $(g \circ f)(1) = g(4) = 1$; $(g \circ f)(2) = g(1) = 2$; $(g \circ f)(3) = g(2) = 3$; $(g \circ f)(4) = g(3) = 4$
21. (a) $f^{-1}(b) = b^2 - 1$ (b) $f^{-1}(b) = \sqrt[3]{b-1}$
22. (a) $f^{-1}(b) = \frac{3b+1}{2}$ (b) $f^{-1} = \{(3, 1), (2, 2), (4, 3), (5, 4), (1, 5)\}$
23. (a) Suppose $f(x, y) = f(w, z)$. Then $2x - y = 2w - z$ and $x - 2y = w - 2z$. Solving this system of equations gives $x = w$, $y = z$. Therefore, f is one to one.
 (b) $f^{-1}(x, y) = \left(\frac{2x-y}{3}, \frac{x-2y}{3}\right)$
24. no $(a, 1), (a, 2) \in f^{-1}$
25. yes
26. $(g \circ f)(a) = \frac{2a+1}{3}$; $(g \circ f)^{-1}(c) = \frac{3c-1}{2}$; $f^{-1}(b) = \frac{b-1}{2}$; $g^{-1}(c) = 3c$;
 $(f^{-1} \circ g^{-1})(c) = f^{-1}(3c) = \frac{3c-1}{2}$
27. n^n
28. $n!$
29. n^m
30. g is one to one; f is one to one.
31. C ; B ; because g is onto; A ; because f is onto.
32. Let $g \circ f$ be one to one. Suppose $f(a) = f(b)$. Then $(g \circ f)(a) = (g \circ f)(b)$ and $a = b$. Hence f is one to one.
33. Suppose $g \circ f$ is onto. Let $c \in C$. Then $\exists a \in A$ such that $(g \circ f)(a) = c$. But $(g \circ f)(a) = g(f(a))$, $f(a) \in B$, so g is onto.
34. Suppose $O(a_1, f) \cap O(a_2, f) \neq \{\}$. Then $f^{k_1}(a_1) = f^{k_2}(a_2)$ for some k_1 and k_2 .
 $(f^{-k_1} \circ f^{k_2})(a_1) = a_1 = f^{k_2-k_1}(a_2)$. Hence $a_1 \in O(a_2, f)$ and $f^n(a_1) \in O(a_2, f)$ for all n .
 Similarly, $f^n(a_2) \in O(a_1, f)$ for all n . Thus $O(a_1, f) = O(a_2, f)$.
35. (a) Let $\text{Dom}(f) = \{a_1, a_2, \dots, a_n\}$. Then $\text{Ran}(f) = \{f(a_1), f(a_2), \dots, f(a_n)\}$. Since f is one to one, the $f(a_i)$ are all distinct, so $\text{Ran}(f)$ has n elements. Thus $m = n$.
 (b) If f is not one to one, some $f(a_i) = f(a_j)$ and $m < n$.
36. Since f is everywhere defined, $\text{Dom}(f) = A$. Suppose f is one to one. Then by Exercise 35, $|\text{Ran}(f)| = |\text{Dom}(f)| = n$. Since $|B| = n$, $\text{Ran}(f) = B$ and f is onto. Next, suppose f is onto. Then $\text{Ran}(f) = B$, $|\text{Ran}(f)| = n$ and $|\text{Dom}(f)| = n$. By Exercise 35, f must be one to one. Since (a) and (b) are equivalent, (a) and (b) are each equivalent to (c).
37. One possible one-to-one correspondence is $f: Z^+ \rightarrow B: z \rightarrow 2z$.
38. One possible one-to-one correspondence is $g: Z^+ \rightarrow B: z \rightarrow 2z + 1$.
39. Yes, the cardinality of a set is determined by creating a one-to-one correspondence between the set and a subset of Z^+ . Here we use all of Z^+ .
40. (a) A counterexample is $f(2+3) = f(5) = 5^2$ or 25, but $f(2) + f(3) = 2^2 + 3^2$ or 14.
 (b) $f(s_1 \cdot s_2) = f(s_1) + f(s_2)$, because catenation does not lengthen or shrink either of the strings used.
41. (a) and (b). Consider the table for $\diamond$.
- | | | |
|------------|---|---|
| $\diamond$ | 0 | 1 |
| 0 | 0 | 1 |
| 1 | 1 | 0 |
- Since $f(0) = \text{true}$ and $f(1) = \text{false}$, we see this is not the table for either $\vee$ or $\wedge$.
42. (a) SKHIEDSFDQKJSI
 (b) TRAPPED CAN NOT RETURN
43. JCCGKFMNUQPJKNFQKJUM
44. The weakness of a substitution code is that fact that a letter is always replaced by the same letter. Analyzing the frequency of appearance of each letter and comparing this to the known frequencies in natural language gives a powerful tool for breaking a substitution code.

Exercise Set 5.2, page 181

1. (a) 7 (b) 8 (c) 3
2. (a) 1 (b) 6 (c) 7
3. (a) 1 (b) 0 (c) 1
4. (a) 0 (b) 1 (c) 0
5. (a) 2 (b) -3 (c) 14 (d) -18 (e) 21
6. (a) 3 (b) -2 (c) 14 (d) -17 (e) 22
7. $n = ak + r$, $0 \leq r < n$. Since $k < 2k < 3k < \dots < ak \leq n$, the number of multiples of k between 1 and n is a . But $\frac{n}{k} = a + \frac{r}{n}$ with $0 \leq \frac{r}{n} < 1$ so $\lfloor \frac{n}{k} \rfloor = a$.
8. If n is odd, then $n = 2k + 1$ for some integer k and $\frac{n^2}{4} = \frac{4k^2 + 4k + 1}{4} = k^2 + k + \frac{1}{4}$.
So $\lceil \frac{n^2}{4} \rceil = k^2 + k + 1 = \frac{4k^2 + 4k + 4}{4} = \frac{n^2 + 3}{4}$
9. (a) 26 (b) 866 (c) 74 (d) 431
10. $g(4) = -3$ $g(14) = -23$ $g(129) = -253$ $g(23) = -41$
11. (a) 2 (b) 8 (c) 32 (d) 1024
12. (a) $f_3(2) = 9$ (b) $f_3(3) = 27$ (c) $f_3(6) = 729$ (d) $f_3(8) = 6561$
13. (a) 4 (b) 7 (c) 9 (d) 10
14. (a) 3; 4 (b) 4; 5
15. (a) 5; 6 (b) 6; 7
16. (a) 8; 8 (b) 8; 9
17. For any 2×2 matrix $\mathbf{M}$, $\mathbf{M}^T$ exists so t is everywhere defined. If $\mathbf{M}$ is a 2×2 matrix, then $t(\mathbf{M}^T) = \mathbf{M}$, so t is onto. Suppose $\mathbf{M}^T = \mathbf{N}^T$. Then $(\mathbf{M}^T)^T = (\mathbf{N}^T)^T$; that is, $\mathbf{M} = \mathbf{N}$ and t is one to one.
18. (a) Each string, including Λ , has a fixed number of elements, so l is everywhere defined for A^* .
(b) $l(abc) = l(dba)$ and $abc \neq dba$.
(c) l is not onto since $\text{Ran}(l) = Z^+ \cup \{0\}$, not Z .
19. Every relation R on A defines a unique matrix $\mathbf{M}_R$ so f is everywhere defined and one to one. Any $n \times n$ Boolean matrix $\mathbf{M}$ defines a relation on A so f is onto.
20. 4; 16
21.

x	y	z	$f(x, y, z)$
T	T	T	T
T	T	F	F
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	T
F	F	T	T
F	F	F	F
22. (a) False (b) False (c) True
23. (a) True (b) False (c) False (d) True
24. (a) 59 (b) 128
25. (a) 31 (b) 0 (c) 36
26. (a) 47 (b) 34 (c) 44
27. $\lfloor \frac{n}{k} + 2 \rfloor$
28. If $|A| = n$, then each subset of A is represented by a string of 0's and 1's of length n . By the multiplication principle, there are 2^n such strings.
29. $f^{-1}(1)$ is the set of elements of A .
30. $2^n C_n$
31. $()$; $(())$; $()()$; $((()))$; $((())())$; $((())())$; $((())())$; $((())())$; $C_1 = 1$, $C_2 = 2$, $C_3 = 5$.
32. There can be only one fewer, since p_i is the first such p_j ; one more; $n - 1$ left and $n + 1$ right
33. because there are more right than left parentheses; one less; n left and n right. $s_1 s_2 \dots s_{2n}$ is not well formed since in $s_1 s_2 \dots s_k$ there are more right than left parentheses.
34. $2^n C_{n-1}$
35. $2^n C_n - 2^n C_{n-1}$

36. $\frac{2nC_n}{n+1} = \frac{(2n)!}{n!(n+1)!}$
37. (a) $0 + 2 + 9 + 0 + 40 + 18 + 7 + 32 + 27 = 135$; $135 + 30 = 165$ or $0 \pmod{11}$.
 (b) i. 9 i. X iii. 5 iv. 3
38. (a) One example is to change 0-183-47381-7 to 0-183-47380-7.
 (b) One example is to change 0-183-47381-7 to 0-382-47381-7.

Exercise Set 5.3, page 187

1. (a) The number of steps remains 1001. (b) The number of steps doubles.
 (c) The number of steps quadruples. (d) The number of steps increases eightfold.
2. (a) The number of steps increases by 1. (b) The number of steps is squared.
 (c) The number of steps is $2n$ plus twice as many.
 (d) The number of steps increases by a factor of 16.
3. $|n!| = |n(n-1)(n-2)\cdots 2 \cdot 1| \leq 1 \cdot |n \cdot n \cdots n|, n \geq 1$.
4. $|1 + 2 + \cdots + n| \leq 1 \cdot |n + n + \cdots + n| = |n \cdot n| = |n^2|, n \geq 1$.
5. $|8n + \lg(n)| \leq |8n + n| = 9|n|, n \geq 1$.
6. $|n^2(7n-2)| = |7n^3 - 2n^2| \leq |7n^3 + 2n^2| \leq |7n^3 + 2n^3| = 9|n^3|, n \geq 1$.
7. $|n \lg(n)| \leq |n \cdot n| = n^2, n \geq 1$. Suppose there exist c and k such that $n^2 \leq c \cdot n \lg(n), n \geq k$. Choose $N > k$ with $N > c \cdot \lg(N)$. Then $N^2 \leq c \cdot N \cdot \lg(N) < N^2$, a contradiction.
8. For $n \geq 1024$, $\lg(n) \leq \lfloor n/100 \rfloor$ and $|n^{100}| = |(2^{\lg(n)})^{100}| \leq (2^{\lfloor n/100 \rfloor})^{100} \leq 2^n$. Suppose there exist c and k such that $2^n \leq c \cdot n^{100}, n \geq k$. Choose $N > \max(k, c, 1024)$ and $\lg(N) < N/101$. $2^N \leq c \cdot N^{100} < N^{101} \leq (2^{\lg(N)})^{101} < 2^N$.
9. $|5n^2 + 4n + 3| \leq |5n^2 + 500n|, n \geq 1$; $|5n^2 + 500n| \leq 5|n^2 + 100n|$
 We have $|n^2 + 100n| = |n^2 + 4 \cdot 25n| \leq |n^2 + 4n^2|, n \geq 25$. But $|5n^2| \leq |5n^2 + 4n + 3|$
10. $|\lg(n^3)| = 3|\lg(n)| \leq 3|\log_5 6 + \lg(n)| \leq 3|\log_5 6 + \lg 5 \cdot \log_5 n|$
 $\leq 3|\lg 5 \cdot \log_5 6 + \lg 5 \cdot \log_5 n| = 3|\lg 5| \log_5(6n)|$
 $|\log_5(6n)| = |\lg 5 \cdot \lg(6n)| \leq \lg 5 |\lg n^3|, n \geq 3$.
11. $\{f_5\}, \{f_6, f_{10}, f_{11}\}, \{f_7\}, \{f_4\}, \{f_8\}, \{f_1\}, \{f_2\}, \{f_3\}, \{f_9\}, \{f_{12}\}$
12. $f_5, f_7, f_4, f_8, (f_6, f_{10}, f_{11}), f_{12}, f_1, f_2, f_3, f_9$
13. $f_1, \Theta(n \lg n); f_2, \Theta(n^2); f_4, \Theta(\lg n); f_5, \Theta(1); f_6, \Theta(n); f_{10}, \Theta(n); f_{11}, \Theta(n)$
14. $f(n) = 2 + 18 \cdot 6 + 1$ or 111 $\Theta(1)$
15. $f(n) = 2 + 4 \cdot 5 + 1$ or $f(n) = 23$ $\Theta(1)$
16. $f(n) = 2 + n \cdot 5 + 1$ or $f(n) = 3 + 5n$ $\Theta(n)$
17. $f(n) = 1 + n \cdot 2$ $\Theta(n)$
18. $f(n) = 2 + 4 \lg(n)$ $\Theta(\lg n)$
19. $f(n) = 2 + 2n + \frac{n(n+1)}{2}$ $\Theta(n^2)$
20. $f(n) = 1 + 2n$ $\Theta(n)$
21. $f(n, m, q) = 1 + nq + 3nmq + 1$ Let $N = \max(n, m, q)$, then f is $\Theta(N^3)$.
22. $\Theta(2^n)$
23. (a) $P_n = P_{n-1} + (n-2) + (n-3), P_3 = 1, P_4 = 4$.
 (b) $\Theta(n^2)$
24. Suppose $a < b$. Then $n^a \leq n^b, n \geq 1$. Suppose $n^b \leq c \cdot n^a$, for $n \geq k$. Choose $N > k$ with $N^{b-a} > c$. Then $N^b \leq c \cdot N^a < N^{b-a} \cdot N^a = N^b$, a contradiction.
 Suppose $\Theta(n^a)$ is lower than $\Theta(n^b)$. Then $n^a \leq c \cdot n^b, n \geq k$. Suppose $a > b$. Choose $N > k$ with $N^{a-b} > c$. Then $N^a \leq c \cdot N^b < N^{a-b} \cdot N^b = N^a$, a contradiction. Hence $a < b$.
25. Suppose $h(n) > 0, \forall n, \Theta(f)$ lower than (or the same as) $\Theta(g)$. $|f(n)| \leq c \cdot |g(n)|, n \geq k$
 (and $|g(n)| \leq d \cdot |f(n)|, n \geq l$). $h(n) |f(n)| \leq c \cdot h(n) \cdot |g(n)|, n \geq k$
 (and $h(n) |g(n)| \leq d \cdot h(n) \cdot |f(n)|, n \geq l$). Hence $|f(n) \cdot h(n)| \leq c \cdot |g(n) \cdot h(n)|, n \geq k$
 (and $|g(n) \cdot h(n)| \leq d \cdot |f(n) \cdot h(n)|, n \geq l$). Hence $\Theta(fh)$ is lower than (or the same as) $\Theta(gh)$.
 Note that if $\Theta(f)$ is strictly lower than $\Theta(g)$ then $\Theta(fh)$ must be strictly lower than $\Theta(gh)$.
26. There exist c_1, k_1, c_2, k_2 such that $|f(n) + g(n)| \leq |f(n)| + |g(n)| \leq c_1|h(n)| + c_2|h(n)|,$
 $n \geq \max\{k_1, k_2\}$. Thus, $f + g$ is $O(h)$.

27. There exist c_1, k_1 such that $|f(n)| \leq c_1|g(n)|, \forall n \geq k_1$, so $|cf(n)| = |c| \cdot |f(n)| \leq c_1|c| \cdot |g(n)|, \forall n \geq k_1$. Also, there exist c_2, k_2 such that $|g(n)| \leq c_2|f(n)|, \forall n \geq k_2$ and so $|g(n)| \leq c_2|f(n)| = \frac{c_2}{|c|} \cdot |cf(n)|, \forall n \geq k_2$.
28. For $n \geq 1, a^n < b^n$ if and only if $0 < a < b$ so a^n is $O(b^n)$. Suppose that there exist c, k such that $b^n \leq c \cdot a^n, n \geq k$. We can choose $N > k$ such that $\left(\frac{b}{a}\right)^N > c$ if and only if $0 < a < b$. But then $b^N \leq c \cdot a^N < \left(\frac{b}{a}\right)^N a^N = b^N$, a contradiction so $\Theta(a^n)$ is lower than $\Theta(b^n)$ if and only if $0 < a < b$.
29. $|rf(n)| \leq |r| \cdot |f(n)|, n \geq 1$ so rf is $O(f)$. Choose c such that $c \cdot |r| \geq 1$, then $|f(n)| \leq c \cdot |r| \cdot |f(n)| = c \cdot |rf(n)|, n \geq 1$ and f is $O(rf)$.

Exercise Set 5.4, page 195

1. (a) Yes (b) No
2. (a) Yes (b) No
3. (a) Yes (b) No
4. (a) No (b) Yes
5. (a) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 2 & 6 & 5 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 5 & 6 & 3 & 1 & 4 \end{pmatrix}$
6. (a) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 1 & 6 & 3 & 4 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 1 & 2 & 6 & 3 \end{pmatrix}$
7. (a) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 3 & 2 & 5 & 4 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 4 & 6 & 2 & 5 \end{pmatrix}$
8. (a) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 4 & 3 & 2 & 1 & 5 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 2 & 5 & 1 & 3 & 4 \end{pmatrix}$
9. (a) (1, 5, 7, 8, 3, 2) (b) (2, 7, 8, 3, 4, 6)
10. (a) (1, 5, 6, 7, 4, 2) (b) (1, 2, 3, 8, 5, 6, 7, 4)
11. (a) $(a, f, g) \circ (b, c, d, e)$ (b) $(a, c) \circ (b, g, f)$
12. (a) $(6, 8) \circ (2, 3) \circ (1, 4, 5)$ (b) $(5, 7, 8, 6) \circ (1, 2, 3, 4)$
13. (a) (1, 6, 3, 7, 2, 5, 4, 8) (b) $(5, 6, 7, 8) \circ (1, 2, 3)$
14. (a) (a, g, e, c, b, d) (b) (a, d, b, e, g, c)
15. (a) $(2, 6) \circ (2, 8) \circ (2, 5) \circ (2, 4) \circ (2, 1)$ (b) $(3, 6) \circ (3, 1) \circ (4, 5) \circ (4, 2) \circ (4, 8)$
16. UYROEHWRAEE
17. I AM NOT AT HOME
18. (a) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 & 17 & 18 & 19 & 20 & 21 & 22 & 23 & 24 & 25 & 26 & 27 & 28 & 29 & 30 \\ 7 & 19 & 13 & 1 & 25 & 8 & 20 & 14 & 2 & 26 & 9 & 21 & 15 & 3 & 27 & 10 & 22 & 16 & 4 & 28 & 11 & 23 & 17 & 5 & 29 & 12 & 24 & 18 & 6 & 30 \end{pmatrix}$
(b) $(1\ 7\ 20\ 28\ 18\ 16\ 10\ 26\ 12\ 21\ 11\ 9\ 2\ 19\ 4) \circ (3\ 13\ 15\ 27\ 24\ 5\ 25\ 29\ 6\ 8\ 14) \circ (17\ 22\ 23)$
19. (a) EOXMFEKNRAAEMFX
(b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\ 4 & 10 & 7 & 1 & 13 & 5 & 11 & 8 & 2 & 14 & 6 & 12 & 9 & 3 & 15 \end{pmatrix}$
20. (a) Even (b) Even
21. (a) Even (b) Odd
22. Suppose p_1 is the product of $2k_1$ transpositions and p_2 is the product of $2k_2$ transpositions. Then $p_2 \circ p_1$ can be written as the product of $2(k_1 + k_2)$ transpositions. By Theorem 3, $p_2 \circ p_1$ is even.
23. Suppose p_1 is the product of $2k_1 + 1$ transpositions and p_2 is the product of $2k_2 + 1$ transpositions. Then $p_2 \circ p_1$ can be written as the product of $2(k_1 + k_2) + 2$ transpositions. By Theorem 3, $p_2 \circ p_1$ is even.
24. Suppose p_1 is the product of $2k_1 + 1$ transpositions and p_2 is the product of $2k_2$ transpositions. Then $p_2 \circ p_1$ can be written as the product of $2(k_1 + k_2) + 1$ transpositions. By Theorem 3, $p_2 \circ p_1$ is odd. The case of p_1 even and p_2 odd is similar.
25. (a) (1, 5, 2, 3, 4) (b) (1, 4, 2, 5, 3)
26. By Exercises 30 and 31, Section 5.1, $p \circ p$ is one to one and onto. Since p is everywhere defined on A so is p . Hence p^2 is a permutation of A .

27. (a) $(1, 2, 4)$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 3 & 2 & 5 & 6 \end{pmatrix}$
 (c) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 3 & 2 & 5 & 6 \end{pmatrix}$ (d) 3
28. (a) $(1, 4) \circ (2, 3, 5)$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 2 & 1 & 3 & 6 \end{pmatrix}$
 (c) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 5 & 2 & 4 & 3 & 6 \end{pmatrix}$ (d) 6
29. (a) Basis step: $n = 1$ If p is a permutation of a finite set A , then p^1 is a permutation of A is true.
 Induction Step: The argument in Exercise 26 also shows that if p^{n-1} is a permutation of A , then $p^{n-1} \circ p$ is a permutation of A . Hence p^n is a permutation of A .
 (b) If $|A| = n$, then there are $n!$ permutations of A . Hence, the sequence $1_A, p, p^2, p^3, \dots$ is finite and $p^i = p^j$ for some $i \neq j$. Suppose $i < j$, then $p^{-i} \circ p^i = 1_A = p^{-i} \circ p^j$. So $p^{j-i} = 1_A$, $j-i \in \mathbb{Z}$.
30. $p^0(a) = a$, $a \in A$, so aRa , $a \in A$. If aRb , then $p^n(a) = b$, for some n . Then $p^{-n}(p^n(a)) = p^{-n}(b)$ and $a = p^{-n}(b)$ so bRa . Suppose $aRb \wedge bRc$. Then $p^n(a) = b \wedge p^m(b) = c$, for some n and m . Hence $p^{m+n}(p^n(a)) = p^m(b) = c$ so $p^{m+n}(a) = c$ and aRc . This shows that R is an equivalence relation. Write p as a product of disjoint cycles. The elements of each cycle form an equivalence class of R .
31.

	1_A	p_1	p_2	p_3	p_4	p_5
1_A	1_A	p_1	p_2	p_3	p_4	p_5
p_1	p_1	1_A	p_4	p_5	p_2	p_3
p_2	p_2	p_3	1_A	p_1	p_5	p_4
p_3	p_3	p_2	p_5	p_4	1_A	p_1
p_4	p_4	p_5	p_1	1_A	p_3	p_2
p_5	p_5	p_4	p_3	p_2	p_1	1_A
32. For the row labeled p , find the label of the column containing 1_A . This label is p^{-1} .
33. $\{1_A\}, \{1_A, p_1\}, \{1_A, p_2\}, \{1_A, p_5\}, \{1_A, p_3, p_4\}, \{1_A, p_1, p_2, p_3, p_4, p_5\}$
34.

Element	1_A	p_1	p_2	p_3	p_4	p_5
Period	1	2	2	3	3	2
- It appears that the order of p equals the cardinality of the subset to which p belongs in Exercise 33. Also, elements of the subset are $1_A, p, p^2, p^3, \dots, p^{\text{order}-1}$.
35. There is exactly one of each kind.
36. 10
37. (a) 3 (b) 6
38. 10
39. For each increasing sequence of length $\lceil \frac{n}{2} \rceil$, there is exactly one associated up-down permutation of A , because there is just one way to arrange the remaining elements of A in decreasing order and insert them to fill the even positions.

Review Questions, page 197

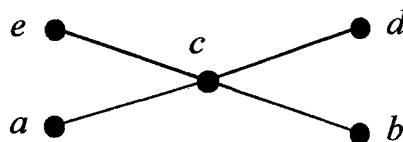
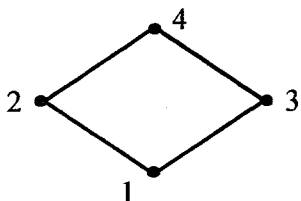
- Let $f : A \rightarrow B$ be a function, then $|f(a)| \leq 1$, $a \in A$.
- Assume that $f(a_1) = f(a_2)$ and show that $a_1 = a_2$.
- Let $f : A \rightarrow B$. Choose $b \in B$ and find $a \in A$ such that $f(a) = b$.
- A hashing function is designed to assign items to a limited number of storage places, and any mod- n function has only n outputs.
- The Θ -class of a function f represents an approximation of how values $f(n)$ grow as n grows.

Chapter 6 Answers

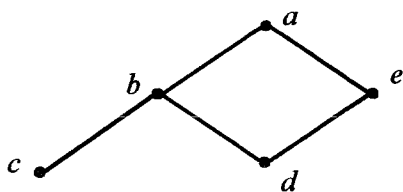
Exercise Set 6.1, page 209

- (a) No (b) No
- (a) Yes (b) Yes

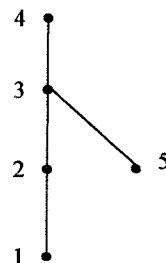
3. (a) Yes (b) Yes
 4. (a) No if $|A| \geq 2$ (b) No
 5. $\{(a, a), (b, b), (c, c), (a, b)\}, \{(a, a), (b, b), (c, c), (a, b), (a, c)\},$
 $\{(a, a), (b, b), (c, c), (a, b), (c, b)\}, \{(a, a), (b, b), (c, c), (a, b), (b, c), (a, c)\}$
 $\{(a, a), (b, b), (c, c), (a, b), (c, b), (c, a)\}, \{(a, a), (b, b), (c, c), (a, c), (c, b), (a, b)\}$
 6. R must be the equality relation on A .
 7. The structure of the proof is to check directly each of the three properties required for a partial order.
 8. The proof is by contradiction based on assuming there is a cycle of length greater than or equal to 2.
 9. 10.



11. $\{(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$
 12. $\{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$
 13.



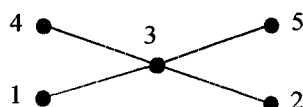
14.



15.



16.

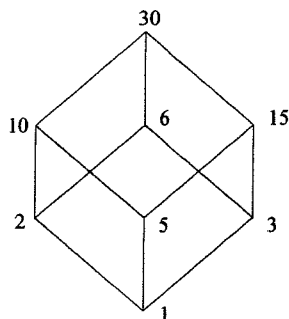


17.
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

18.
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

19. ACE, BASE, CAP, CAPE, MACE, MAP, MOP, MOPE
 20. (a) true (b) true (c) false (d) false

21.



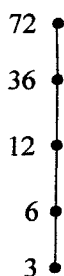
22.

linear

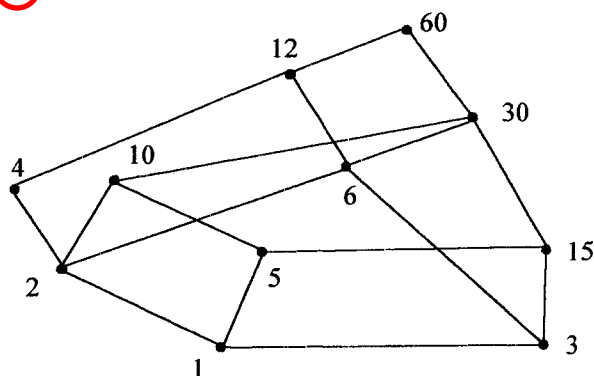


23.

linear



24.



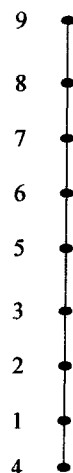
25. If the main diagonal of $\mathbf{M}_R$ is all 1's, then R is reflexive. If $\mathbf{M}_R \odot \mathbf{M}_R = \mathbf{M}_R$, then R is transitive. If $a_{ij} = 1$, $i \neq j$ in $\mathbf{M}_R$, then a_{ji} must be 0 in order for R to be antisymmetric.

26. Possible answers are (a) $\{\{1, 2, 4, 12\}, \{3\}\}$ (b) $\{\{a, d, f\}, \{b, e\}, \{c\}\}$
 (c) $\{\{1, 2, 3, 4\}\}$ (d) $\{\{1, 2, 6, 7, 8\}, \{3, 4, 5\}\}$ (e) $\{\{1, 2\}, \{3, 4\}, \{5, 6, 7\}, \{8, 9\}\}$
 27. (a) $\{2, 3\}$ (b) $\{b, c, d\}$ (c) $\{3\}$ (d) $\{2, 3\}$ (e) $\{2, 3, 7, 8\}$

28. The numbers appear to be equal. Incomparable elements would have to belong to disjoint linear orders.

29.

30.



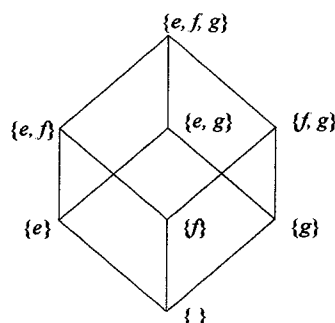
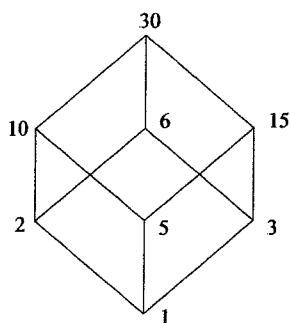
31. $a \leq a$ gives $a \leq' a$ for all $a \in A'$. $\leq'$ is reflexive.

Suppose $a \leq' b$ and $b \leq' a$. Then $a \leq b$, $b \leq a$, and $a = b$. Hence $\leq'$ is antisymmetric.

Suppose $a \leq' b$ and $b \leq' c$. Then $a \leq b$, $b \leq c$, and $a \leq c$. Hence $a \leq' c$ and $\leq'$ is transitive.

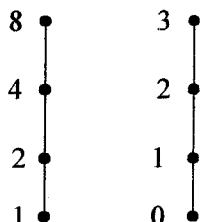
32. For every $a, b \in A$, either aRb or bRa . Hence either $bR^{-1}a$ or $aR^{-1}b$, and R^{-1} is a linear order.

33. Suppose $U \subset T$ and $T \subset V$. Then $U \subset V$ and R is transitive. No set is a proper subset of itself so R is irreflexive. Hence R is a quasiorder.
34. Suppose $x < y, y < z, x, y, z \in A$. Then $x < z$ so $<$ is transitive. Clearly, $x \not< x, x \in A$; $<$ is irreflexive. Thus $<$ is a quasiorder.
35. Suppose $aR^{-1}b$ and $bR^{-1}c$. Then cRb, bRa , and cRa . Hence $aR^{-1}c$ and R^{-1} is transitive. Suppose that $xR^{-1}x$. Then xRx , but this is a contradiction. Hence R^{-1} is irreflexive and a quasiorder.
36. aRb if and only if $a \mid b$ and $a \neq b$.
37. $(a, b) \prec (a, b)$ since $a \mid a$ and $b \leq b$. Thus $\prec$ is reflexive. Suppose $(a, b) \prec (c, d)$ and $(c, d) \prec (a, b)$. Then $a \mid c$ and $c \mid a$. This means $c = ka = k(mc)$ and for a and c in $B, km = 1$ implies $k = m = 1$. Hence $a = c$. Also, $b \leq d$ and $d \leq b$ so $b = d$. Thus $\prec$ is antisymmetric. Suppose $(a, b) \prec (c, d)$ and $(c, d) \prec (e, f)$. Then $a \mid c$ and $c \mid e$. Hence $c = ka, e = mc$, and we have $e = m(ka)$ and $a \mid e$. $b \leq d, d \leq f$ yields $b \leq f$. Hence $(a, b) \prec (e, f)$ and $\prec$ is transitive.
38. R can be viewed as four instances of ordinary $\leq$ applied entry-wise to elements of A . Hence, R will have the same properties as $\leq$.
- 39.



Define F as follows: $F(1) = \{\}$; $F(2) = \{e\}$; $F(5) = \{f\}$; $F(3) = \{g\}$; $F(10) = \{e, f\}$; $F(6) = \{e, g\}$; $F(15) = \{f, g\}$; $F(30) = \{e, f, g\}$.

40. Define G as follows: $G(1) = 0$ $G(2) = 1$ $G(4) = 2$ $G(8) = 3$
The Hasse diagrams confirm that if $a \leq b$, then $G(a) \leq G(b)$.

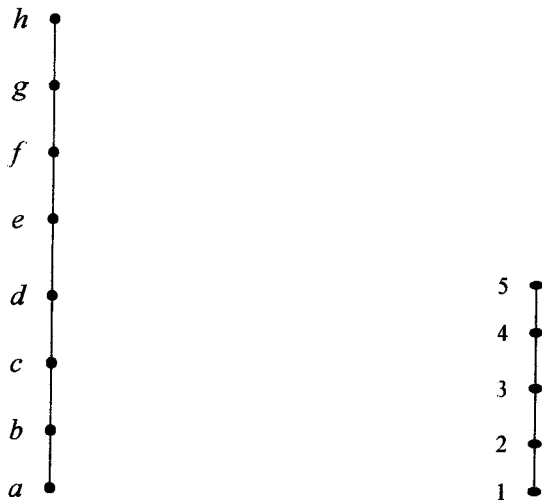


41. Let $U = \{a, b, c, d\}$. For $S \subseteq U$, let f_S be the characteristic function of S relative to U . Define $g(S) = \begin{bmatrix} f_S(a) & f_S(b) \\ f_S(c) & f_S(d) \end{bmatrix}$. This g is a one-to-one correspondence between $P(U)$ and A . If $S \leq T$ in $(P(U), \subseteq)$, then $f_S(x) \leq f_T(x)$ for $x \in U$. Hence $g(S) \leq g(T)$.

Exercise Set 6.2, page 215

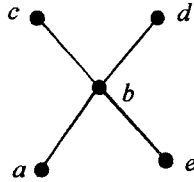
- | | |
|-------------------|-------------------|
| 1. maximal 3, 5 | minimal 1, 6 |
| 2. maximal f, g | minimal a, b, c |
| 3. maximal e, f | minimal a |
| 4. maximal 4, 7 | minimal 1, 8, 9 |
| 5. maximal none | minimal none |
| 6. maximal none | minimal 0 |
| 7. maximal 1 | minimal none |
| 8. maximal 48 | minimal 2, 3 |
| 9. greatest f | least a |

10. greatest e least none
11. no greatest or least
12. greatest 5 least none
13. greatest none least none
14. greatest 1 least 0
15. greatest 72 least 2
16. greatest none least none
17. No, a may be maximal and there be an element of A , b , such that a and b are incomparable.
18. No, a may be minimal and there be an element of A , c , such that a and c are incomparable.
19. (a) True. There can not be $a_1 < a_2 < \dots$ since A is finite.
 (b) False. Not all elements have to be comparable.
 (c) True. There can not be $\dots < a_2 < a_1$ since A is finite.
 (d) False. Not all elements have to be comparable.
20. Suppose a and b are greatest elements of $(A, \leq)$. Then $a \leq b$ and $b \leq a$. Since $\leq$ is antisymmetric, $a = b$. Note: This is a restatement of Theorem 2.
21. Suppose a and b are least elements of $(A, \leq)$. Then $a \leq b$ and $b \leq a$. Since $\leq$ is antisymmetric, $a = b$. Note: This is a restatement of Theorem 2.
22. Suppose b_1, b_2 are each a LUB (or GLB) of B . Then $b_2 \leq b_1$ and $b_1 \leq b_2$ (or $b_1 \leq b_2$ and $b_2 \leq b_1$) by the definition of LUB (or GLB). Since $\leq$ is antisymmetric, $b_1 = b_2$.
23. (a) f, g, h (b) a, b, c (c) f (d) c
24. (a) none (b) none (c) none (d) none
25. (a) d, e, f (b) b, a (c) d (d) b
26. (a) 5 (b) 1, 2, 3 (c) 5 (d) 3
27. (a) none (b) b (c) none (d) b
28. (a) none (b) none (c) none (d) none
29. (a) none (b) 1, 2, 3 (c) none (d) 3
29. (a) $x \in [2, \infty)$ (b) $x \in (-\infty, 1]$ (c) 2 (d) 1
30. (a) $x \in [2, \infty)$ (b) $x \in (-\infty, 1]$ (c) 2 (d) 1
31. (a) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
32. (a) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
33. 34.



35. The least element of A is the label on the row that is all ones. The greatest element of A is the label on the column that is all ones.

36. One possible answer is

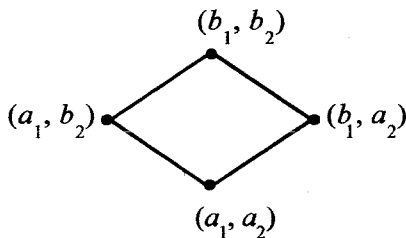


37. (a) 49 (b) $\{2, 4, 8, 16, 32, 64\}$
 38. 25; each prime number less than 100 is minimal.

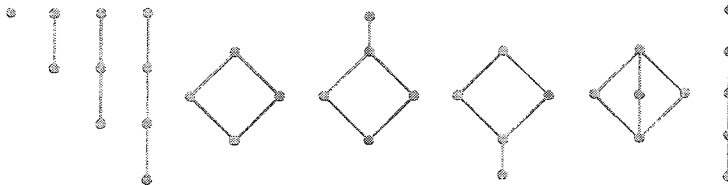
Exercise Set 6.3, page 224

1. Yes, all the properties are satisfied. 2. No, $\text{LUB}(\{e, f\})$ does not exist.
 3. No, $\text{GLB}(\{e, b\})$ does not exist. 4. Yes, all the properties are satisfied.
 5. Yes, all the properties are satisfied.
 6. Yes 7. No
 8. Yes, $\mathbf{M} \vee \mathbf{N} = \text{LUB}(\mathbf{M}, \mathbf{N})$ and $\mathbf{M} \wedge \mathbf{N} = \text{GLB}(\mathbf{M}, \mathbf{N})$.
 9. Yes, $\text{LUB}(\mathbf{M}, \mathbf{N}) = [a_{ij} = \max\{m_{ij}, n_{ij}\}]$ and $\text{GLB}(\mathbf{M}, \mathbf{N}) = [b_{ij} = \min\{m_{ij}, n_{ij}\}]$
 10. For Figure 6.40(c), $f \vee g$ does not exist, because there is no element x with $f \leq x$ and $g \leq x$.
 For Figure 6.40(f), $d \wedge e$ does not exist, because the incomparable elements b and c are both lower bounds of $\{d, e\}$. For Figure 6.40(g), $c \wedge d$ does not exist, because the incomparable elements a and b are both lower bounds of $\{c, d\}$.

11.



12. Since $a_1 \vee a_2$ is $\text{LUB}(\{a_1, a_2\})$ and $b_1 \vee b_2$ is $\text{LUB}(\{b_1, b_2\})$, $(a_1 \vee a_2, b_1 \vee b_2)$ is an upper bound of $\{(a_1, b_1), (a_2, b_2)\}$. If (c, d) is another upper bound of $\{(a_1, b_1), (a_2, b_2)\}$, then $a_1 \leq c, a_2 \leq c, b_1 \leq d, b_2 \leq d$ so $a_1 \vee a_2 \leq c$ and $b_1 \vee b_2 \leq d$. Hence $(a_1 \vee a_2, b_1 \vee b_2) \leq (c, d)$ and $(a_1 \vee a_2, b_1 \vee b_2)$ is the least upper bound of $\{(a_1, b_1), (a_2, b_2)\}$. A similar argument shows that $(a_1 \wedge a_2, b_1 \wedge b_2)$ is the greatest lower bound of $\{(a_1, b_1), (a_2, b_2)\}$.
 13. For each $T_1, T_2 \subseteq T$, $T_1 \cap T_2$ and $T_1 \cup T_2$ are subsets of T so $P(T)$ is a sublattice of $P(S)$.
 14. Let x, y be in $[a, b]$. Then $x \leq x \vee y \leq b$, so $x \vee y \in [a, b]$. $a \leq x \wedge y \leq x \leq b$, so $x \wedge y \in [a, b]$. Hence $[a, b]$ is a sublattice.
 15. For any elements x, y of a linearly ordered poset, $x \leq y$ or $y \leq x$. Say $x \leq y$. Then $x = x \wedge y$ and $y = x \vee y$. Hence any subset of a linearly ordered poset is a sublattice.
 16. $\{1, 2, 3, 6, 12\}, \{1, 2, 3, 6, 12, 24\}, \{1, 2, 6, 12, 24\}, \{1, 3, 6, 12, 24\}, \{2, 4, 6, 8, 12, 24\}, \{1, 2, 4, 12, 24\}, \{1, 2, 3, 4, 6, 12, 24\}, \{1, 2, 3, 4, 6, 12\}, \{1, 2, 4, 8, 24\}, \{2, 4, 8, 12, 24\}, \{2, 4, 6, 12, 24\}, \{1, 2, 4, 8, 12, 24\}, \{1, 2, 4, 6, 12, 24\}, \{1, 2, 4, 6, 12\}, \{1, 2, 4, 8, 12, 24\}, \{1, 2, 3, 4, 6, 8, 24\}, \{1, 2, 3, 4, 6, 8, 12, 24\}$,
 17.



18. Let L be a bounded lattice, $|L| \geq 2$. Suppose $0 = I$. There must be $x \neq 0, x \neq I$. Then $x \vee 0 = x$ and $x \vee 0 = x \vee I = I$. But this is a contradiction; $x \neq I$. Hence $0 \neq I$.
19. Suppose $a \wedge b = a$. $a \leq a \vee b = (a \wedge b) \vee b = (a \vee b) \wedge b \leq b$. Thus $a \leq b$. Suppose $a \leq b$. $a \wedge b \leq a$ and $a \leq a, a \leq b$ gives $a \leq a \wedge b$. Hence $a \wedge b = a$.
20. In D_n , $a \wedge b = \text{GCD}(a, b)$, $a \vee b = \text{LCD}(a, b)$. Hence, $a \wedge (b \vee c) = \text{GCD}(a, b \vee c) = \text{GCD}(a, \text{LCD}(b, c)) = \text{LCM}(\text{GCD}(a, b), \text{GCD}(a, c)) = (a \wedge b) \vee (a \wedge c)$. Similarly, $a \vee (b \wedge c) = \text{LCM}(a, b \wedge c) = \text{LCM}(a, \text{GCD}(b, c)) = \text{GCD}(\text{LCM}(a, b), \text{LCM}(a, c)) = (a \vee b) \wedge (a \vee c)$.
21. (a) 12 (b) Figure 6.44(a) 3 Figure 6.44(b) 3
22. Since they hold for all elements of the lattice, the distributive properties hold for elements of the sublattice.
23. $(a_1, a_2) \wedge ((b_1, b_2) \vee (c_1, c_2)) = (a_1, a_2) \wedge (b_1 \vee c_1, b_2 \vee c_2) = (a_1 \wedge (b_1 \vee c_1), a_2 \wedge (b_2 \vee c_2)) = ((a_1 \wedge b_1) \vee (a_1 \wedge c_1), (a_2 \wedge b_2) \vee (a_2 \wedge c_2)) = ((a_1, a_2) \wedge (b_1, b_2)) \vee ((a_1, a_2) \wedge (c_1, c_2))$. A similar argument establishes the other distributive property.
24. $a \vee (a' \wedge b) = (a \vee a') \wedge (a \vee b) = I \wedge (a \vee b) = a \vee b$.
 $a \wedge (a' \vee b) = (a \wedge a') \vee (a \wedge b) = 0 \vee (a \wedge b) = a \wedge b$.
25. Suppose $a \wedge x = a \wedge y$ and $a \vee x = a \vee y$. Then $y \leq y \vee (y \wedge a) = (y \wedge y) \vee (y \wedge a) = y \wedge (y \vee a) = y \wedge (a \vee x) = (y \wedge a) \vee (y \wedge x) = (a \wedge x) \vee (y \wedge x) = x \wedge (a \vee y) \leq x$. Hence $y \leq x$. A similar argument shows $x \leq y$. Thus $x = y$.
26. (a) Suppose $a \leq c$. Then $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) = (a \vee b) \wedge c$.
 (b) For the lattice given, $a \wedge (b \vee c) = a \wedge I = a$, but $(a \wedge b) \vee (a \wedge c) = 0 \vee 0 = 0$. Hence the lattice is not distributive. For $x \leq y$, the only cases are $0 \leq a, b, c$ or I and a, b , or $c \leq I$.
 $0 \vee (x \wedge y) = x \vee y$ and $(0 \vee x) \wedge y = x \wedge y$, for any $x, y \in \{a, b, c, I\}$. $x \vee (y \wedge I) = x \vee y$ and $(x \vee y) \wedge I = x \vee y$, $x, y \in \{a, b, c\}$. Hence the lattice is modular.
27. $1' = 42, 42' = 1, 2' = 21, 21' = 2, 3' = 14, 14' = 3, 7' = 6, 6' = 7$
28. neither
29. neither
30. distributive, but not complemented
31. distributive, but not complemented
32. If $a \leq c$ and $b \leq c$, then $a \vee b \leq c$ since $a \vee b$ is $\text{LUB}\{a, b\}$. If $a \vee b \leq c$, we have $a \leq a \vee b \leq c$ so $a \leq c$. Similarly, $b \leq c$.
33. If $x = x'$, then $x = x \vee x = I$ and $x = x \wedge x = 0$. But by Exercise 18, $0 \neq I$. Hence, $x \neq x'$.
34. $a' = e, e' = a, b' = c, c' = b, d, d' = a, c$
35. Suppose $\mathcal{P}_1 \leq \mathcal{P}_2$. Then $R_1 \subseteq R_2$. Let $x \in A_i$, then $A_i = \{y \mid xR_1y\}$ and $A_i \subseteq \{y \mid xR_2y\} = B_j$ where $x \in B_j$. Suppose each $A_i \subseteq B_j$, then xR_1y implies xR_2y and $R_1 \subseteq R_2$. Thus $\mathcal{P}_1 \leq \mathcal{P}_2$.
36. $a' \wedge I = a' \wedge (a \vee a'') = (a' \wedge a) \vee (a' \wedge a'') = 0 \vee (a' \wedge a'') = a' \wedge a''$
 $a'' \wedge I = a'' \wedge (a \vee a') = (a'' \wedge a) \vee (a'' \wedge a') = 0 \vee (a'' \wedge a') = a'' \wedge a'$
37. The sublattice $\{a, b, d\}$ of Figure 6.57 is not complemented.
38. The lattice $\{x \mid 0 \leq x \leq 1, x \in \mathbb{R}\}$ is bounded, but the sublattice $\{x \mid 0 \leq x < 1, x \in \mathbb{R}\}$ is not.
39. For any a, b, c in the sublattice with $a \leq c$, $a \vee (b \wedge c) = (a \vee b) \wedge c$, since this is true in the full lattice.
40. Applying Theorem 6 of this section, we see that any linear order is distributive.

Exercise Set 6.4, page 231

- No, it has 6 elements, not 2^n elements.
- No, it has 7 elements, not 2^n elements.
- No, it has 6 elements, not 2^n elements.
- No, it has 5 elements, not 2^n elements.
- Yes, it is B_3 .
- No, e has no complement.
- Yes, it is B_1 .
- Yes, it is B_2 .
- Yes $385 = 5 \cdot 7 \cdot 11$
- No $60 = 2^2 \cdot 3 \cdot 5$

11. No, each Boolean algebra must have 2^n elements.
12. Suppose $a \leq b$. $a = a \wedge b$ so $a' = (a \wedge b)' = a' \vee b' \geq b'$. Conversely, suppose $b' \leq a'$. Then $b' = b' \wedge a'$ so $(b')' = (b' \wedge a')' = b \vee a \geq a$, so $b \geq a$.
13. Suppose $a = b$. $(a \wedge b') \vee (a' \wedge b) = (b \wedge b') \vee (a' \wedge a) = 0 \vee 0 = 0$.
Suppose $(a \wedge b') \vee (a' \wedge b) = 0$. Then $a \wedge b' = 0$ and $a' \wedge b = 0$. We have $I = 0' = (a \wedge b')' = a' \vee b$. So a' is the complement of b ; $b' = a'$.
14. Suppose $a \leq b$. Then $a \leq b \leq b \vee c$ and $c \leq b \vee c$ so $a \vee c \leq b \vee c$.
15. Suppose $a \leq b$. Then $a \wedge c \leq a \leq b$ and $a \wedge c \leq c$ so $a \wedge c \leq b \wedge c$.
16. Suppose $a \vee b = b$. Then $a \leq a \vee b = b$. Hence $a \wedge b = a$.
Suppose $a \wedge b = a$. Then $a' = (a \wedge b)' = a' \vee b'$. So $a' \vee b = (a' \vee b') \vee b = a' \vee (b' \vee b) = a' \vee I = I$.
Suppose $a' \vee b = I$. Then $0 = I' = (a' \vee b)' = a \wedge b'$.
Suppose $a \wedge b' = 0$. Then $a = a \wedge I = a \wedge (b \vee b') = (a \wedge b) \vee (a \wedge b') = (a \wedge b) \vee 0 = a \wedge b \leq b$.
Suppose $a \leq b$. Then $a \vee b = b$.
17. $(a \wedge b) \vee (a \wedge b') = a \wedge (b \vee b') = a \wedge I = a$.
18. $b \wedge (a \vee (a' \wedge (b \vee b'))) = b \wedge (a \vee (a' \wedge I)) = b \wedge (a \vee a') = b \wedge I = b$.
19. $(a \wedge b \wedge c) \vee (b \wedge c) = (a \vee I) \wedge (b \wedge c) = I \wedge (b \wedge c) = b \wedge c$.
20. $((a \vee c) \wedge (b' \vee c))' = ((a \wedge b') \vee c)' = (a \wedge b')' \wedge c' = (a' \vee b) \wedge c'$.
21. Suppose $a \leq b$. Then $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) = b \wedge (a \vee c)$.
22. Example 7 and Example 8 are both cases where a lattice fails to be a Boolean algebra, because not every element has a unique complement.
23. R is reflexive because $m_{ii} = 1$, $i = 1, 2, \dots, 8$. R is antisymmetric since if $m_{ij} = 1$ and $i \neq j$, then $m_{ji} = 0$. R is transitive, because $\mathbf{M}_R \odot \mathbf{M}_R$ shows that $R^2 \subseteq R$.
24. e is the least element. d is the greatest element.
25. Complement pairs are a, h ; b, g ; c, f ; d, e . Since each element has a unique complement, (A, R) is complemented.
26. (A, R) is a Boolean algebra, because its Hasse diagram is isomorphic to B_3 .
27. (A, R) is not a Boolean algebra; complements are not unique.
28. By Exercise 41, Section 6.1, (A, R) is isomorphic to B_4 , so it is a Boolean algebra.
29. (a) $\{a\}, \{b\}, \{c\}$ (b) 2, 3, 5
30. n
31. matrices with exactly one 1
32. (a) $\{a\}, \{b\}, \{c\}, \{a, b\} = \{a\} \vee \{b\}, \{b, c\} = \{b\} \vee \{c\}, \{a, c\} = \{a\} \vee \{c\}, \{a, b, c\} = \{a\} \vee \{b\} \vee \{c\}$
(b) 2, 3, 7, 6 = $2 \vee 3$, 14 = $2 \vee 7$, 21 = $3 \vee 7$, 42 = $2 \vee 3 \vee 7$
(c) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$
 $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$
 $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$
 $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$
 $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Exercise Set 6.5, page 236

1.

x	y	z	$x \wedge (y \vee z')$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

$\uparrow$

2.

x	y	z	$(x \vee y) \wedge (z \vee x')$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$\uparrow$

3.

x	y	z	$(x \vee y')$	$\vee$	$(y \wedge (x' \vee y))$
0	0	0	0	0	1
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	1
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	0	1	1
1	1	1	0	1	1

(1) (4) (3) (2)

4.

x	y	z	$(x \wedge y)$	$\vee$	$(x' \wedge (y \wedge z'))$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	1	1	0
1	1	1	1	1	0

(1) (4) (3) (2)

5. $(x \vee y) \wedge (x' \vee y) = (x \wedge x') \vee y = 0 \vee y = y$

6. $x \wedge (y \vee (y' \wedge (y \vee y'))) = x \wedge (y \vee (y' \wedge I)) = x \wedge (y \vee y') = x \wedge I = x$

7. $(z' \vee x) \wedge ((x \wedge y) \vee z) \wedge (z' \vee y) = (z' \vee (x \wedge y)) \wedge ((x \wedge y) \vee z)$
 $= (x \wedge y) \vee (z' \wedge z) = (x \wedge y) \vee 0 = x \wedge y.$

8. $((x \wedge z) \vee (y' \vee z')) \vee ((y \wedge z) \vee (x \wedge z')) = ((x \wedge z) \vee (x \vee z')) \vee ((y \wedge z') \vee (y \wedge z))$
 $= (x \wedge (z \vee z')) \vee (y \wedge (z' \vee z)) = (x \wedge I) \vee (y \wedge I) = x \vee y$

9. $(x' \vee y)' \vee z \vee (x \wedge ((y \wedge z) \vee (y' \wedge z'))) = (x \wedge y') \vee z \vee (x \wedge y \wedge z) \vee (x \wedge y' \wedge z')$
 $= ((x \wedge y') \wedge (1 \vee z')) \vee z \wedge (1 \vee (x \wedge y)) = (x \wedge y') \vee z$

10. $(x \wedge z) \vee (y' \vee (y' \wedge z)) \vee ((x \wedge y') \wedge z') = (x \wedge z) \vee y' \vee (x \wedge y' \wedge z') = (x \wedge z) \vee y'$

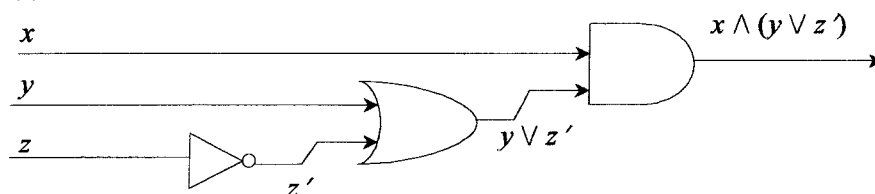
11. $x \wedge z$

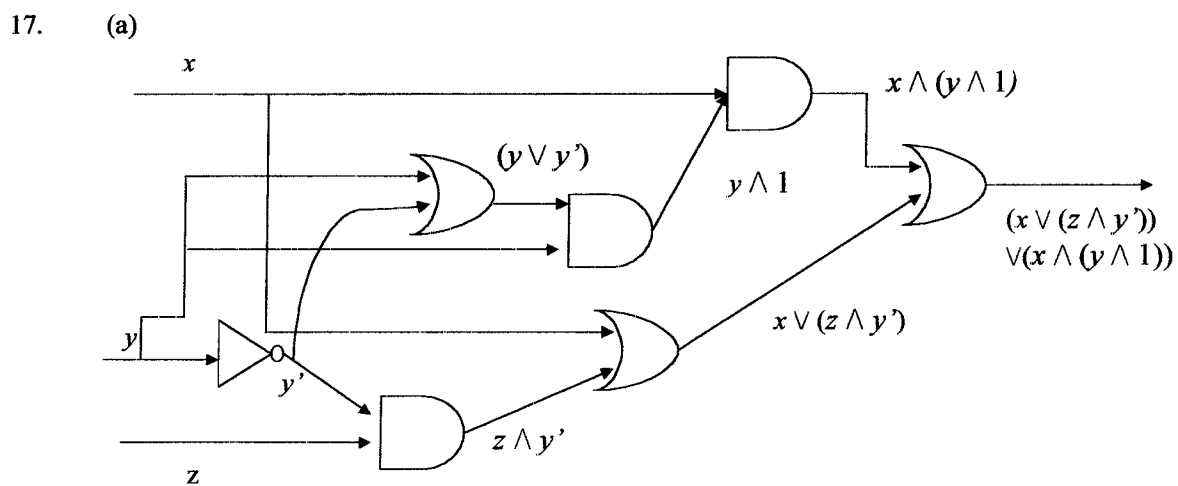
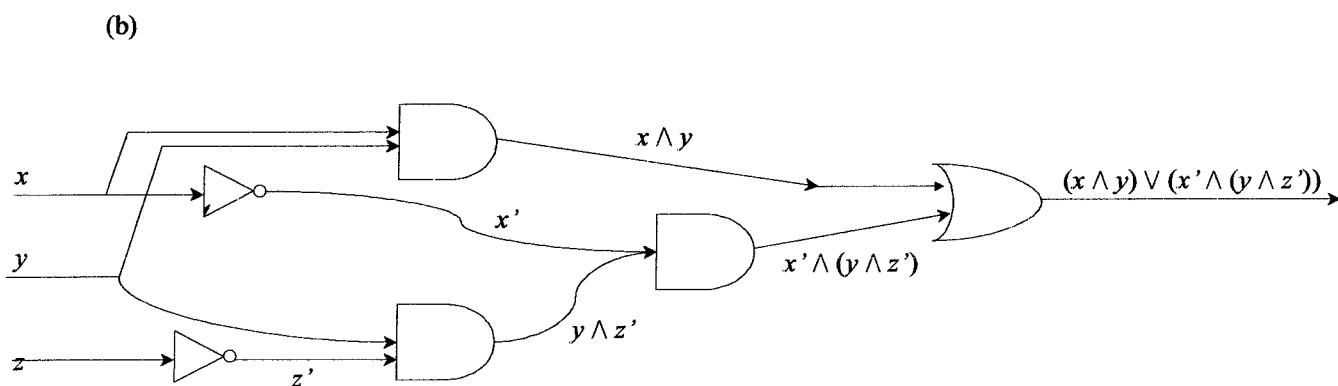
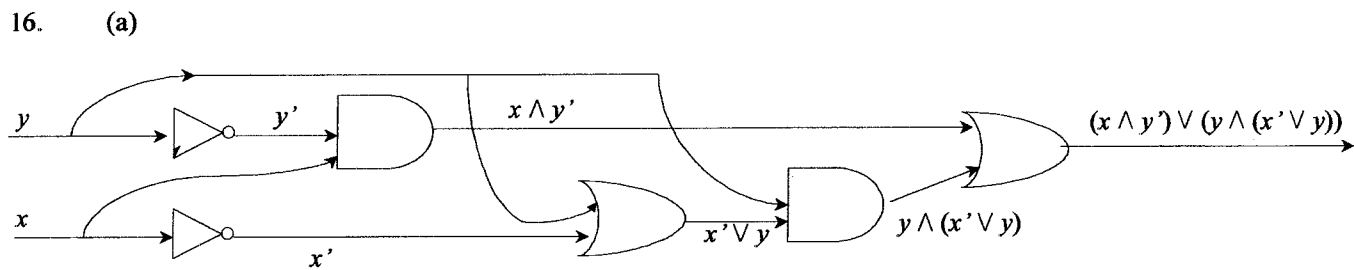
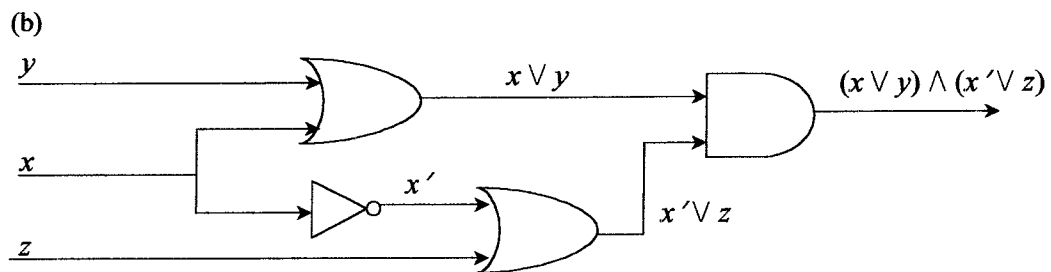
12. z

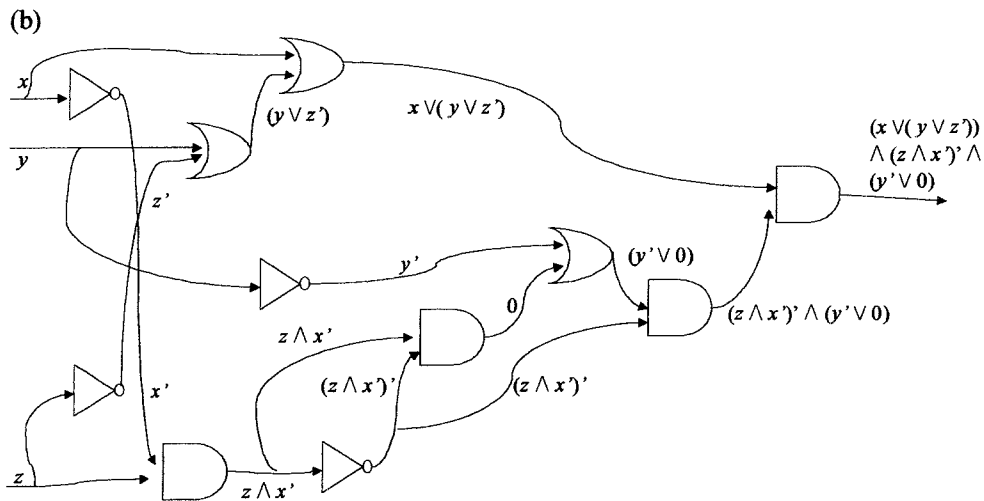
13. $y \vee x'$

14. $x' \wedge z'$

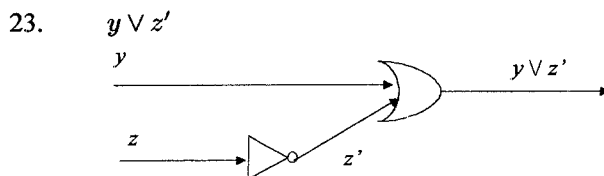
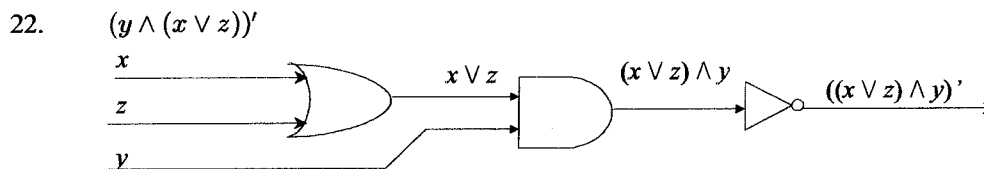
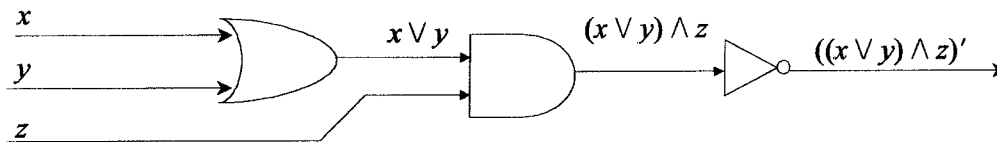
15. (a)







18. $(x \vee (y \wedge z))' \vee z'$
 19. $((x \wedge y) \vee (y \wedge z))'$
 20. $((x \wedge x')' \vee ((w \wedge y) \vee z')) \vee (w' \wedge y)$
 21. $((x \vee y) \wedge z)'$



Exercise Set 6.6, page 246

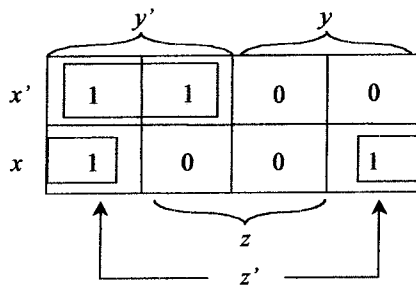
1.

	y'	y
x'	1	0
x	0	1

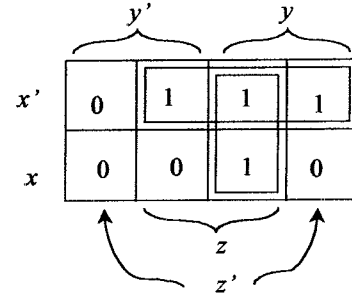
2.

	y'	y
x'	1	0
x	1	0

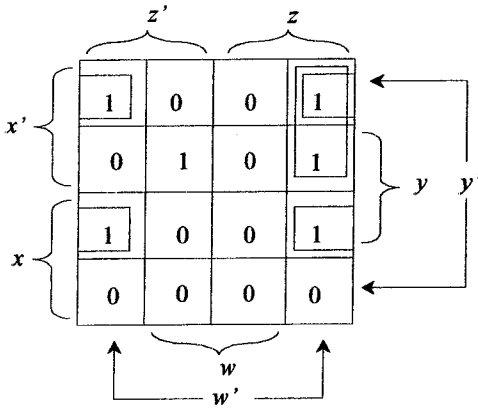
3.



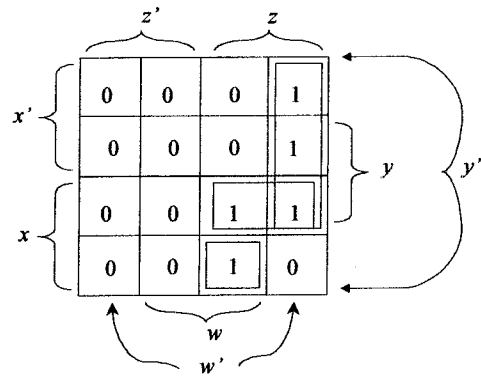
4.



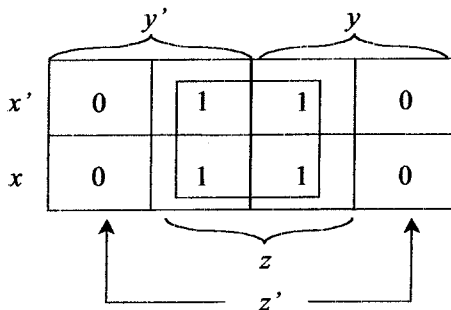
5.



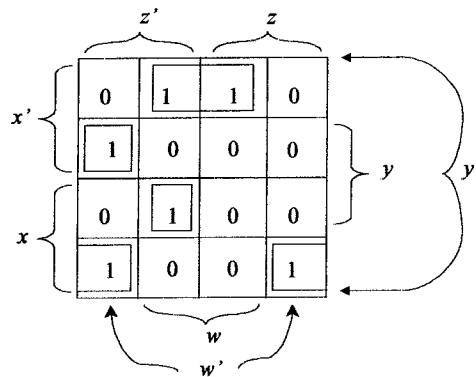
6.



7.



8.



9. $(x' \wedge y') \vee (x \wedge y)$
10. $x' \vee (x \wedge y')$
11. $z' \vee (x' \wedge z)$
12. $(z' \wedge x') \vee (z \wedge y') \vee (x \wedge y \wedge z')$
13. $(z' \wedge y) \vee (x \wedge y') \vee (y' \wedge z)$
14. $x' \vee (z \wedge y)$
15. $(z \wedge x') \vee (w' \wedge x \wedge y) \vee (w \wedge x \wedge y')$
16. $(x' \wedge z') \vee (x' \wedge z \wedge w') \vee (x \wedge y' \wedge w')$
17. $(x' \wedge y') \vee (x \wedge y)$
18. y'
19. $(x' \wedge y') \vee (x \wedge z')$
20. $(x' \wedge y) \vee (y \wedge z) \vee (x' \wedge y' \wedge z)$

21. z
22. $(z \wedge x' \wedge w') \vee (z \wedge x \wedge y) \vee (z \wedge x \wedge w)$
23. $(x' \wedge y' \wedge w') \vee (y \wedge z \wedge w') \vee (x' \wedge z' \wedge y \wedge w)$
24. $(x' \wedge y' \wedge w) \vee (y' \wedge w' \wedge x) \vee (x' \wedge z' \wedge y \wedge w') \vee (x \wedge z' \wedge y \wedge w)$
25. (a) $x' \wedge y', x' \wedge y, x \wedge y'$
 (b) Since $\wedge$ is commutative and associative, we need only consider the case $(w_1 \wedge w_2 \wedge \dots \wedge w_n \wedge y) \vee (w_1 \wedge w_2 \wedge \dots \wedge w_n \wedge y')$. But this is equivalent to $w_1 \wedge w_2 \wedge \dots \wedge w_n$.
26. (a) x', y' (b) A simple check of the values of $f(x, y)$ will verify this.
27. $x' \wedge z', y' \wedge z', y \wedge z', x \wedge z', x \wedge y$
28. z'
29. (a) $(x \wedge y) \vee z'$ (b) A simple check of the values of $f(x, y, z)$ will verify this.

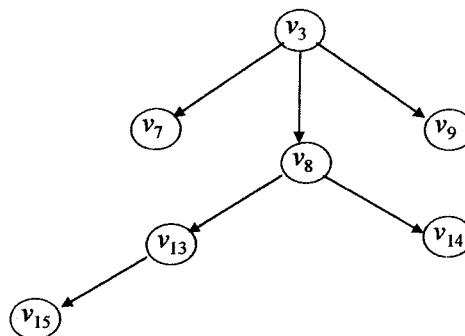
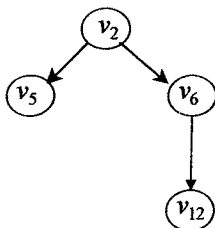
Review Questions, page 249

1. Partial order is a generalization of less than or equal for the real numbers.
2. In a partial order not every pair of elements must have a least upper bound and a greatest lower bound.
3. A model for any Boolean algebra is the power set of a set and the subset relation.
4. The Karnaugh map of a function is an $n \times n$ array of 0's and 1's used to create a Boolean expression that produces the function.
5. Every Boolean function can be produced by a Boolean expression; every Boolean expression produces a Boolean function.

Chapter 7 Answers

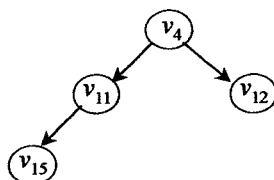
Exercise Set 7.1, page 258

1. Yes, the root is b .
2. No
3. Yes, the root is f .
4. Yes, the root is 6.
5. No
6. No
7. Yes, the root is t .
8. No
9. (a) $v_{12}, v_{10}, v_{11}, v_{13}, v_{14}$ (b) $v_{10}, v_{11}, v_5, v_{12}, v_7, v_{15}, v_{14}, v_9$
10. (a) v_7, v_9 (b) v_{13}, v_{14}, v_{15}
11. (a) (b)

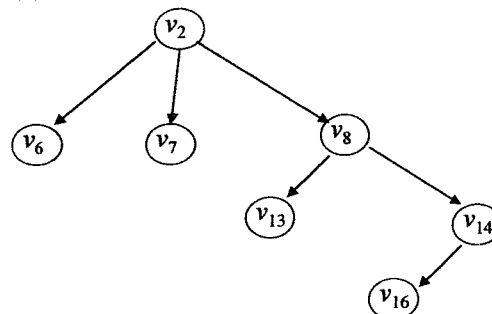


12. (a) 4 (b) 3
13. (T, v_0) may be an n -tree for $n \geq 3$. It is not a complete 3-tree.
14. (a) v_{15}, v_{16} (b) $v_{15}, v_{12}, v_5, v_6, v_7, v_{13}, v_{16}, v_{10}$
15. (a) v_1, v_3 (b) $v_6, v_7, v_8, v_{13}, v_{14}, v_{16}, v_{10}$

16. (a)



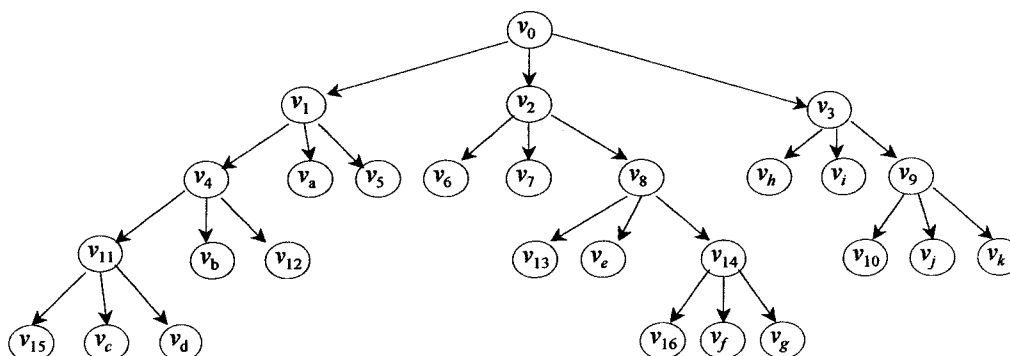
(b)



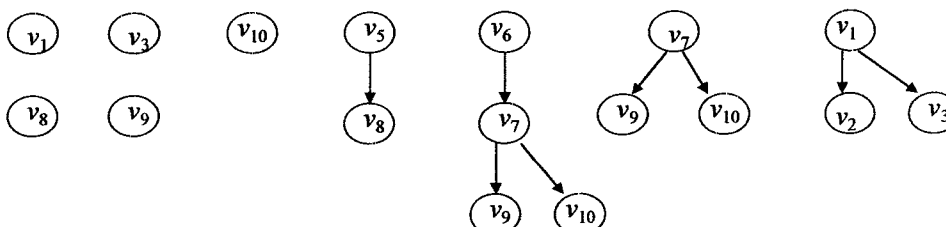
17. (a) 4

(b) 2

18. 11



19.



20. The maximum number of vertices in a binary tree of height n is

$$1 + 2 + 2^2 + 2^3 + \cdots + 2^n = 2^{n+1} - 1. \text{ (See Section 3.5.)}$$

21. Basis Step: $n = 1$ An n -tree of height 1 can have at most n leaves by definition.

Induction Step: Use $P(i)$: An n -tree of height i has at most n^i leaves to show $P(i+1)$: An n -tree of height $i+1$ has at most n^{i+1} leaves. The leaves of a tree T of height $i+1$ belong to the subtrees of T whose roots are at level 1. Each of these subtrees has height at most i , and there are at most n of them. Hence the maximum number of leaves of T is $n \cdot n^i$ or n^{i+1} .

22. Each of the n vertices at level i has either 0 or n offspring since T is complete. At least one of the level 1 vertices has offspring, because there are three levels. Therefore, at level 2, there are jn , $1 \leq j \leq n$, vertices. The total number of vertices is $1 + n + jn$ or $1 + kn$, $2 \leq k \leq n+1$.

23. The total number of vertices is $1 + kn$, where k is the number of non-leaves and 1 counts the root, because every vertex except the root is an offspring. Since $l = m - k$, $l = 1 + kn - k$ or $1 + k(n-1)$.

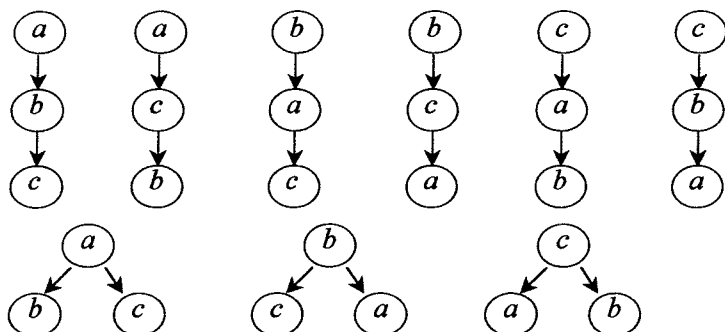
24. Suppose vTv . Then there is a cycle in T . But this is a contradiction. Hence $v \not T v$ for all v . T is irreflexive.

25. If both vTu and uTv , then v, u, v is a cycle in T . Thus vTu implies $u \not T v$. T is asymmetric.

26. Suppose $(a, b) \in T$ and $(b, c) \in T$. Assume $(a, c) \in T$. Then there are two paths from a to c and so two paths from the root to c . This is impossible. Hence, $(a, c) \notin T$.

27. Each vertex except the root has in-degree 1. Thus $s = r - 1$.

28.



29. 4 The tree of maximum height has one vertex on each level.

30. 2

31. Assume that the in-degree of $v_0 \neq 0$. Then there is a cycle that begins and ends at v_0 . This is impossible. Hence the in-degree of v_0 must be 0.

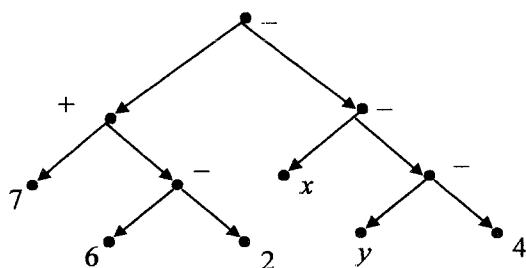
32. (a) Yes (b) Yes (c) No (d) No

33. (a) $2 \leq n$ (b) $1 \leq k \leq 7$

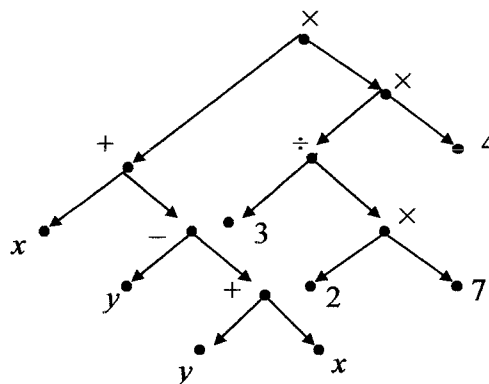
34. n must be 4.

Exercise Set 7.2, page 263

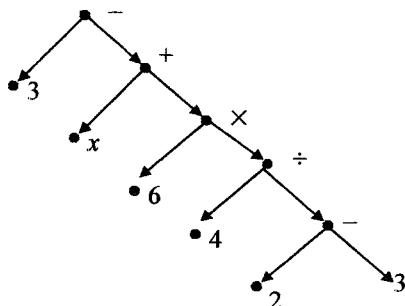
1.



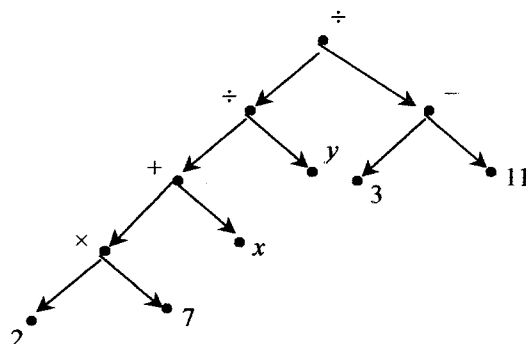
2.



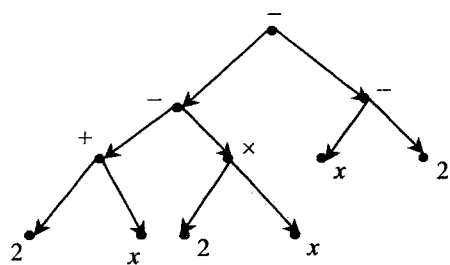
3.



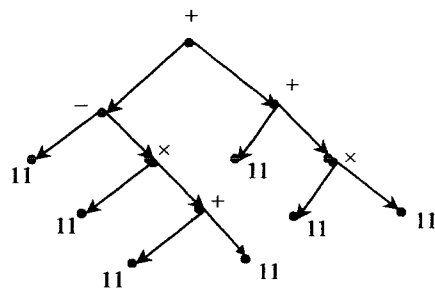
4.



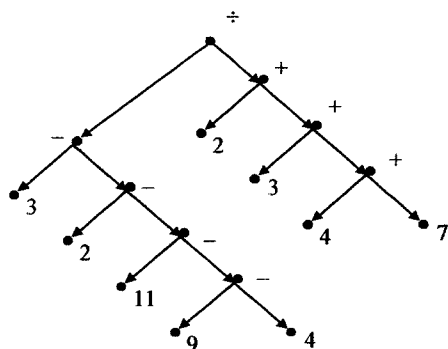
5.



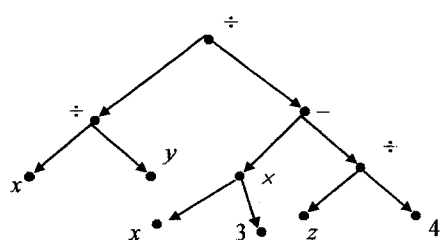
6.



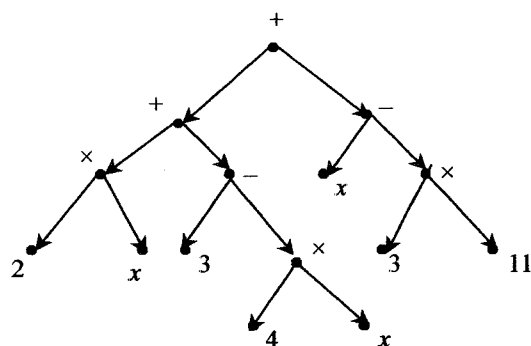
7.



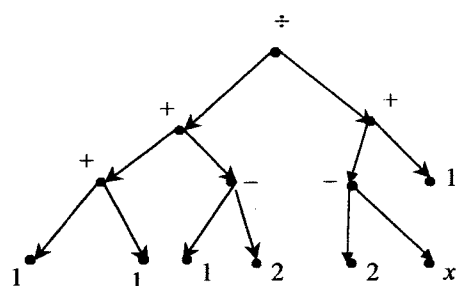
8.



9.



10.



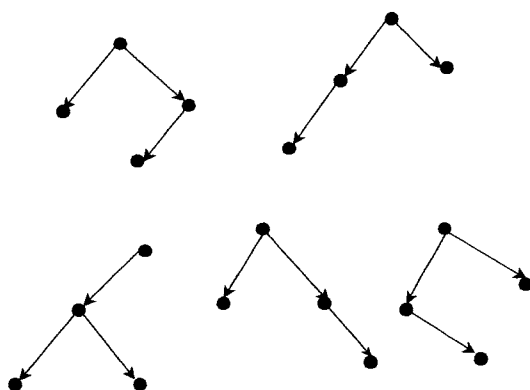
11.

6

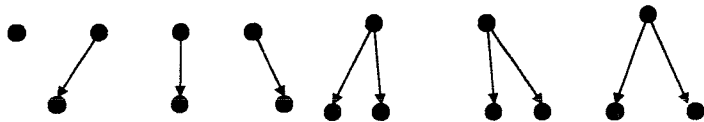
12.

Answers will vary, but there should be at least 4 levels.

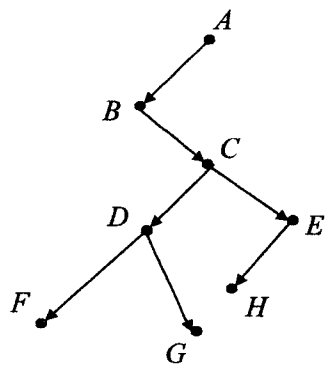
13.



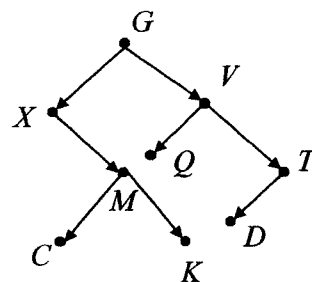
14. 21
15. 721
16.



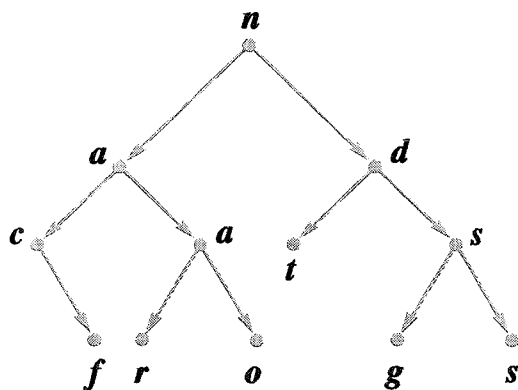
17.



18.



19.



20.	LEFT	DATA	RIGHT
	2		0
	3	1	4
	5	2	6
	0	3	10
	0	4	7
	8	5	0
	0	7	0
	0	8	9
	0	11	0
	11	6	12
	0	9	0
	13	10	0
	0	12	0

21.	LEFT	DATA	RIGHT
	2		0
	3	—	4
	5	+	6
	9	—	10
	0	7	0
	7	—	8
	0	6	0
	0	2	0
	0	x	0
	11	—	12
	0	y	0
	0	4	0

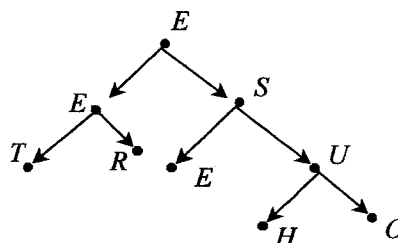
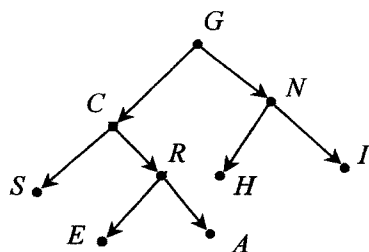
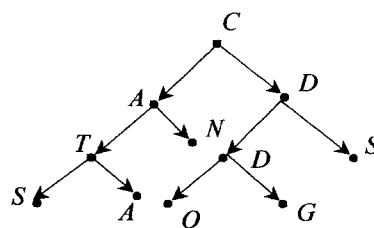
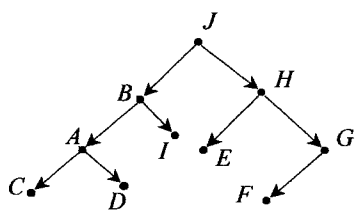
22.	LEFT	DATA	RIGHT
	2		0
	3	$\div$	4
	5	$\div$	6
	7	—	8
	9	+	10
	0	y	0
	0	3	0
	0	11	0
	11	$\times$	12
	0	x	0
	0	2	0
	0	7	0

23.	LEFT	DATA	RIGHT
	2		0
	3	—	4
	5	—	6
	9	—	10
	7	+	8
	11	$\times$	12
	0	2	0
	0	x	0
	0	x	0
	0	2	0
	0	2	0
	0	x	0

24.	LEFT	DATA	RIGHT
	2		0
	3	$\div$	4
	7	$\div$	9
	5	—	6
	8	$\times$	10
	11	$\div$	12
	0	x	0
	0	x	0
	0	y	0
	0	3	0
	0	z	0
	0	4	0

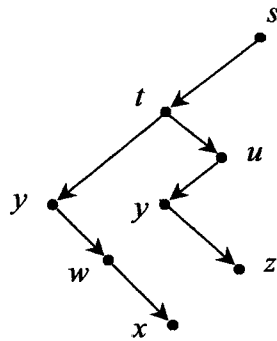
-

1. $xy sz tv$
2. $\times - + 2312$
3. $abcghidke jf$
4. 1246357
5. $TSAMZWEDQMLCKFNTRGJ$
6. $ysxzvut$
7. $2+3-1 \times 2$
8. $gchbiakd jef$
9. 6421357
10. $AZMWSDEQTCLKNFTMGJR$
11. $syvutzx$
12. $23+1-2 \times$
13. $ghcibkjfed a$
14. 6427531
15. $ZWMADQESCNTFKLJGRMT$
16. ONE COW NEVER I PURPLE A SAW SE
17. NEVER I COW A SAW PURPLE ONE SE
18. I NEVER SAW A PURPLE COW I HOPE
19. 4
21. $\frac{15}{16}$
23. $\frac{8}{6}$
- 25.

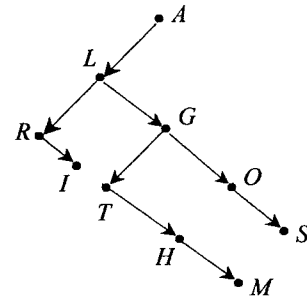


29. (a) The root must be labeled J; if J has a left offspring, it must be labeled B, otherwise the right offspring is labeled B.
 (b) The root must be labeled G; if G has a left offspring, it must be labeled N, otherwise the right offspring is labeled N.
30. Select an element of the string ABCDEF. We can place the elements to the left of it in a left subtree, the elements to the right of it in a right subtree, and label the root with our choice. This process can be repeated to construct the subtrees.

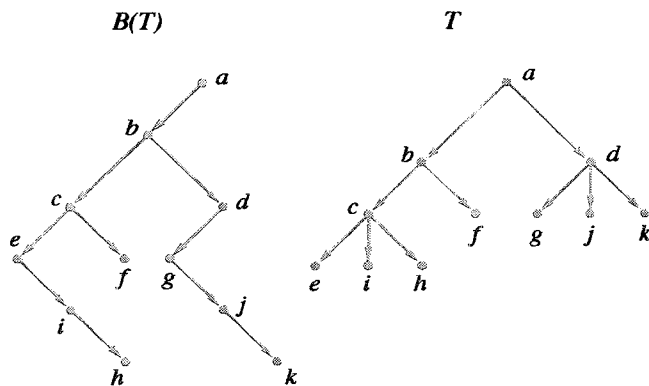
31.



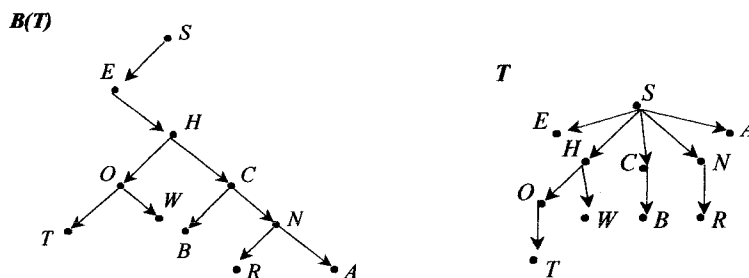
32.



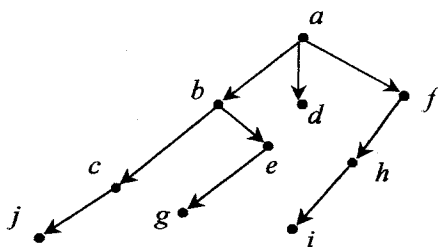
33.



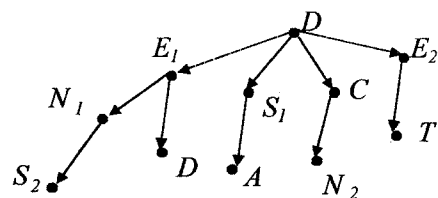
34.



35.

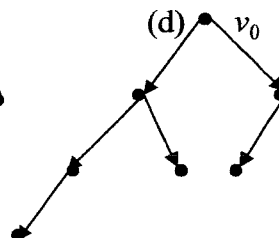
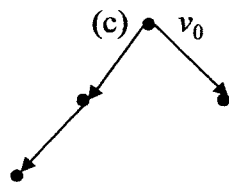
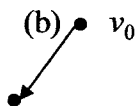


36.



37.

(a) $\bullet v_0$

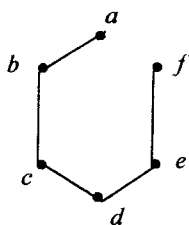


38. Yes

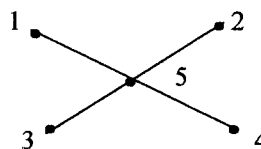
39. $AVL_n = AVL_{n-1} + AVL_{n-2} + 1$. Let v be a new root. Let $T(v_L)$ be an AVL tree of height $n - 1$ using a minimal number of vertices. Let $T(v_R)$ be an AVL tree of height $n - 2$ using a minimal number of vertices. Then $T(v)$, where the left offspring of v is v_L and its right offspring is v_R , is an AVL tree of height n using a minimal number of vertices.

Exercise Set 7.4, page 278

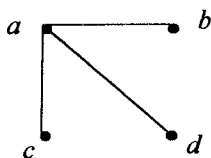
1.



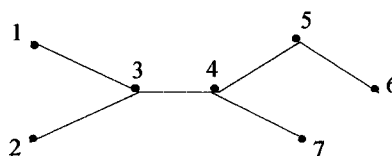
2.



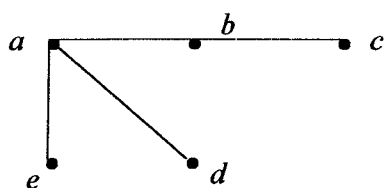
3.



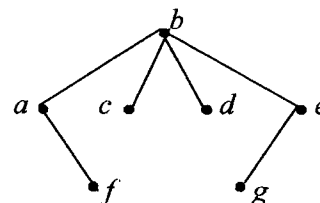
4.



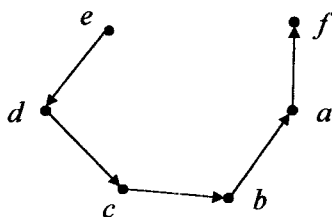
5.



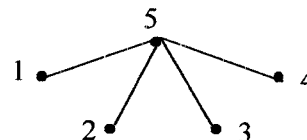
6.



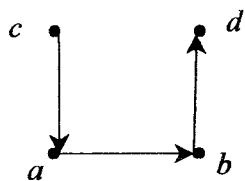
7.



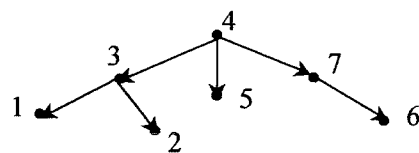
8.



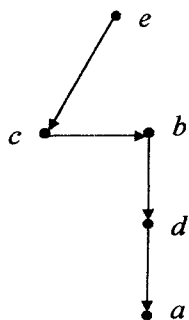
9.



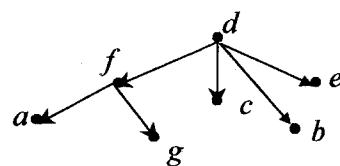
10.



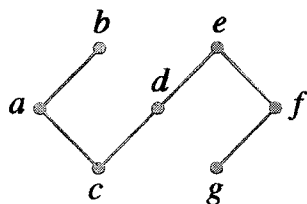
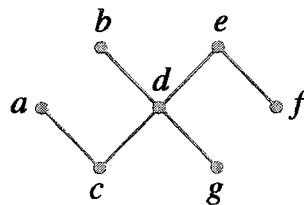
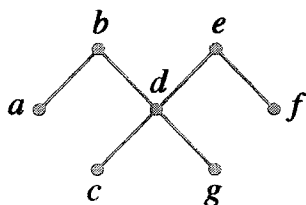
11.



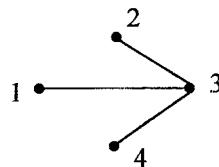
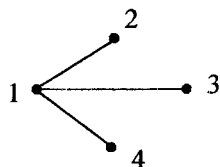
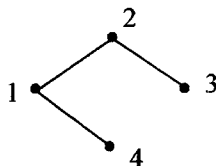
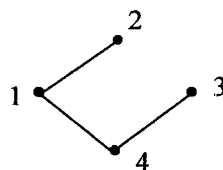
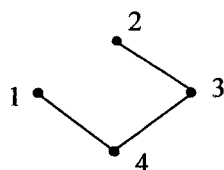
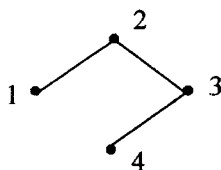
12.



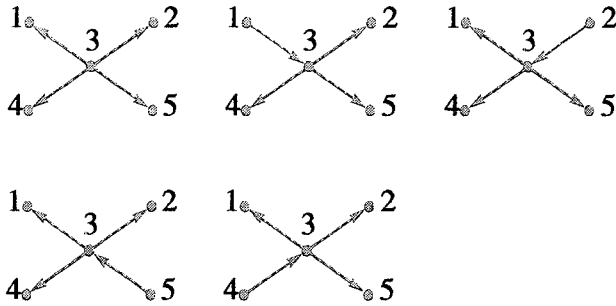
13.



14.

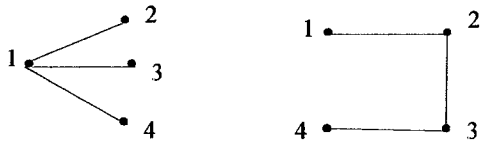


15.



There are 5 spanning trees.

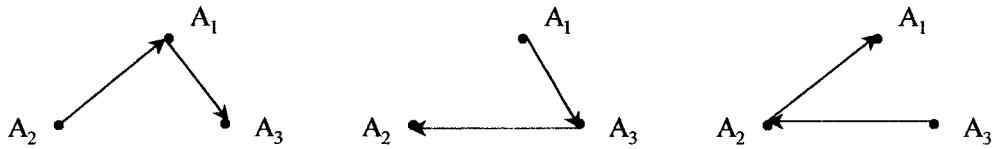
16.



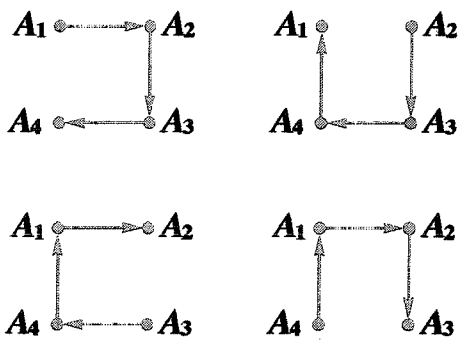
17.



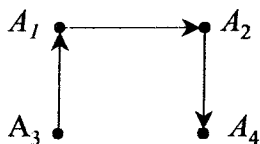
18.



19.

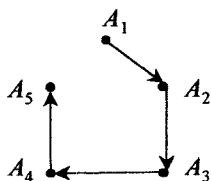


20. All four graphs in the answer to Exercise 19 are isomorphic to each other.



21. Five

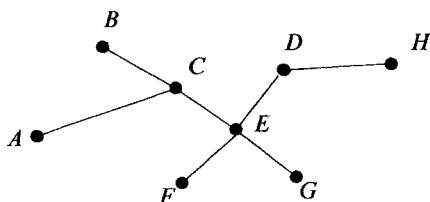
22. All five are isomorphic to each other.



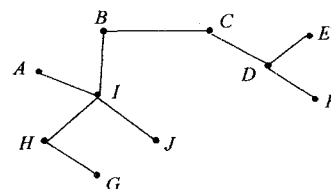
23. n
 24. If the cycle $A_1, A_2, \dots, A_n, A_1$ includes each edge and each vertex, except A_1 , exactly once, then there are n spanning trees.
 Proof: G has n edges. Each edge may be omitted to form a spanning tree. Thus the number of spanning trees is at least n . No two edges can be omitted at the same time. Hence the number of spanning trees is exactly n .
 25. There is only one. Any spanning tree is formed by omitting one edge from the graph. A clockwise shift of the labels gives an isomorphism between any spanning tree and the tree formed by omitting the edge (v_n, v_1) .
 26. Prim's algorithm will produce a spanning tree; its symmetric closure is an undirected spanning tree for the relation.

Exercise Set 7.5, page 285

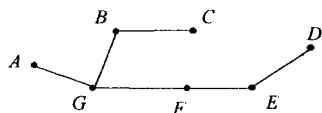
1.



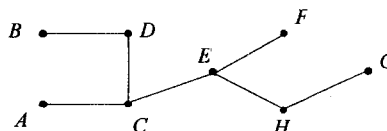
2.



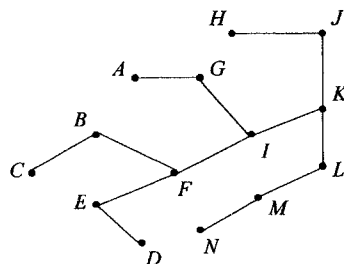
3.



4.

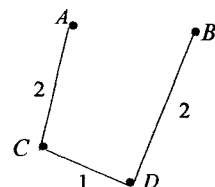


5.

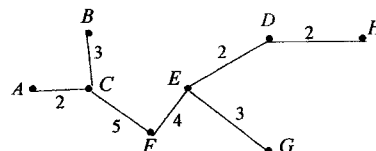


6. The minimal spanning tree is the same as that in Exercise 5.

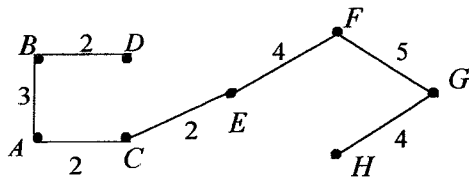
7.



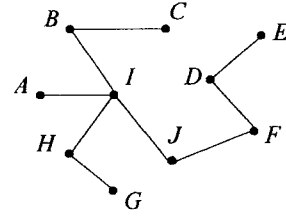
8.



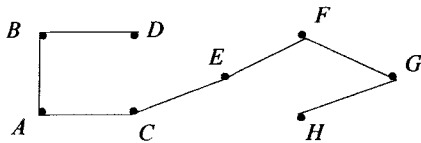
9.



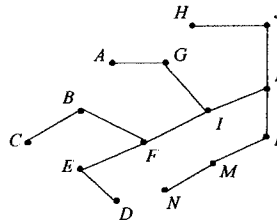
10.



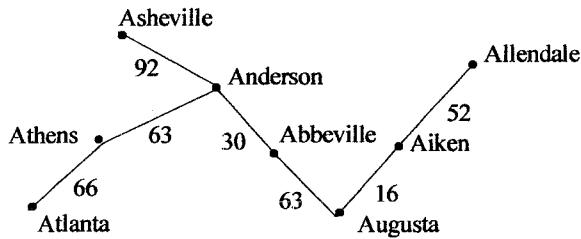
11.



12.

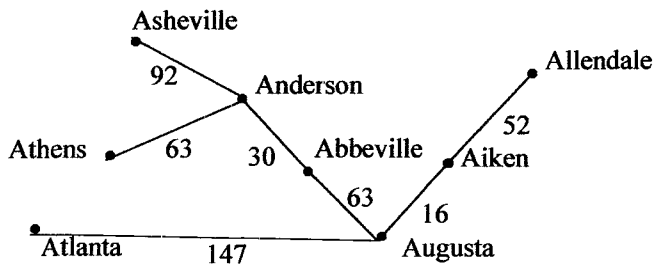


13.



14. Suppose e_r is the required edge. Let $S = \{e_1, e_2, \dots, e_k\} - \{e_r\}$ and $E = \{e_r\}$. Now apply Kruskal's algorithm to S and E .

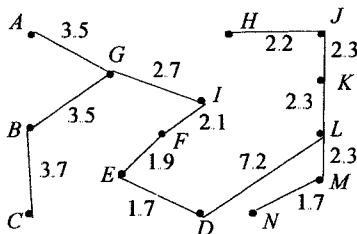
15.



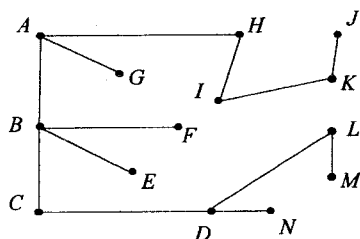
It would be 81 miles longer.

16. Suppose (v_r, v_s) is the edge to be included.
Step 1. Let $V = \{v_r\}$ and $E = \{(v_r, v_s)\}$. Steps 2 and 3 are unchanged.

17.



18. Change Step 1 to read "Choose an edge e_1 in S of greatest weight."
 19.



20. Define v to be a farthest vertex of $V = \{v_1, v_2, \dots, v_k\}$ if v is adjacent to some v_i in V and no other vertex is joined to a member of V by an edge of greater weight than (v, v_i) . In Step 2 replace "nearest" with "farthest."
 21. The maximal spanning tree is the same as that in Exercise 19.
 22. If each edge has a distinct weight, there will be a unique minimal spanning tree since only one choice can be made at each step.
 23. If each edge has a distinct weight, there will be a unique maximal spanning tree since only one choice can be made at each step.
 24. Let (v, w) be the edge of unique minimal weight. Suppose there is a minimal spanning tree that does not contain (v, w) . Adding (v, w) to this tree will produce a cycle. This cycle contains either (w, x) or (y, v) ; each of these edges would have weight greater than the weight of (v, w) . Deleting (w, x) or (y, v) from the cycle would produce a spanning tree with smaller total weight; this is a contradiction.
 25. One example is the sum 50. The greedy algorithm would select four 11-xebec coins and then six 1-xebec coins for a total of 10 coins. But the amount can be made from seven 7-xebec coins and one 1-xebec coin, a total of eight coins.
 26. (a) T is a tree with n vertices. (Theorem 3, Section 7.4)
 (b) $T' \cup \{t_{k+1}\}$ would have n edges and n distinct edges. (Theorem 2, Section 7.4)
 (c) it is connected and acyclic (Theorem 1, Section 7.4)
 (d) T is a spanning tree for R .

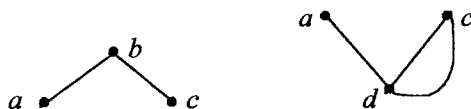
Review Questions, page 287

1. A tree is a relation with certain conditions.
 2. The only change in performing a preorder, inorder, or postorder search is when the root is visited.
 3. The other three sequences are (i) right, root, left, (ii) right, left, root, and (iii) root, right, left.
 4. A complete n -tree usually has smaller height than a general n -tree with the same number of vertices. This would shorten a search of the tree that begins at the root.
 5. An estimate of how many vertices and edges the graph has would help in the decision process.

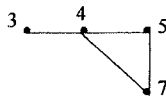
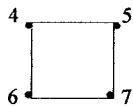
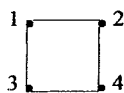
Chapter 8 Answers

Exercise Set 8.1, page 295

1. $V = \{a, b, c, d\}$ $E = \{\{a, b\}, \{b, c\}, \{b, d\}, \{c, c\}\}$
 2. $V = \{a, b, c, d, e\}$ $E = \{\{a, c\}, \{b, c\}, \{c, d\}, \{e, c\}\}$
 3. $V = \{a, b, c, d\}$ $E = \{\{a, b\}, \{b, c\}, \{d, a\}, \{d, c\}\}$ All edges are double edges.
 4. $V = \{1, 2, 3, 4, 5, 6, 7\}$ $E = \{\{1, 2\}, \{2, 4\}, \{1, 3\}, \{3, 4\}, \{4, 5\}, \{5, 7\}, \{4, 6\}, \{6, 7\}, \{4, 7\}\}$
 5. Possible answers are

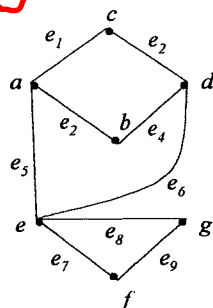
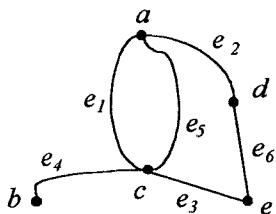


6. Possible answers



7.

8.



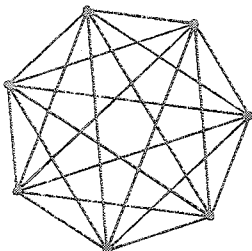
9. degree of a is 2; degree of b is 3; degree of c is 3; degree of d is 1.

10. Each vertex has degree 4.

11. a, c ; a, b, c ; a, c, d ; a, c, e

12. 5, 7, 4, 5; 5, 7, 6, 4, 5; 5, 4, 2, 1, 3, 4, 5 are possible answers.

13.



14. Each vertex has degree $n - 1$.

15. Only the graph given in Exercise 3 is regular.

16. Many such graphs are possible.

17. One possible solution is

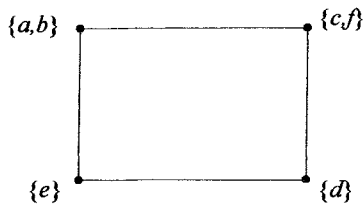


18.

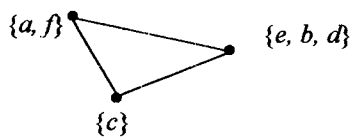
Answers will vary.

19.

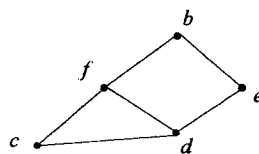
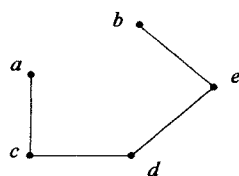
20.



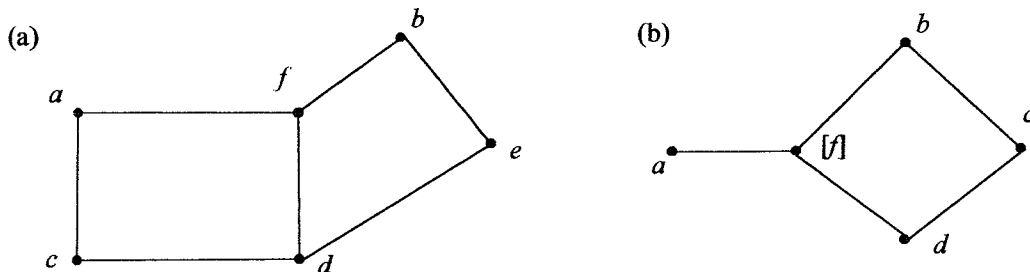
(b)



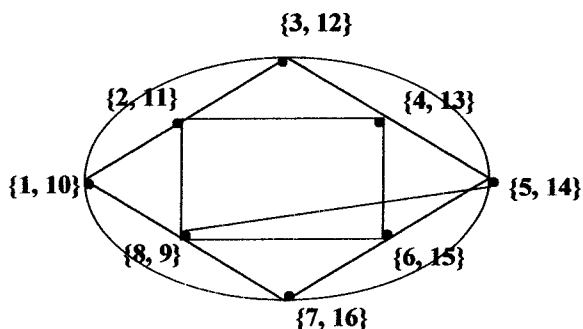
21. (a)



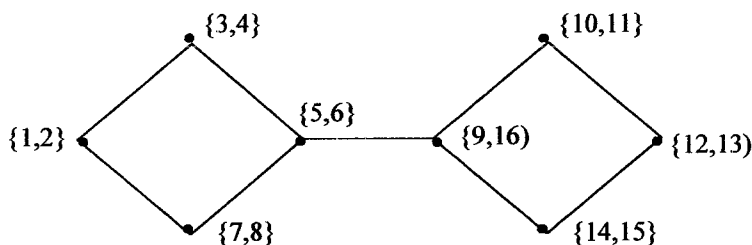
22.



23.



24.

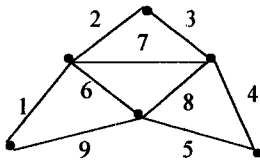


25. $n - 1$ The two "end points" have degree 1; the other $n - 2$ vertices each have degree 2. Hence the number of edges is $\frac{2(1)+2(n-2)}{2}$ or $n - 1$, since each edge is counted twice in the sum of the degrees.
26. Each vertex in K_n has degree $n - 1$. Thus the total number of edges is $\frac{n(n-1)}{2}$ since each edge is counted twice.
27. The graphs in Figures 8.24a and 8.24b are not isomorphic, because that in Figure 8.24a has two vertices of degree one and that in Figure 8.24b has only one vertex of degree one.
28. The graphs in Figures 8.24a and 8.24c are not isomorphic, because one has a vertex of degree four and the other does not.
29. Define a function f as follows: $f(a) = 1$, $f(b) = 3$, $f(c) = 5$, $f(d) = 2$, $f(e) = 4$. F is a one-to-one correspondence between the vertices of the two graphs with the property that (u, v) is an edge in Figure 8.24a if and only if $(f(u), f(v))$ is an edge in Figure 8.24d. Thus, the graphs are isomorphic.
30. In a digraph there are no multiple edges between vertices. In a graph, the edges are not directed.
31. If G has no loops or multiple edges, then each edge contributes 1 to the degree of each of its end points. Thus, the sum of the degrees of all vertices is twice the number of edges.
32. The sum of the degrees of all vertices with even degree is clearly even. Thus the sum of the degrees of all vertices of odd degree must also be even (using Exercise 31). But if there were an odd number of vertices of odd degree, the sum of their degrees would be odd, a contradiction.

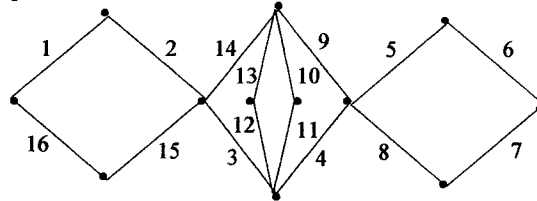
Exercise Set 8.2, page 302

1. neither
 2. neither
 3. Euler circuit
 4. Euler path, no Euler circuit
 5. Euler path only, since exactly two vertices have odd degree.
 6. Euler path only, since exactly two vertices have odd degree.
 7. neither The graph is disconnected.
 8. Euler circuit
 9. Yes, all vertices have even degree.
 10. Yes, all vertices have even degree.
- There are 4 vertices of odd degree.
There are 4 vertices of odd degree.
All vertices have even degree.
Exactly two vertices have odd degree.
All vertices have even degree.

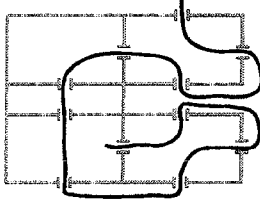
11. One possible answer



12. One possible answer



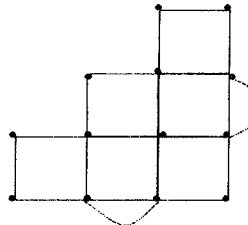
13. Yes. Note that if a circuit is required, it is not possible.



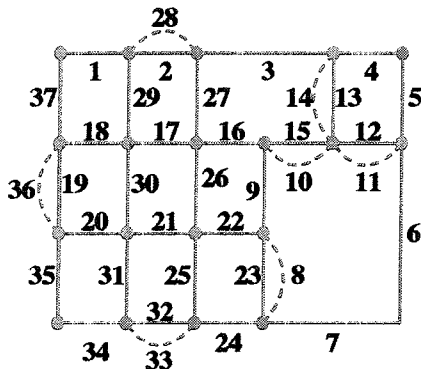
14. Yes. The corresponding graph has six vertices, each with even degree.

15. One more edge

16.

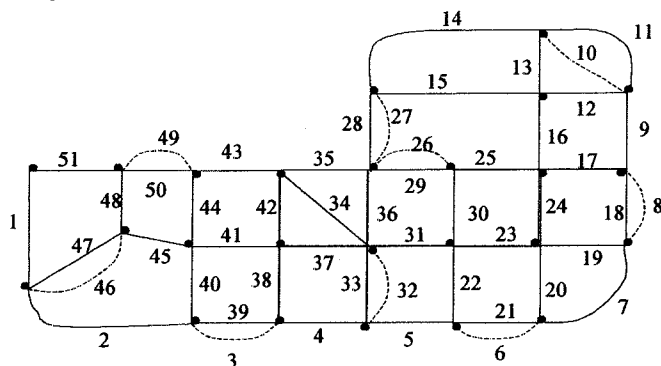


17. 7 edges



one possible solution

18. 9 edges

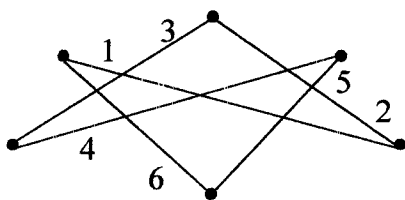


one possible solution

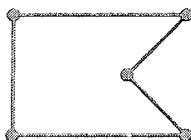
19. See the solution for Exercise 17. The consecutively numbered edges are one possible circuit.
20. See the solution for Exercise 18. The consecutively numbered edges are one possible circuit.
21. If n is odd, each vertex in K_n has degree $n - 1$, an even number. In this case, K_n has an Euler circuit. If n is even, then each vertex of K_n has odd degree; K_n does not have an Euler circuit.
22. If G and H are isomorphic using the function f , then an Euler circuit in G , $\pi : v_1, v_2, \dots, v_n$ yields an Euler circuit in H , $\pi' : f(v_1), f(v_2), \dots, f(v_n)$.
23. Suppose the strings $a_1a_2\dots a_n$ and $b_1b_2\dots b_n$ differ only in the i^{th} position. Then a_i or b_i is 1 (and the other is 0); say $a_i = 1$. Let A be the subset represented by $a_1a_2\dots a_n$ and B , the one represented by $b_1b_2\dots b_n$. Then B is a subset of A , and there is no proper subset of A that contains B . Hence there is an edge in B_n between these vertices.
Suppose there is an edge between the vertices labeled $a_1a_2\dots a_n$ and $b_1b_2\dots b_n$. Then $a_i \leq b_i, 1 \leq i \leq n$ (or vice versa). Hence there are at least as many 1's in $b_1b_2\dots b_n$ as in $a_1a_2\dots a_n$. If the strings differ in two or more positions, say $a_j = a_k = 0$ and $b_j = b_k = 1$, consider the label $c_1c_2\dots c_n$ with $c_i = b_i, i \neq j$, and $c_j = 0$. Then $c_1c_2\dots c_n$ represents a subset between those represented by $a_1a_2\dots a_n$ and $b_1b_2\dots b_n$. But this is not possible if there is an edge between the vertices labeled $a_1a_2\dots a_n$ and $b_1b_2\dots b_n$.
24. (a) Yes (b) No (c) Yes (d) No
25. If n is even, there is an Euler circuit. Each vertex is labeled with a string of even length. Hence it must have even degree as the value of each position could be changed in turn to create the label of a vertex connected to the original one. If n is odd, by the same reasoning every vertex has odd degree and there is no Euler circuit.

Exercise Set 8.3, page 306

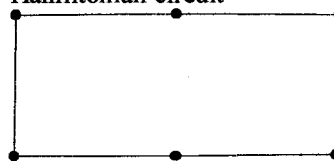
1. Neither
2. Hamiltonian circuit a, b, c, d, a
3. Hamiltonian path, but no Hamiltonian circuit
4. Hamiltonian circuit



5. Hamiltonian circuit



6. Hamiltonian circuit



7.



8. $A, B, D, F, G, H, E, C, A$

9. $A, B, D, H, G, E, F, C, A$

10. $A, B, C, E, D, F, J, G, H, I, A$

11. A, G, B, C, F, E, D, A

12. $C, A, B, D, F, G, H, E, C$

13. $A, B, D, H, G, E, F, C, A$

14. $I, H, G, J, F, D, E, C, B, A, I$

15. A, G, B, C, F, E, D, A

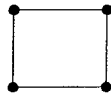
16. $D, B, A, C, E, H, G, F, D$

17. $F, E, G, H, D, B, A, C, F$

18. (a) There must be n edges. (b) $(n-1)!$

19. Choose any vertex, v_1 , in K_n , $n \geq 3$. Choose any one of the $n-1$ edges with v_1 as an end point. Follow this edge to v_2 . Here we have $n-2$ edges from which to choose. Continuing in this way we see there are $(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1$ Hamiltonian circuits we can choose.

20. A simple example is



21. One example is



22. G is connected if and only if each entry of $\mathbf{M}_{R^\infty}$ is a one.

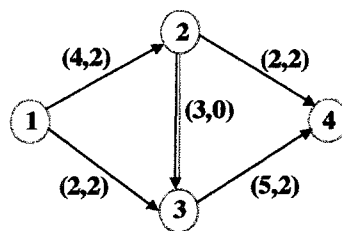
23. One solution is 000, 100, 110, 111, 101, 001, 011, 010, 000.

24. One solution is 0000, 0001, 0101, 0111, 0110, 0100, 1100, 1110, 1111, 1101, 1001, 1000, 1010, 1011, 0011, 0010, 0000.

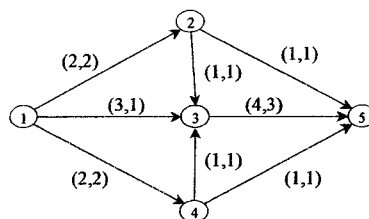
25. Construct a sequence of $2^n + 1$ strings of 0's and 1's of length n such that the first and the last terms are the same, and consecutive terms differ in exactly one position.

Exercise Set 8.4, page 314

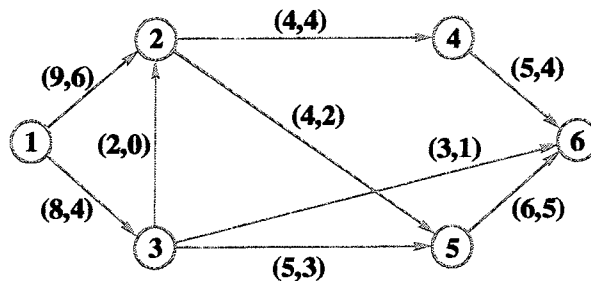
1. One possible solution



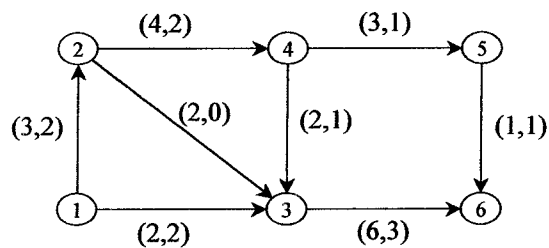
2. One possible solution



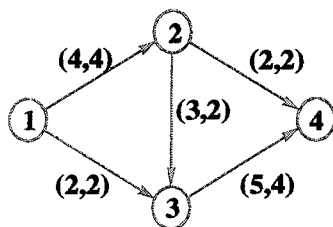
3. One possible solution



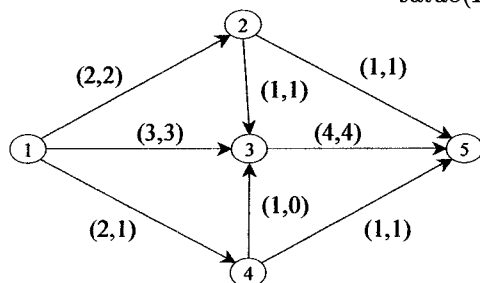
4. One possible solution



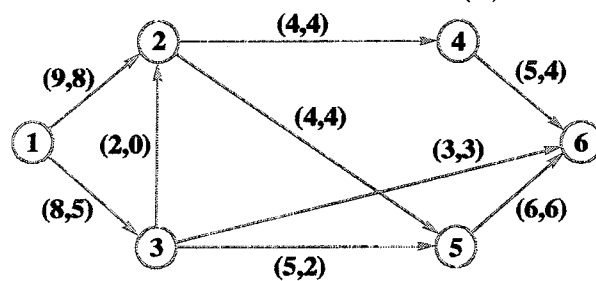
5. $value(F) = 6$



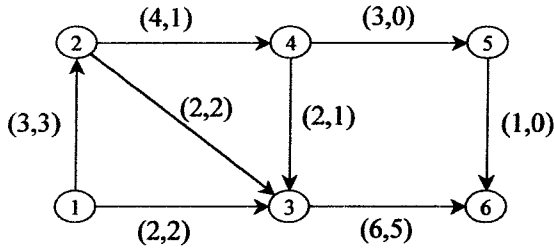
6. $value(F) = 6$



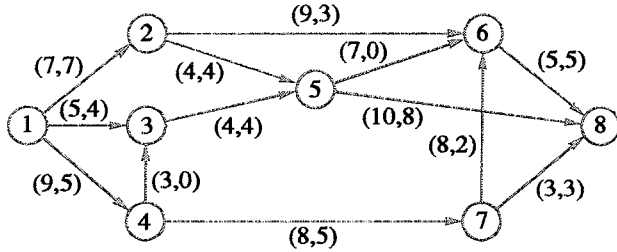
7. $value(F) = 13$



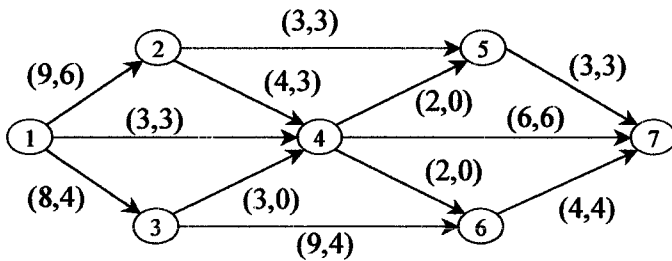
8. $value(F) = 5$



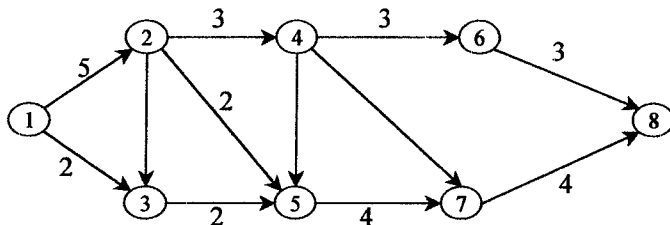
9. $value(F) = 16$



10. $value(F) = 13$



11. $value(F) = 7$

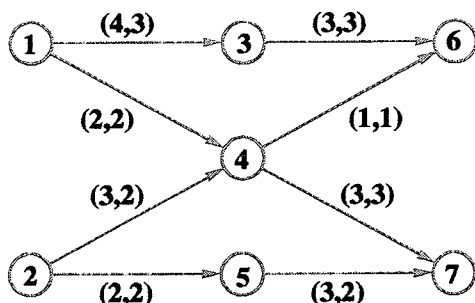


12. Possible cuts include $C_1 = \{(3, 5), (3, 6), (2, 5), (2, 4)\}$ with capacity 16 and $C_2 = \{(1, 3), (1, 2)\}$ with capacity 17.
13. Possible cuts include $C_1 = \{(2, 6), (5, 6), (5, 8), (4, 7)\}$ with capacity 34 and $C_2 = \{(6, 8), (7, 8)\}$ with capacity 8.
14. Possible cuts include $C_1 = \{(4, 6), (4, 7), (5, 7)\}$ with capacity 13, $C_2 = \{(2, 4), (2, 5), (3, 5)\}$ with capacity 10, and $C_3 = \{(6, 8), (7, 8)\}$ with capacity 8.
15. $\{(1, 2), (1, 3)\}$
16. $\{(2, 5), (3, 5), (4, 5)\}$
17. $\{(2, 4), (5, 6), (3, 6)\}$
18. $\{(1, 2), (1, 3)\}$
19. $\{(2, 5), (3, 5), (6, 8), (7, 8)\}$
20. $\{(5, 7), (4, 7), (6, 7)\}$
21. $\{(2, 4), (5, 7)\}$ with capacity 7.

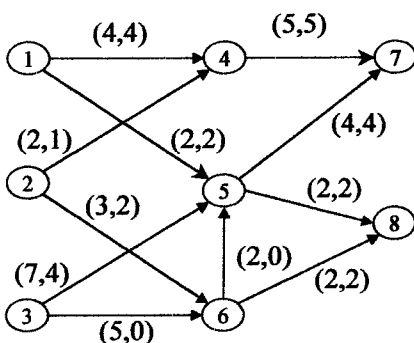
Exercise Set 8.5, page 319

1. Yes, $M(s_2) = b_1$, $M(s_3) = b_3$, $M(s_4) = b_4$, $M(s_5) = b_2$
2. < no solution given >

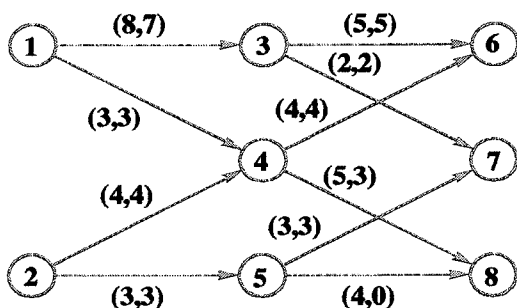
3. $value(F) = 9$



4. $value(F) = 13$



5. $value(F) = 17$

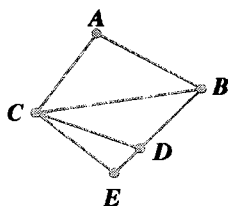


6. $\{(a, 1), (b, 3), (d, 4)\}$ is a maximal matching.
7. $\{(a, 1), (b, 2), (c, 4), (d, 3)\}$ is a maximal matching.
8. $\{(a, 1), (b, 2), (c, 3), (d, 4)\}$ is a maximal matching.
9. $\{(a, 5), (b, 2), (c, 3), (d, 1), (e, 4)\}$ is a maximal matching.
10. $\{(a, 3), (b, 2), (c, 1), (d, 4), (e, 5), (f, 6)\}$ is a maximal matching.
11. The matchings in Answers 7, 9, and 10 are complete.
12. Yes, one way gives Ann, the painting labeled E, Bing, B, Carmen, A, D'Andrea, D, Emilio, C, and Henri, F.
13. (a) One set of pairings is Sam - Jane, Kip - Gail, Homer - Rufus, and Kirk - Stacey.
(b) Yes, one such pairing is Sam - Stacey, Kip - Rufus, Homer - Jane, and Kirk - Gail.
14. Let S be any subset of A and E the set of edges that begin in S . Then $|E| = k|S|$. Each edge in E must terminate in a node of $R(S)$, so there must be at least $k|S|$ such nodes. But the number of edges terminating at nodes of $R(S)$ is $k|R(S)|$. Hence $k|S| \leq k|R(S)|$ and $|S| \leq |R(S)|$. By Hall's Marriage theorem, there is a complete matching for A , B , and R .
15. Let S be any subset of A and E the set of edges that begin in S . Then $k|S| \leq |E|$. Each edge in E must terminate in a node of $R(S)$. There are at most $j|R(S)|$ such nodes. Since $j \leq k$, $j|S| \leq k|S| \leq j|R(S)|$ and $|S| \leq |R(S)|$. By Hall's Marriage theorem, there is a complete matching for A , B , and R .

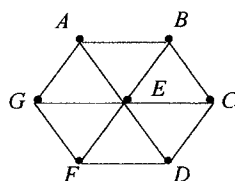
16. Consider the symmetric closure of the network representing A , B , and R . By the corollary to Theorem 1, Section 8.3, this graph has a Hamiltonian circuit. The edges of the circuit that belong to the original network give a complete matching.
17. (a) No, vertices 1, 2, and 6 must be in different subsets, but there are only two sets in the partition.
(b) Yes, $\{\{1, 3, 5, 7\}, \{2, 4, 6, 8\}\}$.
18. An edge in B_n connects vertices whose labels differ in exactly one position. Partition the vertices by their labels; one set contains the vertices whose labels have an odd number of 1's and the other vertices whose labels have an even number of 1's.
19. A triangle is formed by edges (u, v) , (v, w) , and (w, u) . If u and v are placed in different subsets by a two-set partition, neither subset can contain w .

Exercise Set 8.6, page 324

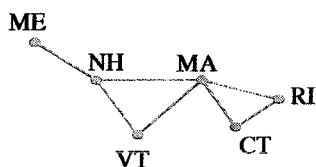
1.



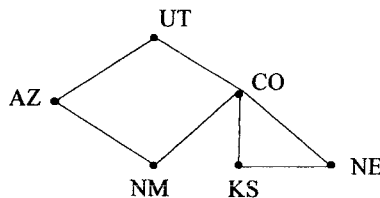
2.



3.



4.



5. 2

6. 3

7. 2

8. 2 if n is even, 3 if n is odd.

9. $P_G(x) = x(x-1)(x^2-3x+3)$

$$P_G(0) = P_G(1) = 0, P_G(2) = 2$$

10. $P_G(x) = x(x-1)(x-2)(x^2-2x+2)$

$$P_G(0) = P_G(1) = P_G(2) = 0, P_G(3) = 30$$

11. $P_G(x) = x(x-1)(x^4-5x^3+10x^2-10x+5)$

$$P_G(0) = P_G(1) = 0, P_G(2) = 2$$

12. $P_G(x) = x(x-1) \sum_{j=0}^{n-2} {}_{n-1}C_j (-1)^j x^{(n-2)-j}$ If n is even, $P_G(0) = P_G(1) = 0, P_G(2) = 1$;

if n is odd, $P_G(0) = P_G(1) = P_G(2) = 0, P_G(3) \neq 0$.

13. $P_G(x) = x(x-1)(x-2)^2$

$$\chi(G) = 3$$

14. $P_G(x) = x(x-1)(x-2)(x-3)$

$$\chi(G) = 4$$

15. $P_G(x) = x(x-1)(x-2)(x-3)$

$$\chi(G) = 4$$

16. $P_G(x) = x(x-1)(x-2)^2(x^2-2x+2)$

$$\chi(G) = 3$$

17. $P_G(x) = x(x-1)(x-2)^3$

$$\chi(G) = 3$$

18. $P_G(x) = x(x-1)^3(x-2)^2$

$$\chi(G) = 3$$

19. $P_G(x) = x(x-1)^2(x-2)(x^2-3x+3)$

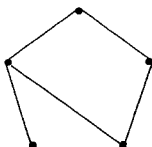
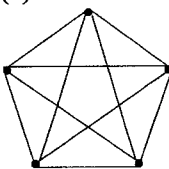
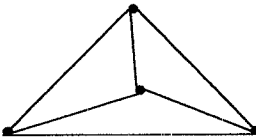
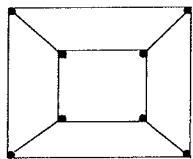
$$\chi(G) = 3$$

20. $P_G(x) = x(x-1)(x-2)(x^2-5x+7)$

$$\chi(G) = 3$$

21. $\chi(G) = 2$. Let A and B be the subsets that partition the vertices. Color vertices in A one color and those in B the other. It is easy to see that this will give a proper coloring of G .

22. Suppose that $\chi(G) \geq n+2$. Then there are $n+2$ distinct vertices, $x_1, x_2, \dots, x_{n+2}$, that are each adjacent to every other x_i . Now $x_1, x_2, \dots, x_{n+2}, x_1$ is a circuit of length $n+2$. This contradicts the fact that n is the length of the longest cycle of odd length.

23. (outline) Basis Step: $n = 1$ $P(1)$: $P_{L_1}(x) = x$ is true, because L_1 consists of a single vertex.
 Induction Step: We use $P(k)$ to show $P(k + 1)$. Let $G = L_{k+1}$ and e be an edge $\{u, v\}$ with $\deg(v) = 1$.
 Then G_e has two components, L_k and v . Using Theorem 1 and $P(k)$, we have $P_{G_e}(x) = x \cdot x(x - 1)^{k-1}$.
 Merging v with u gives $G^* = L_k$. Thus $P_{G^*}(x) = x(x - 1)^{k-1}$. By Theorem 2,
 $P_{L_{k+1}}(x) = x^2(x - 1)^{k-1} - x(x - 1)^{k-1} = x(x - 1)^{k-1}(x - 1)$ or $x(x - 1)^k$.
24. These are possible answers.
- (a) 
- (b) 
25. These are possible answers.
- (a) 
- (b) 
26. (a) courses with a student in common (b) time slots
 (c) the minimum number of time slots needed
27. Label the vertices with the fish species. An edge connects two vertices if one species eats the other.
 The colors represent the tanks, so $\chi(G)$ will be the minimum number of tanks needed.

Review Questions, page 326

1. A quotient graph is a quotient set of the vertices with a graph structure of inherited edges.
2. An Euler circuit must use all edges exactly once; a Hamiltonian circuit must use all vertices exactly once except for the starting vertex which is also the ending vertex.
3. The subset of edges can not contain edges whose endpoints do not belong to the subset of vertices.
4. Any matching problem can be converted to a maximum flow problem by adding to the graph a supersource and a supersink, and giving each edge capacity 1.
5. Since the chromatic polynomial counts the number of ways the graph can be colored with x colors, the smallest value of x for which $P(x) \neq 0$ is the minimum number of colors needed; that is, the chromatic number.

Chapter 9 Answers

Exercise Set 9.1, page 333

1. Yes
2. No
3. No
4. Yes
5. No
6. No
7. No
8. Yes
9. commutative, associative
10. commutative, associative
11. not commutative, associative
12. not commutative, not associative
13. commutative, associative
14. not commutative, associative
15. commutative, associative

16. not commutative, not associative
 17. commutative, associative
 18. commutative, associative
 19. commutative, not associative
 20. The operation has the idempotent property, because $\text{GCD}(a, a) = a$.
 21. The operation has the idempotent property, because $a * a = \frac{a+a}{2} = a$.

22.

*	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

23. One solution is

*	a	b	c
a	a	a	c
b	b	c	b
c	c	c	a

24. (a) No, $a * b \neq b * a$ (b) b; b (c) No, $c * (b * a) = c$, but $(c * b) * a = b$.
 25. (a) a, a (b) c, b (c) c, a (d) neither

26.

*	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	a	b
d	d	c	b	a

27.

*	a	b	c	d
a	a	b	c	d
b	b	a	c	d
c	c	d	c	d
d	d	c	c	d

28. n^{n^2} Consider creating the table for any binary operation.

29. $n^{\frac{n(n+1)}{2}}$ commutative operations

30. (a)

<u>a b</u>	<u>a b</u>	<u>a b</u>	<u>a b</u>	<u>a b</u>	<u>a b</u>
a a a	a b a	a a b	a a a	a a a	a b b
b a a	b a a	b a a	b b a	b a b	b a a
(1)	(2)	(3)	(4)	(5)	(6)
<u>a b</u>	<u>a b</u>	<u>a b</u>	<u>a b</u>	<u>a b</u>	<u>a b</u>
a b a	a b a	a a b	a a b	a a a	a b b
b b a	b a b	b b a	b a b	b b b	b b a
(7)	(8)	(9)	(10)	(11)	(12)
<u>a b</u>	<u>a b</u>	<u>a b</u>	<u>a b</u>		
a b b	a b a	a a b	a b b		
b a b	b b b	b b b	b b b		
(13)	(14)	(15)	(16)		

- (b) Commutative : (1), (2), (5), (8), (9), (12), (15), (16)
 31. (a) Associative: (1), (5), (8), (9), (10), (11), (15), (16)
 (b) Idempotent: (5), (10), (11), (15)

32. Since $*$ is idempotent, $a * a = a$ so $a \leq a$. Hence $\leq$ is reflexive. Suppose $a \leq b$ and $b \leq a$. Then $b = a * b = b * a = a$. Hence $\leq$ is antisymmetric. Suppose $a \leq b$, $b \leq c$. Then $b = a * b$, $c = b * c$. $a * c = a * (b * c) = (a * b) * c = b * c = c$, so $a \leq c$. Hence $\leq$ is transitive and $(A, \leq)$ is a poset. Consider $a * (a * b)$. $a * (a * b) = (a * a) * b = a * b$. Thus $a \leq a * b$. Similarly, since $b * (a * b) = b * (b * a) = (b * b) * a = b * a = a * b$, $b \leq a * b$. Hence $a * b$ is an upper bound for $\{a, b\}$. Suppose c is an upper bound for $\{a, b\}$. $(a * b) * c = a * (b * c) = a * c = c$. Thus $a * b \leq c$ and $a * b = \text{LUB}(a, b)$.
33. A binary operation on a set S must be defined for every a, b in S . According to the earlier definition $a * b$ may be undefined for some a, b in S . Any binary operation on a set S is a binary operation in the sense of Section 1.6.
34. $a * (b * c) = a * c = c$ and $(a * b) * c = b * c = c$ so $*$ is associative. Because $a * b = b$ and $b * a = a$, $*$ is not commutative. $a * a = a$, hence $*$ is idempotent.

Exercise Set 9.2, page 340

1. semigroup: (b) monoid: (b)
2. semigroup: (a), (b) monoid: (a), (b)
3. semigroup: (a) monoid: neither
4. monoid; identity is 1; commutative
5. monoid; identity is 1; commutative
6. semigroup; commutative
7. semigroup
8. monoid; identity is 1; commutative
9. monoid; identity is S ; commutative
10. monoid; identity is I ; commutative
11. monoid; identity is 12; commutative
12. monoid; identity is 1; commutative
13. monoid; identity is 0; commutative
14. monoid; identity is 2; commutative
15. monoid; identity $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$; commutative
16. semigroup; commutative
17. neither
18. neither
19.

$*$	a	b	c
a	c	a	b
b	a	b	c
c	b	c	a
20. One solution is

$*$	a	b	c	d
a	c	d	a	b
b	d	a	b	c
c	a	b	c	d
d	b	c	d	a
21. Let $f_1(a) = a$, $f_1(b) = a$; $f_2(a) = a$, $f_2(b) = b$; $f_3(a) = b$, $f_3(b) = a$; $f_4(a) = b$, $f_4(b) = b$. These are the only functions on S . It is not commutative.

$\circ$	f_1	f_2	f_3	f_4
f_1	f_1	f_1	f_4	f_4
f_2	f_1	f_2	f_3	f_4
f_3	f_1	f_3	f_2	f_1
f_4	f_1	f_4	f_4	f_4
22.

$\cup$	$\{\}$	$\{a\}$	$\{b\}$	$\{a, b\}$
$\{\}$	$\{\}$	$\{a\}$	$\{b\}$	$\{a, b\}$
$\{a\}$	$\{a\}$	$\{a\}$	$\{a, b\}$	$\{a, b\}$
$\{b\}$	$\{b\}$	$\{a, b\}$	$\{b\}$	$\{a, b\}$
$\{a, b\}$	$\{a, b\}$	$\{a, b\}$	$\{a, b\}$	$\{a, b\}$

23. (a) $abaccbababc$ (b) $babcabacabac$ (c) $babccbaabac$
24. All products of elements in the subset must belong to the subset.
25. The subset must form a subsemigroup and the identity element must belong to the subset.
26. Let S_1 and S_2 be subsemigroups of $(S, *)$. Suppose $x, y \in S_1 \cap S_2$, then $x * y \in S_1$ and $x * y \in S_2$. Hence $S_1 \cap S_2$ is a subsemigroup of S .
27. By Exercise 26, we need only check that $e \in S_1 \cap S_2$. But $e \in S_1$ and $e \in S_2$, because each is a submonoid of $(S, *)$.
28. No, the catenation of two sequences with an odd number of 1's will have an even number of 1's.
29. Yes. Refer to Exercise 1.
30. Let $T = \{x \mid x \in S \text{ and } x * x = x\}$. Clearly, $e \in T$. Suppose $x, y \in T$. Then $(x * y) * (x * y) = x * y * x * y = x * x * y * y = x * y$. Thus $x * y \in T$.
31. Let $x, y \in S_1$. $(g \circ f)(x * y) = g(f(x * y)) = g(f(x) * f(y)) = g(f(x)) * g(f(y)) = (g \circ f)(x) * (g \circ f)(y)$. Thus $g \circ f$ is a homomorphism from $(S_1, *)$ to $(S_3, *)$.
32. By Exercise 31, we have that $g \circ f$ is a homomorphism. Suppose $(g \circ f)(a) = (g \circ f)(b)$. Then $f(a) = f(b)$ since g is one to one. But then $a = b$ because f is also one to one. Let $s \in S_3$. Then $\exists y \in S_2$ such that $g(y) = s$, because g is onto. There exists $z \in S_1$ such that $f(z) = y$, since f is onto. Hence $(g \circ f)(z) = s$ and $g \circ f$ is onto. Hence $g \circ f$ is an isomorphism from $(S_1, *)$ to $(S_3, *)$.
33. onto; homomorphism
34. The only properties of f used in the proof of Theorem 2 are those assumed in Theorem 3.
35. Let $x, y \in \mathbb{R}^+$. $\ln(x * y) = \ln(x) + \ln(y)$ so $\ln$ is a homomorphism. Suppose $x \in \mathbb{R}$. Then $e^x \in \mathbb{R}^+$ and $\ln(e^x) = x$ so $\ln$ is onto $\mathbb{R}^+$. Suppose $\ln(x) = \ln(y)$, then $e^{\ln(x)} = e^{\ln(y)}$ and $x = y$. Hence $\ln$ is one to one and an isomorphism of $(\mathbb{R}^+, +)$ and $(\mathbb{R}, +)$.
36. (a) Clearly $\hat{A}$ is closed with respect to $*$, so $(\hat{A}, *)$ is a subsemigroup of $(S, *)$.
 (b) Let $(B, *)$ be a subsemigroup of $(S, *)$ such that $A \subseteq B \subseteq \hat{A}$. But then B must contain all finite products of elements in A so $\hat{A} \subseteq B$. Thus, $\hat{A} = B$ is the smallest subsemigroup containing the elements of A .

Exercise Set 9.3, page 346

- Let $(s_1, t_1), (s_2, t_2) \in S \times T$. $(s_1, t_1) *'' (s_2, t_2) = (s_1 * s_2, t_1 *' t_2)$ so $*''$ is a binary operation. Consider $(s_1, t_1) *'' ((s_2, t_2) *'' (s_3, t_3)) = (s_1, t_1) *'' (s_2 * s_3, t_2 *' t_3) = (s_1 * (s_2 * s_3), t_1 *' (t_2 *' t_3)) = ((s_1 * s_2) * s_3, (t_1 *' t_2) *' t_3) = ((s_1, t_1) *'' (s_2, t_2)) *'' (s_3, t_3)$. Thus $(S \times T, *'')$ is a semigroup. $(s_1, t_1) *'' (s_2, t_2) = (s_1 * s_2, t_1 *' t_2) = (s_2 * s_1, t_2 *' t_1) = (s_2, t_2) *'' (s_1, t_1)$. Hence $*''$ is commutative.
- By Exercise 1, we have $(S \times T, *'')$ is a semigroup. Let $(s, t) \in S \times T$. Then $(s, t) *'' (e_S, e_T) = (s * e_S, t *' e_T) = (s, t)$. Similarly, $(e_S, e_T) *'' (s, t) = (s, t)$. Hence, (e_S, e_T) is the identity for the monoid $(S \times T, *'')$.
- Let $(s_1, t_1), (s_2, t_2) \in S \times T$. Then $f((s_1, t_1) *'' (s_2, t_2)) = f(s_1 * s_2, t_1 *' t_2) = s_1 * s_2 = f(s_1, t_1) * f(s_2, t_2)$. f is a homomorphism.
- Define $f: S \times T \rightarrow T \times S$ by $f((s, t)) = (t, s)$. $f((s_1, t_1) *'' (s_2, t_2)) = f((s_1 * s_2, t_1 *' t_2)) = (t_1 *' t_2, s_1 * s_2) = f((s_1, t_1)) *'' f((s_2, t_2))$. Hence f is a homomorphism. If $(t, s) \in T \times S$, clearly $f((s, t)) = (t, s)$ and f is onto. Suppose $f((s_1, t_1)) = f((s_2, t_2))$. Then $(t_1, s_1) = (t_2, s_2)$. But then $t_1 = t_2$ and $s_1 = s_2$ and $(s_1, t_1) = (s_2, t_2)$. f is one to one and thus is an isomorphism from $(S \times T, *'')$ to $(T \times S, *'')$.
- $*''$ is a binary operation, because both $*$ and $*'$ are. Consider $(s_1, t_1) *'' ((s_2, t_2) *'' (s_3, t_3)) = (s_1, t_1) *'' (s_2 * s_3, t_2 *' t_3) = (s_1 * (s_2 * s_3), t_1 *' (t_2 *' t_3)) = ((s_1 * s_2) * s_3, (t_1 *' t_2) *' t_3) = ((s_1, t_1) *'' (s_2, t_2)) *'' (s_3, t_3)$. Thus, $*''$ is associative.
- No
- Yes
- Yes
- Yes
- Yes
- Yes
- No
- Yes

16. R is a known equivalence relation. Suppose $(3k+1)R(3j+1)$ and $(3m+1)R(3n+1)$, then $5 \mid (3k-3j)$ and $5 \mid (3m-3n)$. Consider $(3k+1)(3m+1) - (3j+1)(3n+1) = 9km + 3k + 3m - 9jn - 3j - 3n = 3m(3k-3j) + 3j(3m-3n) + (3k-3j) + (3m-3n)$. Since 5 divides each term, 5 divides the sum and $(3k+1)(3m+1)R(3j+1)(3n+1)$. R is a congruence relation.
17. $S/R = \{[4], [7], [10], [13], [16]\}$
18. Let R_1 and R_2 be congruence relations on S . By Exercise 24, Section 4.5, $R_1 \cap R_2$ is an equivalence relation on S . Suppose $s_1(R_1 \cap R_2)t_1, s_2(R_1 \cap R_2)t_2$. Then $s_1R_1t_1, s_2R_1t_2, s_1R_2t_1, s_2R_2t_2$, and $(s_1 * s_2)R_1(t_1 * t_2), (s_1 * s_2)R_2(t_1 * t_2)$. Hence $(s_1 * s_2)(R_1 \cap R_2)(t_1 * t_2)$. Thus $R_1 \cap R_2$ is a congruence relation on S .
19. By Exercise 23, Section 4.7, we have that the composition of two equivalence relations need not be an equivalence relation.
20. $(S/R, \odot)$ is the collection of rational numbers in lowest terms with $\frac{a}{b} \odot \frac{c}{d} = \frac{ab}{cd}$ in lowest terms.
21. $S/R = \{[0], [1], [2]\}$. $[0] = \{0, \pm 3, \pm 6, \dots\}$, $[1] = \{\pm 1, \pm 4, \pm 7, \dots\}$, $[2] = \{\pm 2, \pm 5, \pm 8, \dots\}$
- | | | | |
|----------|-------|-------|-------|
| $\oplus$ | $[0]$ | $[1]$ | $[2]$ |
| $[0]$ | $[0]$ | $[1]$ | $[2]$ |
| $[1]$ | $[1]$ | $[2]$ | $[0]$ |
| $[2]$ | $[2]$ | $[0]$ | $[1]$ |
22. $S/R = \{[0], [1]\}$. $[0] = \{0, \pm 2, \pm 4, \dots\}$, $[1] = \{\pm 1, \pm 3, \pm 5, \dots\}$ $\oplus$ $[0]$ $[1]$
- | | | |
|-------|-------|-------|
| $[0]$ | $[0]$ | $[1]$ |
| $[1]$ | $[1]$ | $[0]$ |
23. $S/R = \{[0], [1], [2], [3], [4]\}$, $[a] = \{z \mid z = 5k + a, k \in \mathbb{Z}\}$, $a = 0, 1, 2, 3, 4$
- | | | | | | |
|----------|-------|-------|-------|-------|-------|
| $\oplus$ | $[0]$ | $[1]$ | $[2]$ | $[3]$ | $[4]$ |
| $[0]$ | $[0]$ | $[1]$ | $[2]$ | $[3]$ | $[4]$ |
| $[1]$ | $[1]$ | $[2]$ | $[3]$ | $[4]$ | $[0]$ |
| $[2]$ | $[2]$ | $[3]$ | $[4]$ | $[0]$ | $[1]$ |
| $[3]$ | $[3]$ | $[4]$ | $[0]$ | $[1]$ | $[2]$ |
| $[4]$ | $[4]$ | $[0]$ | $[1]$ | $[2]$ | $[3]$ |
24. (a) $\otimes$ $[a]$ $[c]$
- | | | |
|-------|-------|-------|
| $[a]$ | $[a]$ | $[c]$ |
| $[c]$ | $[c]$ | $[a]$ |
- (b) $f_R(a) = [a]$, $f_R(b) = [a]$, $f_R(c) = [c]$, $f_R(d) = [c]$
25. (a) $\otimes$ $[a]$ $[b]$
- | | | |
|-------|-------|-------|
| $[a]$ | $[a]$ | $[b]$ |
| $[b]$ | $[b]$ | $[b]$ |
- (b) $f_R(e) = [a] = f_R(a)$, $f_R(b) = [b] = f_R(c)$
26. (a) Let $\alpha, \beta \in A^*$. $f(\alpha, \beta)$ is the number of digits in $\alpha \cdot \beta$, but this is the number of digits in α plus the number of digits in β or $f(\alpha) + f(\beta)$.
- (b) $\alpha R \beta$ if and only if $\text{length}(\alpha) = \text{length}(\beta)$. R is clearly an equivalence relation.
- (c) Each equivalence class $[\alpha]$ consists of strings with the same length. Define $g: A^*/R \rightarrow \mathbb{N}$ by $g([\alpha]) = \text{length}(\alpha)$. Clearly, g is onto. If $g([\alpha]) = g([\beta])$, then $\text{length}(\alpha) = \text{length}(\beta)$ and $[\alpha] = [\beta]$. Thus g is one to one. We have A^*/R is isomorphic to $\mathbb{N}$.
27. An examination of the two multiplication tables shows that they are identical and so $\mathbb{Z}_2$ is isomorphic to S/R .
28. The semigroups are not isomorphic. In the table for the semigroup of Exercise 24, the upper left 2×2 square and the lower right 2×2 square are identical and contain only two distinct elements. No relabeling of the elements of $\mathbb{Z}_4$ gives a table that matches this pattern.
29. This is a direct proof. For part (a), we check the three properties for an equivalence relation and then the property for a congruence relation. In part (b), we first check $\bar{f}$ is a function and then the properties of an isomorphism are confirmed.
30. Let R be an equivalence relation on S . If aRb and cRd , then since $a * c = c$ and $b * d = d$, we have $(a * c)R(b * d)$.

Exercise Set 9.4, page 357

1. No
2. No
3. Yes; Abelian; identity is 0; a^{-1} is $-a$.
4. No
5. No
6. Yes; Abelian; identity is -2 ; a^{-1} is $-4 - a$.
7. No
8. Yes; Abelian; identity is 0; a^{-1} is $\frac{-a}{1+a}$.
9. No
10. Yes; Abelian; identity is $m \times n$ zero matrix; a^{-1} is $-a$.
11. Yes; Abelian; identity is $\{ \}$; a^{-1} is a .
12.

$\circ$	f_1	f_2	f_3	f_4	f_5	f_6
f_1	f_1	f_2	f_3	f_4	f_5	f_6
f_2	f_2	f_1	f_5	f_6	f_3	f_4
f_3	f_3	f_4	f_1	f_2	f_6	f_5
f_4	f_4	f_3	f_6	f_5	f_1	f_2
f_5	f_5	f_6	f_2	f_1	f_4	f_3
f_6	f_6	f_5	f_4	f_3	f_2	f_1

Inspection of the table shows that f_1 is the identity and each element has an inverse.
Function composition is associative.
Thus G is a group.
13. Since g_1, g_2, g_3 in S_3 each have order 2, they must be paired somehow with f_2, f_3, f_4 of Example 12 if the groups are isomorphic. But no rearrangement of the columns and rows labeled f_2, f_3, f_4 in Example 12 will give the "block" pattern shown by g_1, g_2, g_3 in the table for S_3 . Hence the groups are not isomorphic.
14.
$$f(p \circ q) = \begin{cases} 0 & p \circ q \text{ even permutation} \\ 1 & p \circ q \text{ odd permutation} \end{cases}$$

If p and q are both even (or odd), then $f(p) + f(q) = 0$ and $f(p \circ q) = f(p) + f(q)$. If p and q have opposite parity, then $f(p) + f(q) = 1$ and $f(p \circ q) = f(p) + f(q)$. Hence f is a homomorphism.
15. (a) $\frac{8}{3}$ (b) $\frac{-4}{5}$
16. It is easy to check that $(S, *)$ is closed. It inherits associativity and commutativity from the complex numbers and multiplication. The identity is 1, and each element has an inverse.
17. $H_1 = \{1\}$, $H_2 = \{1, -1\}$, $H_3 = \{1, -1, i, -i\}$
18. Let $a, b \in G$. $x^2 = e \Rightarrow x = x^{-1} \forall x \in G$. Thus $abab = e$ and $a^{-1}bab^{-1} = e$. Hence $bab^{-1} = ae$ and $ba = ab$.
19.

$\circ$	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
f_1	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
f_2	f_2	f_3	f_4	f_1	f_8	f_7	f_5	f_6
f_3	f_3	f_4	f_1	f_2	f_6	f_5	f_8	f_7
f_4	f_4	f_1	f_2	f_3	f_7	f_8	f_6	f_5
f_5	f_5	f_7	f_6	f_8	f_1	f_3	f_2	f_4
f_6	f_6	f_8	f_5	f_7	f_3	f_1	f_4	f_2
f_7	f_7	f_6	f_8	f_5	f_4	f_2	f_1	f_3
f_8	f_8	f_5	f_7	f_6	f_2	f_4	f_3	f_1
20. $a^{-1}(aa) = a^{-1}a$ implies $a = e$
21. Consider the sequence $e, a, a^2, a^3, \dots$. Since G is finite, not all terms of this sequence can be distinct; that is, for some $i \leq j$ $a^i = a^j$. Then $(a^{-1})^i a^i = (a^{-1})^j a^j$ and $e = a^{j-i}$. Note $j - i \geq 0$.
22. No
23. Yes
24. For $x, y \in H$, $(xy)^2 = xyxy = x^2y^2 = ee = e$. Hence $xy \in H$. Clearly $e \in H$, so H is a subgroup of G .
25. Clearly, $e \in H$. Let $a, b \in H$. Consider $(ab)y = a(by) = a(yb) = (ay)b = (ya)b = y(ab) \forall y \in G$. Hence H is closed under multiplication and is a subgroup of G .
26. Clearly, $e \in H_a$. Let $x, y \in H_a$. Consider $(xy)a = x(ya) = x(ay) = (xa)y = (ax)y = a(xy)$. Hence $xy \in H_a$ and H_a is a subgroup of G .

27. The identity permutation is an even permutation. If they are even permutations, then each of p_1 and p_2 can be written as the product of an even number of transpositions. Then $p_1 \circ p_2$ can be written as the product of these representations of p_1 and p_2 . But this gives $p_1 \circ p_2$ as the product of an even number of transpositions. Thus $p_1 \circ p_2 \in A_n$ and A_n is a subgroup of S_n .
28. (a) $e \in H$ and $e \in K$ implies $e \in H \cap K$. If $x, y \in H \cap K$, then $x, y \in H$ and $x, y \in K$. Hence $xy \in H$ and $xy \in K$, so $xy \in H \cap K$. $H \cap K$ is a subgroup of G .
 (b) Let $G = S_3$, $H = \{f_1, g_1\}$, $K = \{f_1, g_2\}$. Then $H \cup K = \{f_1, g_1, g_2\}$, but $g_1 \circ g_2 = f_2$.
 29. $\{f_1\}$, $\{f_1, f_2, f_3, f_4\}$, $\{f_1, f_3, f_5, f_6\}$, $\{f_1, f_3, f_7, f_8\}$, $\{f_1, f_5\}$, $\{f_1, f_6\}$, $\{f_1, f_3\}$, $\{f_1, f_7\}$, $\{f_1, f_8\}$, D_4
30. By Exercise 37, $(ab)^n = a^n b^n$. Thus $f(ab) = f(a)f(b)$.
31. $|xy| = |x| \cdot |y|$. Thus $f(xy) = f(x)f(y)$
32. $f(ab) = e = ee = f(a)f(b)$
33. Suppose $f: G \rightarrow G: a \rightarrow a^2$ is a homomorphism. Then $f(ab) = f(a)f(b)$ or $(ab)^2 = a^2 b^2$. Hence $a^{-1}(abab)b^{-1} = a^{-1}(a^2 b^2)b^{-1}$ and $ba = ab$.
 Suppose G is abelian. By Exercise 37, $f(ab) = f(a)f(b)$.
34. Suppose $f: G \rightarrow G: a \rightarrow a^{-1}$ is an isomorphism. Let $a, b \in G$. $f(a^{-1}b^{-1}) = (a^{-1}b^{-1})^{-1} = ba$. But $f(a^{-1}b^{-1}) = f(a^{-1})f(b^{-1}) = ab$. Hence $ab = ba$.
 Suppose G is abelian. Then $f(ab) = (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1} = f(a)f(b)$. Thus f is a homomorphism. Suppose $a \in G$. Then $f(a^{-1}) = a$ so f is onto. Suppose $f(a) = f(b)$. Then $a^{-1} = b^{-1}$. But then $(a^{-1})^{-1} = (b^{-1})^{-1}$ or $a = b$ so f is one to one. Hence f is an isomorphism.
35. Let $x, y \in G$. $f_a(xy) = axya^{-1} = axa^{-1}aya^{-1} = f_a(x)f_a(y)$. f_a is a homomorphism. Suppose $x \in G$. Then $f_a(a^{-1}xa) = aa^{-1}xaa^{-1} = x$ so f_a is onto. Suppose $f_a(x) = f_a(y)$, then $axa^{-1} = aya^{-1}$. Now $a^{-1}(axa^{-1})a = a^{-1}(aya^{-1})a$ and $x = y$. Thus f_a is one to one and an isomorphism.
36. Define $f: G \rightarrow Z_6: a^i \rightarrow i$. We consider e to be a^0 . $f(a^i a^j) = f(a^i) = r = (i + j) \pmod{6}$
 $= i \pmod{6} + j \pmod{6} = i + j = f(a^i) + f(a^j)$. Suppose that $x \in Z_6$, then $f(a^x) = x$. Thus f is onto. Suppose $f(a^i) = f(a^j)$, then $i = j$ and $a^i = a^j$. f is one to one.
37. (outline) Basis Step: $n = 1$ $P(1): (ab)^1 = a^1 b^1$ is true.
 Induction Step: LHS of $P(k+1): (ab)^{k+1} = (ab)^k ab = a^k b^k ab = a^k ab^k b = a^{k+1} b^{k+1}$ the RHS of $P(k+1)$
38. Theorem 2(a) guarantees that $a(a^{-1}b) = b$, so every element b appears in the row labeled a .
 Theorem 2(b) guarantees that $(ba^{-1})a = b$, so every element b appears in the column labeled a .
39. One possible table is
- | | a | b | c |
|-----|-----|-----|-----|
| a | b | a | c |
| b | c | b | a |
| c | a | c | b |
- * has no identity element.

Exercise Set 9.5, page 362

- 1.
- | | $(\bar{0}, \bar{0})$ | $(\bar{0}, \bar{1})$ | $(\bar{0}, \bar{2})$ | $(\bar{1}, \bar{0})$ | $(\bar{1}, \bar{1})$ | $(\bar{1}, \bar{2})$ |
|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| $(\bar{0}, \bar{0})$ | $(\bar{0}, \bar{0})$ | $(\bar{0}, \bar{1})$ | $(\bar{0}, \bar{2})$ | $(\bar{1}, \bar{0})$ | $(\bar{1}, \bar{1})$ | $(\bar{1}, \bar{2})$ |
| $(\bar{0}, \bar{1})$ | $(\bar{0}, \bar{1})$ | $(\bar{0}, \bar{2})$ | $(\bar{0}, \bar{0})$ | $(\bar{1}, \bar{1})$ | $(\bar{1}, \bar{2})$ | $(\bar{1}, \bar{0})$ |
| $(\bar{0}, \bar{2})$ | $(\bar{0}, \bar{2})$ | $(\bar{0}, \bar{0})$ | $(\bar{0}, \bar{1})$ | $(\bar{1}, \bar{2})$ | $(\bar{1}, \bar{0})$ | $(\bar{1}, \bar{1})$ |
| $(\bar{1}, \bar{0})$ | $(\bar{1}, \bar{0})$ | $(\bar{1}, \bar{1})$ | $(\bar{1}, \bar{2})$ | $(\bar{0}, \bar{0})$ | $(\bar{0}, \bar{1})$ | $(\bar{0}, \bar{2})$ |
| $(\bar{1}, \bar{1})$ | $(\bar{1}, \bar{1})$ | $(\bar{1}, \bar{2})$ | $(\bar{1}, \bar{0})$ | $(\bar{0}, \bar{1})$ | $(\bar{0}, \bar{2})$ | $(\bar{0}, \bar{0})$ |
| $(\bar{1}, \bar{2})$ | $(\bar{1}, \bar{2})$ | $(\bar{1}, \bar{0})$ | $(\bar{1}, \bar{1})$ | $(\bar{0}, \bar{2})$ | $(\bar{0}, \bar{0})$ | $(\bar{0}, \bar{1})$ |
2. Let $(x_1, y_1), (x_2, y_2) \in G \times G'$. Then $(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2, y_1 y_2) = (x_2 x_1, y_2 y_1)$
 $= (x_2, y_2) \cdot (x_1, y_1)$. Hence $G \times G'$ is also abelian.
3. Define $f: G_1 \rightarrow G_2: (g_1, g_2) \rightarrow (g_2, g_1)$. By Exercise 4, Section 9.3, f is an isomorphism.
4. By Exercise 3, Section 9.3, f is a homomorphism.
- 5.
- | | $[0]$ | $[1]$ | $[2]$ |
|-------|-------|-------|-------|
| $[0]$ | $[0]$ | $[1]$ | $[2]$ |
| $[1]$ | $[1]$ | $[2]$ | $[0]$ |
| $[2]$ | $[2]$ | $[0]$ | $[1]$ |
6. Let $(a_1, b_1), (a_2, b_2) \in Z \times Z$. $f((a_1, b_1) * (a_2, b_2))$
 $= f((a_1 + a_2, b_1 + b_2)) = a_1 + a_2 + b_1 + b_2 = a_1 + b_1 + a_2 + b_2 = f((a_1, b_1)) * f((a_2, b_2)).$

7. $\text{Ker}(f) = \{(e_1, g_2), e_1, \text{identity of } G_1, g_2 \in G_2\}$.
8. $\text{Ker}(f) = \{(a, -a), a \in Z\}$
9. $\{[0]\}, \{[1]\}, \{[2]\}, \{[3]\}$
10. $\{[0], [2]\}, \{[1], [3]\}$
11. $\{[0], [1], [2], [3]\}$
12. (a) 1 is the identity of G . 1 and -1 are each their own inverse. H is closed with respect to multiplication.
(b) $H, iH = \{i, -i\} = -iH$
13. The groups are isomorphic. Define $f: G \rightarrow Z_4$ by $f(1) = [0], f(i) = [1], f(-1) = [2], f(-i) = [3]$. A comparison of the multiplication tables shows that f preserves the operations.
14. $\{f_1, g_1\}, \{f_2, g_3\}, \{f_3, g_2\}$
15. $\{f_1, g_3\}, \{f_2, g_2\}, \{f_3, g_1\}$
16. $\{f_1, f_2, f_3\}, \{g_1, g_2, g_3\}$
17. $\{f_1\}, \{f_2\}, \{f_3\}, \{g_1\}, \{g_2\}, \{g_3\}$
18. $\{f_1, f_2, f_3, g_1, g_2, g_3\}$
19. $\{[0], [4]\}, \{[1], [5]\}, \{[2], [6]\}, \{[3], [7]\}$
20. $\{[0], [2], [4], [6]\}, \{[1], [3], [5], [7]\}$
21. $\{(m+x, n+x) \mid x \in Z\}$ for $(m, n) \in Z \times Z$
22. Suppose N is a normal subgroup of G . Then $aN = Na, \forall a \in G$. Let $a^{-1}na \in a^{-1}Na$. Then $na = an'$ for some $n' \in N$. $a^{-1}na = a^{-1}an' = n' \in N$. Hence $a^{-1}Na \subseteq N \forall a \in G$.
Let $n \in N$. Then for all $a \in G$ there exists an $n' \in N$ with $an = n'a$. Hence $n = a^{-1}n'a \in a^{-1}Na$.
So $N \subseteq a^{-1}Na$ and $a^{-1}Na = N$ for all $a \in G$.
Conversely, suppose $a^{-1}Na = N, \forall a \in G$. Let $g \in G, x \in gN$. Then $x = gn, n \in N$, and $xg^{-1} = gng^{-1} \in g^{-1}Ng = N$ and $x = n'g \in Ng$. Hence $gN \subseteq Ng$. Similarly, $Ng \subseteq gN$. Hence $gN = Ng, \forall g \in G$ and N is a normal subgroup of G .
23. If N is a normal subgroup of G , Exercise 22 shows that $a^{-1}Na \subseteq N$ for all $a \in G$.
Suppose $a^{-1}Na \subseteq N$ for all $a \in G$. Again the proof in Exercise 22 shows that N is a normal subgroup of G .
24. $\{f_1\}, \{f_1, f_2, f_3\}, S_3$
25. $\{f_1\}, \{f_1, f_3\}, \{f_1, f_3, f_5, f_6\}, \{f_1, f_2, f_3, f_4\}, \{f_1, f_3, f_7, f_8\}, D_4$
26. By Exercise 25, Section 9.4, we have H is a subgroup of G . Choose $a \in G$. Let $x \in aH$. Then $x = ah = ha \in Ha$. Thus $aH \subseteq Ha$. Similarly, $Ha \subseteq aH$ for all $a \in G$. Hence H is a normal subgroup of G .
27. Suppose $f_a(h_1) = f_a(h_2)$. Then $ah_1 = ah_2$ and $a^{-1}(ah_1) = a^{-1}(ah_2)$. Hence $h_1 = h_2$ and f_a is one to one. Let $x \in aH$. Then $x = ah, h \in H$ and $f_a(h) = x$. Thus f_a is onto and since it is everywhere defined as well, f_a is a one-to-one correspondence between H and aH . Hence $|H| = |aH|$.
28. By Exercise 28, Section 9.4, we have $H \cap K$ is a subgroup of G . Let $x \in a^{-1}(H \cap K)a$. Then $x = a^{-1}ya, y \in H \cap K$. Thus by Exercise 23 $x = a^{-1}ya \in H$ and $x = a^{-1}ya \in K$. Hence $x \in H \cap K$. Thus $a^{-1}(H \cap K)a \subseteq H \cap K$ for all $a \in G$. By Exercise 13, $H \cap K$ is normal.
29. Suppose $f(aH) = f(bH)$. Then $Ha^{-1} = Hb^{-1}$ and $a^{-1} = hb^{-1}, h \in H$. Hence $a = bh^{-1} \in bH$ so $aH \subseteq bH$. Similarly, $bH \subseteq aH$ so $aH = bH$. This means f is one to one. If Hc is a right coset of H , then $f(c^{-1}H) = Hc$ so f is also onto.
30. $\text{Ker}(f) = \{(g_1, e_2) \mid g_1 \in G_1\}$
31. Consider $f(aba^{-1}b^{-1}) = f(a)f(b)f(a^{-1})f(b^{-1}) = f(a)f(a^{-1})f(b)f(b^{-1}) = f(a)(f(a))^{-1}f(b)(f(b))^{-1}$ (by Theorem 5, Section 9.4) $= ee = e$. Hence $\{aba^{-1}b^{-1} \mid a, b \in G_1\} \subseteq \text{ker}(f)$.
32. Let $[x], [y] \in G/N$. $[x] * [y] = [xy]$ by definition. Since G is Abelian, $[xy] = [yx]$. So G/N is also Abelian.
33. Let $a \notin H$. The left cosets of H are H and aH . The right cosets are H and Ha .
 $H \cap aH = H \cap Ha = \{e\}$ and $H \cup aH = H \cup Ha$. Thus $aH = Ha$. Since $a \in H \Rightarrow aH = H$, we have $xH = Hx \forall x \in G$. H is a normal subgroup of G .
34. By Exercise 28, Section 9.4, $H \cap N$ is a subgroup of G . Since $H \cap N \subseteq H, H \cap N$ is a subgroup of H . Let $h \in H, x \in H \cap N$. Then $h^{-1}xh \in H$. Since N is normal in $G, h^{-1}xh \in N$. Thus $h^{-1}(H \cap N)h \subseteq H \cap N, \forall h \in H$. By Exercise 23, $H \cap N$ is normal in H .
35. Suppose $f: G \rightarrow G'$ is one to one. Let $x \in \text{ker}(f)$. Then $f(x) = e' = f(e)$. Thus $x = e$ and $\text{ker}(f) = \{e\}$.
Conversely suppose $\text{ker}(f) = \{e\}$. If $f(g_1) = f(g_2)$ then $f(g_1g_2^{-1}) = f(g_1)f(g_2^{-1}) = f(g_1)(f(g_2))^{-1} = f(g_1)(f(g_1))^{-1} = e$. Hence $g_1g_2^{-1} \in \text{ker}(f)$. Thus $g_1g_2^{-1} = e$ and $g_1 = g_2$. Hence f is one to one.

36. (a)

$\times$	1	3	7	9
1	1	3	7	9
3	3	9	1	7
7	7	1	9	3
9	9	7	3	1
- (b) $H, 3H = \{3, 7\} = 7H$
37. Since H is a subgroup of G , the identity element e belongs to H . For any $g \in G$, $g = g * e \in gH$, so every element of G belongs to some left coset of H . If aH and bH are distinct left cosets of H , this means that $aH \cap bH = \{ \}$. Hence the set of distinct left cosets of H forms a partition of G .
38. Let k be the number of distinct left cosets of H . Then $|G| = k \cdot |H|$.

Exercises 9.6, page 367

1. non-commutative ring with identity
2. commutative ring with identity
3. not a ring
4. commutative ring with identity
5. commutative ring with identity
6. commutative ring with identity
7. This is a ring from Exercise 1. An example of zero divisors are the matrices $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ and $\begin{bmatrix} -2 & -2 \\ 1 & 1 \end{bmatrix}$.
8. Z_n is a ring, $\forall n$. In Z_{10} , $\bar{2}$ and $\bar{5}$ are a pair of zero divisors.
9. $\bar{1}, \bar{3}$
10. $\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}$
11. $\bar{1}, \bar{3}, \bar{7}, \bar{9}$
12. $\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}, \bar{8}, \bar{9}, \bar{10}$
13. This subset is a subgroup with respect to $+$ since it contains the zero element and $\mathbf{A} - \mathbf{B}$ belongs to the subset if $\mathbf{A}$ and $\mathbf{B}$ do. This subset is a subsemigroup with respect to $*$ since if $\mathbf{A}$ and $\mathbf{B}$ are in the subset so is $\mathbf{A} * \mathbf{B}$.
14. If $a, b \in Z$, then $a - b$ and $a * b$ are also elements of Z . Also, $0 \in Z$. Hence, $(Z, +)$ is a subgroup of $(S, +)$ and $(Z, *)$ is a subsemigroup of $(S, *)$.
15. The structures in Exercises 1 and 2 are not fields, because they lack multiplicative inverses. The structure in Exercise 3 is not a ring so it is not a field. The structures in Exercises 4, 5, and 6 are fields.
16. (a) $\bar{4}$ (b) $\bar{5}$ (c) $\bar{1}$ (d) $\bar{4}$ (e) $\bar{2} * \bar{5} = \bar{3}$
17. $\bar{55} * \bar{57} = \bar{1}$ in Z_{196} .
18. $\bar{27} * \bar{29} = \bar{1}$ in Z_{196} .
19. $(\bar{4}, \bar{0})$
20. $(\bar{3}, \bar{6})$
21. (a) $\bar{2}, \bar{3}$ (b) There are no solutions.
22. (a) There are no solutions. (b) $\bar{2}, \bar{4}$
23. 9
24. 11
25. (a) $(x \vee y) \wedge \sim (x \wedge y)$ (b) $x \wedge y$
26. Suppose F is a field with zero divisors a and b . Then $a * b = 0$, but a^{-1} exists and $a^{-1} * (a * b) = a^{-1} * 0 = 0$ and $a^{-1} * (a * b) = (a^{-1} * a) * b = 1 * b = b$. But $b \neq 0$.
27. The set of units must contain all non-zero elements of R .
28. If n is not prime, then n is the product of integers j and k with $1 < j, k < n$. In Z_n , $\bar{j} * \bar{k} = 0$ and $\bar{j}, \bar{k}$ are a pair of zero divisors.
29. The statement Z_n is a field implies n is a prime is proven in Exercise 28. The converse is shown by using Theorem 4(a), Section 1.4. Any $a \in Z_n$ is relatively prime to n so we have $1 = sa + tn$ for some integers s, t and $\bar{s}$ is the multiplicative inverse of a .

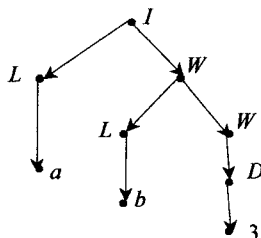
Review Questions, page 369

1. A set is closed with respect to a binary operation if using the operation with any two elements of the set yields an element that belongs to the set.
2. An isomorphism between two semigroups must preserve the operations, but an isomorphism between posets must preserve the orders. In each case the mapping must be a one-to-one correspondence and preserve the defining structure, multiplications for groups and orders for posets.
3. The relation must be reflexive, symmetric, transitive, and preserve the binary operations.
4. Groups are said to have more structure than semigroups, because they must satisfy more conditions.
5. A field must contain a multiplicative inverse for each non-zero element; this is not required in a ring.

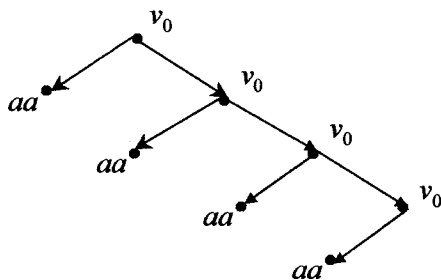
Chapter 10 Answers

Exercise Set 10.1, page 379

1. $\{x^m y^n z, m \geq 0, n \geq 1\}$
 2. $\{a^{2^n}, n \geq 1\}$
 3. $\{a^{2^{n+1}}, n \geq 0\} \cup \{a^{2^n} b, n \geq 0\}$
 4. $\{x^{a_1} y^{a_2} x^{a_3} y^{a_4} \dots x^{a_{k-1}} y^{a_k} z, \text{ each } a_i \geq 0\}$
 5. $\{\underbrace{((\dots (a + a + \dots + a) \dots))}_k, k \geq 0, n \geq 3\}$
 $\qquad \qquad \qquad k \qquad \qquad \qquad n \text{ a's} \qquad \qquad \qquad k$
 6. $\{(ab)^n aa, n \geq 0\}$
 7. $\{x^m y z^n, m \geq 1, n \geq 0\}$
1. Type 3 2. Type 3 3. Type 3 4. Type 3
5. Type 3 6. Type 3 7. Type 1
9. (a), (c), (e), (h), (i)
- 10.



11. $L(G)$ is the set of strings from $\{a, b, c, 1, 2, \dots, 9, 0\}^*$ that begin with a, b , or c .
12. (a), (d)
13. $L(G) = \{(aa)^n bc^k (bb)^j b^k, n \geq 0, k \geq 1, j \geq 0\}$
- 14.

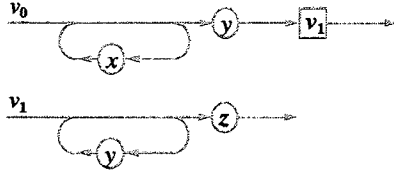


- | | | |
|-----|---------------|---------------|
| 15. | v_0 | v_0 |
| | $v_0 v_1$ | $v_0 v_1$ |
| | $v_0 v_1 v_1$ | $v_2 v_0 v_1$ |
| | $v_2 v_0 z$ | xyv_1 |
| | xyz | xyz |

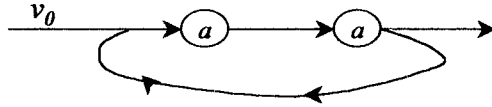
16. v_0 Only one derivation is possible.
 (v_0)
 $((v_0))$
 $((a + v_1))$
 $((a + a + v_2))$
 $((a + a + a))$
17. I I
 LW LW
 aW LDW
 aDW $LDDW$
 $a1W$ $LDDD$
 $a1DW$ $aDDD$
 $a1DD$ $a1DD$
 $a10D$ $a10D$
 $a100$ $a100$
18. $L(G) = \{\text{strings with an even number of } a\text{'s}\}$.
19. The languages are not the same; here $L(G)$ contains $aabba$.
20. $G = (V, S, v_0, \mapsto), V = \{v_0, a, b\}, S = \{a, b\}$
 $\mapsto : v_0 \mapsto av_0b, v_0 \mapsto ab$
21. $G = (V, S, v_0, \mapsto), V = \{v_0, v_1, 0, 1\}, S = \{0, 1\}$
 $\mapsto : v_0 \mapsto 0v_11, v_0 \mapsto 1v_10, v_1 \mapsto 0v_11, v_1 \mapsto 1v_10, v_1 \mapsto 01, v_1 \mapsto 10$
22. $G = (V, S, v_0, \mapsto), V = \{v_0, v_1, v_2, a, b\}, S = \{a, b\}$
 $\mapsto : v_0 \mapsto v_1v_2, v_1 \mapsto av_1, v_1 \mapsto a, v_2 \mapsto bv_2, v_2 \mapsto b$
23. $G = (V, S, v_0, \mapsto), V = \{v_0, v_1, a, b\}, S = \{a, b\}$
 $\mapsto : v_0 \mapsto aav_1bb, v_1 \mapsto av_1b, v_1 \mapsto ab$
24. $G = (V, S, v_0, \mapsto), V = \{v_0, v_1, a, b\}, S = \{a, b\}$
 $\mapsto : v_0 \mapsto v_1bb, v_1 \mapsto av_1, v_1 \mapsto v_1b, v_1 \mapsto ab$
25. $G = (V, S, v_0, \mapsto), V = \{v_0, x, y\}, S = \{x, y\}$
 $\mapsto : v_0 \mapsto v_0yy, v_0 \mapsto xv_0, v_0 \mapsto xx$
26. $G = (V, S, v_0, \mapsto), V = \{v_0, v_1, v_2, x, y\}, S = \{x, y\}$
 $\mapsto : v_0 \mapsto v_1v_2, v_0 \mapsto v_2, v_1 \mapsto xxv_1, v_1 \mapsto xx, v_2 \mapsto yyv_2, v_2 \mapsto y$
27. $v_0 \mapsto av_0a$ $v_0 \mapsto bv_0b$ $v_0 \mapsto a$ $v_0 \mapsto aav_0 \mapsto b$ $v_0 \mapsto bb$
28. $v_0 \mapsto x_i, i = 1, 2, \dots, n$ $v_1 \mapsto x_i, i = 1, 2, \dots, n$ $v_2 \mapsto x_i, i = 1, 2, \dots, n$
 $v_0 \mapsto x'_i, i = 1, 2, \dots, n$ $v_1 \mapsto x'_i, i = 1, 2, \dots, n$ $v_2 \mapsto x'_i, i = 1, 2, \dots, n$
 $v_0 \mapsto 0$ $v_1 \mapsto 0$ $v_2 \mapsto 0$
 $v_0 \mapsto 1$ $v_1 \mapsto 1$ $v_2 \mapsto 1$
 $v_0 \mapsto (v_1 \vee v_2)$ $v_1 \mapsto (v_1 \vee v_2)$ $v_2 \mapsto (v_1 \vee v_2)$
 $v_0 \mapsto (v_1 \wedge v_2)$ $v_1 \mapsto (v_1 \wedge v_2)$ $v_2 \mapsto (v_1 \wedge v_2)$
 $v_0 \mapsto (v_1 \vee v_2)'$ $v_1 \mapsto (v_1 \vee v_2)'$ $v_2 \mapsto (v_1 \vee v_2)'$
 $v_0 \mapsto (v_1 \wedge v_2)'$ $v_1 \mapsto (v_1 \wedge v_2)'$ $v_2 \mapsto (v_1 \wedge v_2)'$
29. by using production rules 1, 2, and 4; rule 4 to $a^{3k-1}v_2$; 3; $a^{3k}av_3 \Rightarrow a^{3k+1}v_1 \Rightarrow a^{3k+2}v_2 \Rightarrow a^{3(k+1)}$
30. a is the only terminal; $0 < i \leq 3$; If a^i is in $L(G)$ and $3k < i \leq 3(k+1)$, then $i = 3(k+1)$. If a^i is in $L(G)$ and $3(k+1) < i \leq 3((k+1)+1)$, then $i = 3(k+2)$. a^{i-1} ; 4; 2; production rule 1 requires v_0 ; $a^{i-3}v_3$ by rule 5; $i-3 = 3(k+1)$ and thus $i = 3(k+2)$.
31. Let $G_1 = (V_1, S_1, v_0, \mapsto_1)$ and $G_2 = (V_2, S_2, v'_0, \mapsto_2)$. Define $G = (V_1 \cup V_2, S_1 \cup S_2, v_0, \mapsto)$ as follows. If $v_i \mapsto_1 wv_k$, then $v_i \mapsto wv_k$. If $v_i \mapsto_1 w$ where w consists of terminal symbols, then $v_i \mapsto wv'_0$. All productions in $\mapsto_2$ become productions in $\mapsto$.
32. $a^i v_j$, because G is regular; $a^i v_j$ and $a^k v_j$; by the pigeonhole principle (n non-terminals, i, k can be any integers ≥ 1 ; i, k).
33. $v_0 \mapsto av_0$ $v_1 \mapsto av_1$ The language is in fact regular.
 $v_0 \mapsto bv_1$ $v_1 \mapsto a$

Exercise Set 10.2, page 389

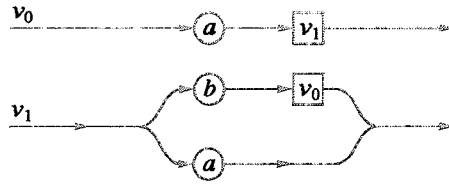
1. $\langle v_0 \rangle ::= x \langle v_0 \rangle \mid y \langle v_1 \rangle \quad \langle v_1 \rangle ::= y \langle v_1 \rangle \mid z$



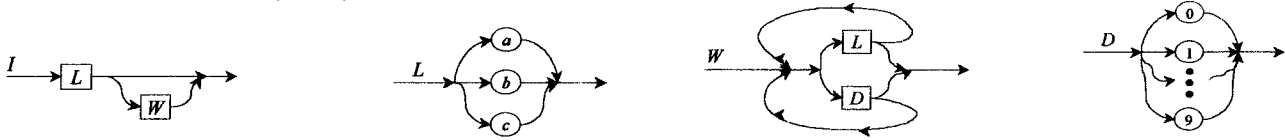
2. $\langle v_0 \rangle ::= aa \langle v_0 \rangle \mid aa$



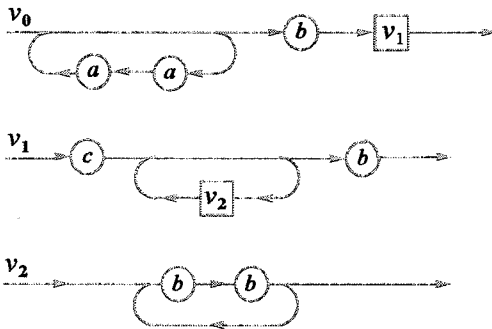
3. $\langle v_0 \rangle ::= a \langle v_1 \rangle$
 $\langle v_1 \rangle ::= b \langle v_0 \rangle \mid a$



4. $\langle I \rangle ::= \langle L \rangle \langle W \rangle \mid \langle L \rangle$
 $\langle W \rangle ::= \langle L \rangle \langle W \rangle \mid \langle D \rangle \langle W \rangle \mid \langle L \rangle \mid \langle D \rangle$
 $\langle L \rangle ::= a \mid b \mid c$
 $\langle D \rangle ::= 0 \mid 1 \mid 2 \mid \dots \mid 9$

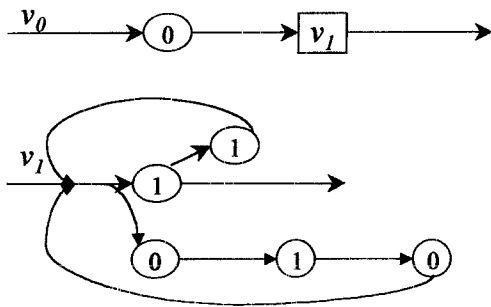


5. $\langle v_0 \rangle ::= aa \langle v_0 \rangle \mid b \langle v_1 \rangle$
 $\langle v_1 \rangle ::= c \langle v_2 \rangle \mid cb$
 $\langle v_2 \rangle ::= bb \langle v_2 \rangle \mid bb$



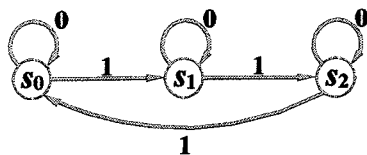
6. $\langle v_0 \rangle ::= aa \langle v_0 \rangle \mid a \mid b$
7. $\langle v_0 \rangle ::= x \langle v_0 \rangle \mid y \langle v_0 \rangle \mid z$
8. $\langle v_0 \rangle ::= (\langle v_0 \rangle) \mid a + \langle v_1 \rangle$
 $\langle v_1 \rangle ::= a + \langle v_2 \rangle$
 $\langle v_2 \rangle ::= a + \langle v_2 \rangle \mid a$
9. $\langle v_0 \rangle ::= a \langle v_1 \rangle$
 $\langle v_1 \rangle ::= b \langle v_0 \rangle \mid a$

10. $\langle v_0 \rangle ::= 0 \langle v_1 \rangle$
 $\langle v_1 \rangle ::= 11 \langle v_1 \rangle \mid 010 \langle v_1 \rangle \mid 1$
11. $\langle v_0 \rangle ::= ab \langle v_1 \rangle$
 $\langle v_1 \rangle ::= c \langle v_1 \rangle \mid \langle v_2 \rangle$
 $\langle v_2 \rangle ::= dd \langle v_2 \rangle \mid d$
12. $\langle v_0 \rangle ::= ab \langle v_1 \rangle \mid ab$
 $\langle v_1 \rangle ::= d \langle v_1 \rangle \langle v_2 \rangle \mid d \langle v_1 \rangle \mid d$
 $\langle v_2 \rangle ::= c \mid d$
13. $\langle v_0 \rangle ::= a \langle v_1 \rangle$
 $\langle v_1 \rangle ::= a \langle v_2 \rangle$
 $\langle v_2 \rangle ::= a \langle v_1 \rangle \mid a$
14. $\langle v_0 \rangle ::= ab \langle v_1 \rangle$
 $\langle v_1 \rangle ::= ab \langle v_2 \rangle \mid b$
 $\langle v_2 \rangle ::= c \langle v_1 \rangle$
15. $\langle v_0 \rangle ::= b \langle v_0 \rangle \mid a \langle v_1 \rangle \mid b$
 $\langle v_1 \rangle ::= a \langle v_0 \rangle \mid b \langle v_1 \rangle \mid a$
16. $\langle v_0 \rangle ::= a \langle v_1 \rangle \mid a \langle v_2 \rangle$
 $\langle v_1 \rangle ::= a \langle v_2 \rangle \mid b \langle v_0 \rangle \mid b$
 $\langle v_2 \rangle ::= b \langle v_0 \rangle \mid b$
17. $(aa)^*aa$
18. $(ab)^*(a \vee b)$
19. $(())^*(a + a + (a +)^*a())^*$ Note: Right and left parentheses must be matched.
20. $(ab)^*aa$
21. $(a \vee b \vee c)(a \vee b \vee c \vee 0 \vee 1 \vee \dots \vee 9)^*$
22. $abc^*d(dd)^*$
23. $ab(d \vee (d(c \vee d)d))^*$
24. $a^{2n+1}, n \geq 1$
25. $ab(abc)^nb, n \geq 1$
26. $(aa \vee b \vee aba)^*$
27. $(aab \vee ab)^*$
28. $0(11 \vee 010)^*1$
- 29.

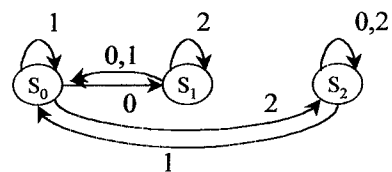


Exercise Set 10.3, page 394

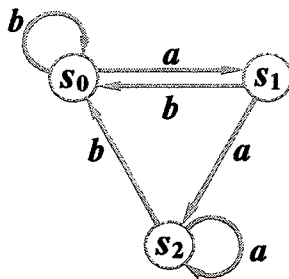
1.



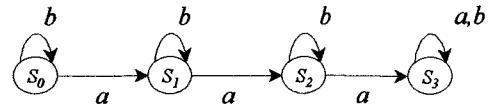
2.



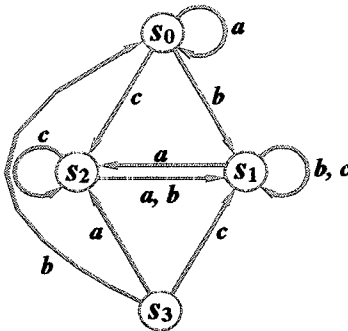
3.



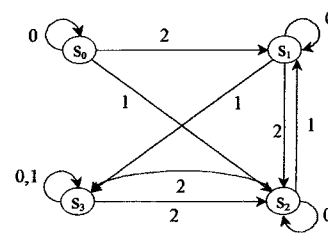
4.



5.



6.



7.

	a	b
s_0	s_1	s_1
s_1	s_1	s_2
s_2	s_0	s_2

8.

	0	1
s_0	s_0	s_1
s_1	s_2	s_1
s_2	s_2	s_3
s_3	s_3	s_3

9.

	T	F
s_0	s_1	s_0
s_1	s_1	s_1
s_2	s_1	s_2

10.

	a	b
s_0	s_0	s_1
s_1	s_2	s_1
s_2	s_0	s_1

11.

	a	b	c
s_0	s_0	s_1	s_2
s_1	s_2	s_1	s_3
s_2	s_3	s_3	s_1
s_3	s_3	s_3	s_2

12.

	0	1
s_0	s_1	s_0
s_1	s_1	s_2
s_2	s_3	s_2
s_3	s_3	s_4
s_4	s_4	s_2

13. Let $x \in I$. Certainly $f_x(s) = f_x(s)$ for all $s \in S$. Thus xRx and R is reflexive.
 Suppose xRy . Then $f_x(s) = f_y(s) \forall s \in S$. But then yRx and R is symmetric.
 Suppose xRy, yRz . Then $f_x(s) = f_y(s) = f_z(s), \forall s \in S$. Hence xRz and R is transitive.
14. Suppose xRy and yRz . Then $f_t(x) = y$ for some $t \in R$, and $f_s(y) = z$ for some $s \in R$. $y = t * x$ and $z = s * y$ so $z = s * (t * x) = (s * t) * x$. Hence $f_u(x) = z$ for $u = s * t$. Thus xRz .
15. Using Exercise 14, we need only show R is reflexive and symmetric. Let $s \in S$. $s = e * s$ so $f_e(s) = s$ and sRs . Suppose xRy . Then $f_z(x) = y$ for some $z \in S$. $y = z * x \Rightarrow z^{-1} * y = x$ and thus $f_{z^{-1}}(y) = x$. Hence yRx and R is symmetric.

	0	1		0	1		0	1		0	1
a	a	a	a	b	a	a	a	b	a	a	a
b	a	a	b	a	a	b	a	a	b	b	a

$\frac{0}{a}$	$\frac{1}{a}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{a}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{a}$
$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{a}$	$\frac{1}{b}$	$\frac{0}{b}$	$\frac{1}{a}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{b}$
$\frac{0}{a}$	$\frac{1}{a}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{a}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{a}$	$\frac{1}{a}$	$\frac{0}{a}$	$\frac{1}{a}$	$\frac{1}{b}$
$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{b}$	$\frac{0}{b}$	$\frac{1}{a}$	$\frac{1}{b}$	$\frac{0}{b}$	$\frac{1}{b}$	$\frac{1}{b}$	$\frac{0}{b}$	$\frac{1}{b}$
$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{a}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{b}$
$\frac{1}{b}$	$\frac{0}{b}$	$\frac{1}{b}$	$\frac{1}{b}$	$\frac{0}{a}$	$\frac{1}{b}$	$\frac{1}{b}$	$\frac{0}{b}$	$\frac{1}{a}$	$\frac{1}{b}$	$\frac{0}{b}$	$\frac{1}{b}$

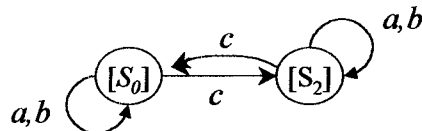
17. (a) Inspection of $\mathbf{M}_R$ shows that R is reflexive and symmetric. Since $\mathbf{M}_R \odot \mathbf{M}_R = \mathbf{M}_R$, R is transitive. Thus R is an equivalence relation. The table below shows that it is a machine congruence.

(b)

$\frac{0}{[1]}$	$\frac{1}{[1]}$	$\frac{1}{[1]}$
$\frac{1}{[2]}$	$\frac{0}{[2]}$	$\frac{1}{[2]}$

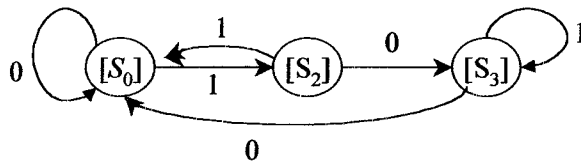
18. (a) Inspection of $\mathbf{M}_R$ shows that R is reflexive and symmetric. Since $\mathbf{M}_R \odot \mathbf{M}_R = \mathbf{M}_R$, R is transitive. Thus R is an equivalence relation. A case-by-case check shows that $xRy \Rightarrow f_i(x)Rf_i(y)$ is true for $i = a, b$, or c . Hence R is machine congruence.

(b)



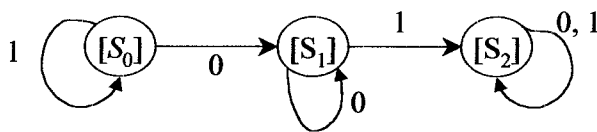
19. (a) R is clearly an equivalence relation. s_0Rs_1 and $f_0(s_0) = s_1$, $f_0(s_1) = s_0$, $f_1(s_0) = s_2 = f_1(s_1)$ so $f_x(s_0)Rf_x(s_1)$, $\forall x \in I$.

(b)

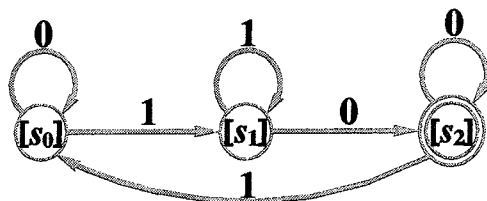


20. (a) Using $\mathbf{M}_R$, it is easy to see that R is an equivalence relation whose partition is $\{\{s_0\}, \{s_1\}, \{s_2, s_3, s_4\}\}$. An examination of Figure 10.38 shows that $f_x(s_i) \in \{s_2, s_3, s_4\}$, $i = 2, 3, 4$, so R is a machine congruence.

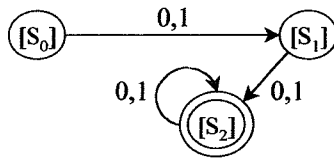
(b)



21. Inspection of $\mathbf{M}_R$ shows that R is reflexive and symmetric. Since $\mathbf{M}_R \odot \mathbf{M}_R = \mathbf{M}_R$, R is transitive. Thus R is an equivalence relation. The digraph below shows that it is a machine congruence.



22. R is clearly reflexive and symmetric. Since $\mathbf{M}_R \odot \mathbf{M}_R = \mathbf{M}_R$, R is transitive. The equivalence classes of R are $\{s_0\}$, $\{s_1\}$, $\{s_2, s_3, s_4\}$. It is easy to see from the digraph that R is a machine congruence.



23. $f_x, x \in I$, is a function on S so f_x is a state transition function. $T \subseteq S$ so the conditions for a Moore machine are met.
24. $f_x, x \in I$, is a function on S so f_x is a state transition function. $T \subseteq S$ so the conditions for a Moore machine are met.

25.

	0	1
(s_0, s_0)	(s_0, s_0)	(s_1, s_1)
(s_0, s_1)	(s_0, s_2)	(s_1, s_1)
(s_0, s_2)	(s_0, s_2)	(s_1, s_3)
(s_0, s_3)	(s_0, s_3)	(s_1, s_3)
(s_1, s_0)	(s_1, s_0)	(s_2, s_1)
(s_1, s_1)	(s_1, s_2)	(s_2, s_1)
(s_1, s_2)	(s_1, s_2)	(s_2, s_3)
(s_1, s_3)	(s_1, s_3)	(s_2, s_3)
(s_2, s_0)	(s_2, s_0)	(s_0, s_1)
(s_2, s_1)	(s_2, s_2)	(s_0, s_1)
(s_2, s_2)	(s_2, s_2)	(s_0, s_3)
(s_2, s_3)	(s_2, s_3)	(s_0, s_3)

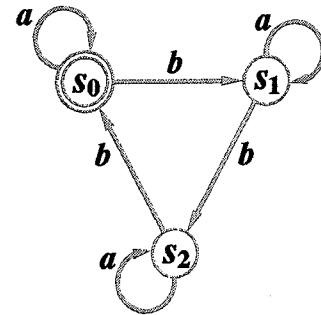
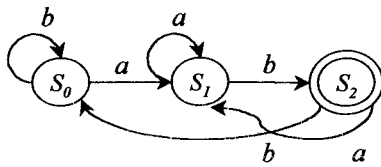
Exercise Set 10.4, page 399

- $f_w(s_0) = s_2, f_w(s_1) = s_3, f_w(s_2) = s_0, f_w(s_3) = s_1$
- $f_w(s_0) = s_3, f_w(s_1) = s_0, f_w(s_2) = s_1, f_w(s_3) = s_2$
- The number of 1's in w is divisible by 4.
- The number of 1's in w is $1 + 4k, k \geq 0$.
- The number of 1's in w is $2 + 4k, k \geq 0$.
- $f_w(s_0) = s_1, f_w(s_1) = s_1, f_w(s_2) = s_1$
- $f_w(s_0) = s_0, f_w(s_1) = s_0, f_w(s_2) = s_0$
- all words ending in 2 or more a 's.
- all words ending in b .
- $w = a$ or w ends in ba .
- strings of 0's and 1's with $3 + 5k$ 1's, $k \geq 0$
- strings of 0's and 1's with $3 + 4k$ 1's, $k \geq 0$
- strings of 0's and 1's that end in 0
- strings of 0's and 1's that contain 010
- strings of a 's and b 's that do not contain bb
- strings of 0's and 1's with $1 + 2k$ 1's, $k \geq 0$
- strings of 0's and 1's that end in 01
- strings of 0's and 1's that end in 1
- strings xy and yz
- strings that have no consecutive x 's
- strings that begin 1 or 00
- strings beginning with n 0's, $n \geq 0$, followed by 1's
- Let $w, u \in L(M)$. Then $f_w(s_0) = s \in T, f_u(s) \in T$. Hence $f_{w \cdot u}(s_0) \in T$, so $w \cdot u \in L(M)$.
- $w \in L(M') \cap L(M'')$ if and only if $w \in L(M')$ and $w \in L(M'')$ if and only if $f'_w(s'_0) \in T'$ and $f''_w(s''_0) \in T''$ if and only if $f_w(s'_0, s''_0) = (f'_w(s'_0), f''_w(s''_0)) \in T$.
- $w \in L(M') \cup L(M'')$ if and only if $w \in L(M')$ or $w \in L(M'')$. Say $w \in L(M'')$. But $w \in L(M'')$ if and only if $f''_w(s''_0) \in T''$ if and only if $f_w(s'_0, s''_0) = (f'_w(s'_0), f''_w(s''_0)) \in T$.

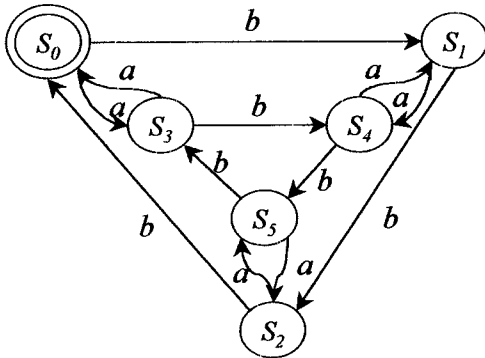
26. For M' , change each acceptance state to a non-acceptance state and vice versa.

Exercise Set 10.5, page 406

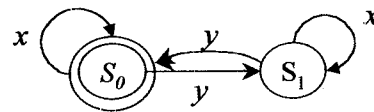
1. $G = (V, I, s_0, \mapsto), V = \{s_0, s_1, s_2, s_3, 0, 1\}, I = \{0, 1\}$
 $\mapsto : s_0 \mapsto 0s_0, s_0 \mapsto 1s_1, s_1 \mapsto 0s_1, s_1 \mapsto 1s_2, s_2 \mapsto 0s_2, s_2 \mapsto 1s_3, s_3 \mapsto 1,$
 $s_3 \mapsto 0s_3, s_3 \mapsto 0, s_3 \mapsto 1s_0$
2. $(0 \vee 1)^*0$
3. $(0 \vee 1)^*1$
4. $G = (V, I, s_0, \mapsto), V = \{s_0, s_1, s_2, s_3, 0, 1\}, I = \{0, 1\}$
 $\langle s_0 \rangle ::= 1 \langle s_0 \rangle \mid 0 \langle s_1 \rangle$
 $\langle s_1 \rangle ::= 0 \langle s_1 \rangle \mid 1 \langle s_2 \rangle$
 $\langle s_2 \rangle ::= 0 \langle s_3 \rangle \mid 1 \langle s_0 \rangle \mid 0$
 $\langle s_3 \rangle ::= 0 \langle s_3 \rangle \mid 1 \langle s_3 \rangle \mid 0 \mid 1$
5. $G = (V, I, s_0, \mapsto), V = \{s_0, s_1, s_2, a, b\}, I = \{a, b\}$
 $\langle s_0 \rangle ::= a \langle s_0 \rangle \mid b \langle s_1 \rangle \mid a \mid b$
 $\langle s_1 \rangle ::= a \langle s_0 \rangle \mid b \langle s_2 \rangle \mid a$
 $\langle s_2 \rangle ::= a \langle s_2 \rangle \mid b \langle s_2 \rangle$
- 6.
- 7.



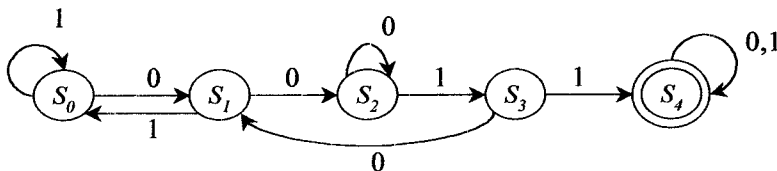
8.



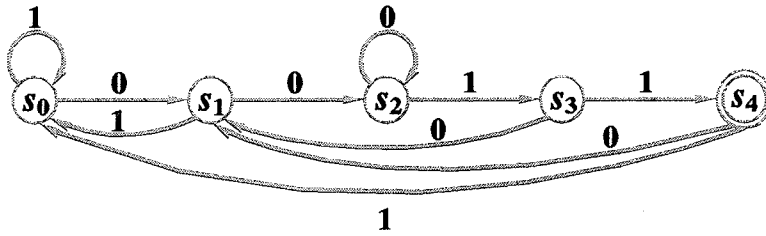
9.



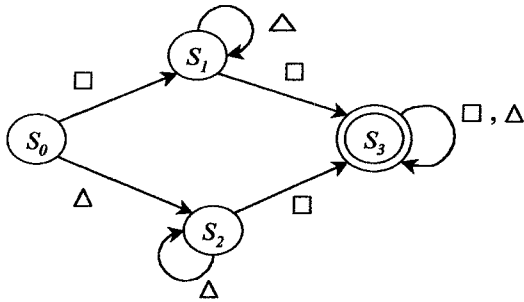
10.



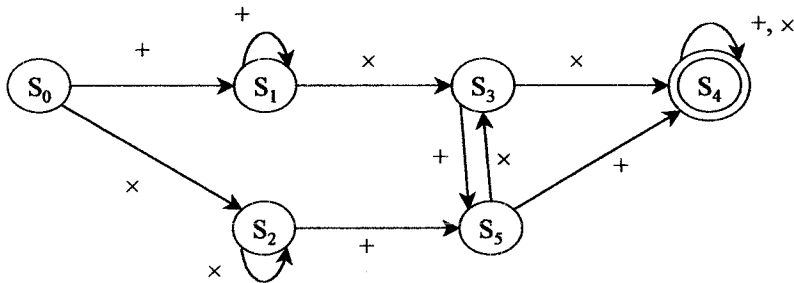
11.



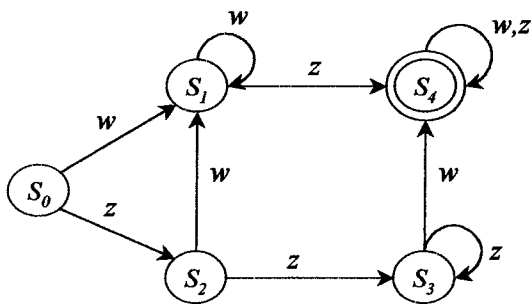
12.



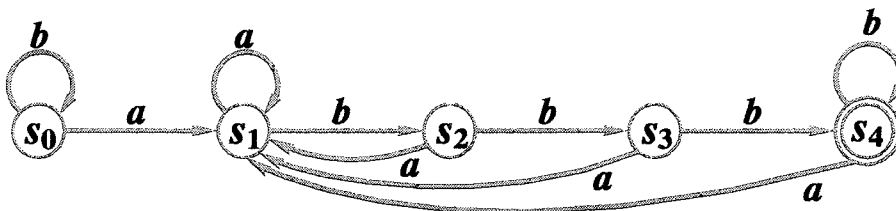
13.



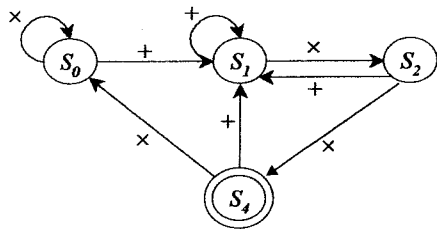
14.



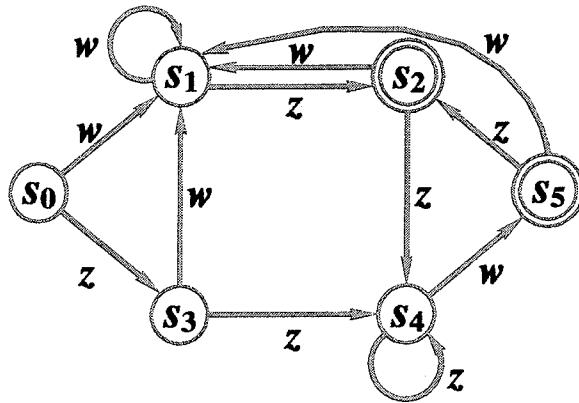
15.



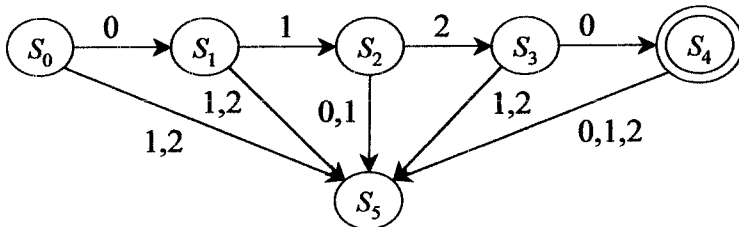
16.



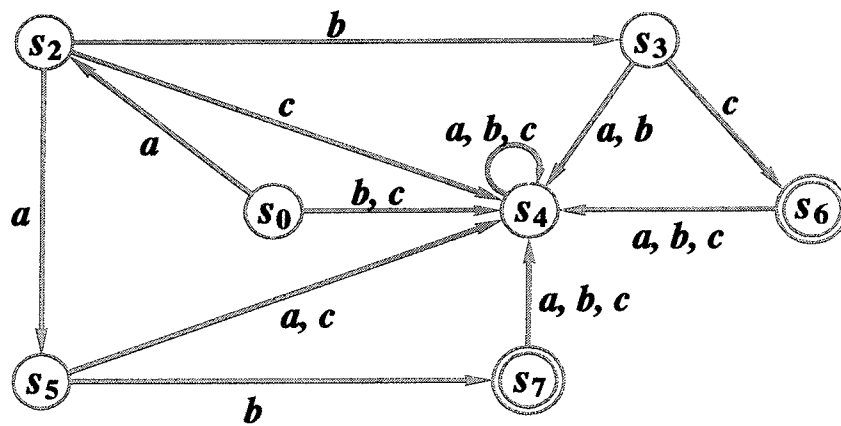
17.



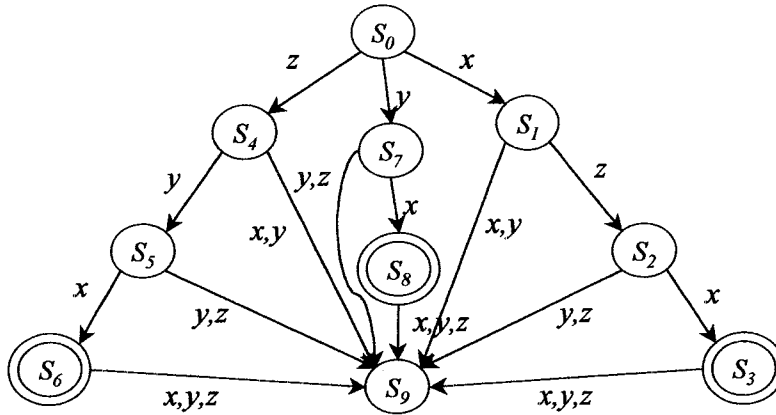
18.



19.



20.



21.

	0	1
s_0	s_1	s_0
s_1	s_1	s_2
s_2	s_3	s_0
s_3	s_1	s_4
s_4	s_3	s_0

 $T = \{s_4\}$

	0	1
s_0	s_1	s_0
s_1	s_1	s_2
s_2	s_3	s_0
s_3	s_1	s_4
s_4	s_3	s_0

22.

	a	b
s_0	s_0	s_1
s_1	s_1	s_2
s_2	s_2	s_3
s_3	s_3	s_0

 $T = \{s_0\}$

	a	b
s_0	s_0	s_1
s_1	s_1	s_2
s_2	s_2	s_3
s_3	s_3	s_0

23.

	x	y
s_0	s_1	s_0
s_1	s_2	s_1
s_2	s_3	s_2
s_3	s_3	s_3

 $T = \{s_2\}$

	x	y
s_0	s_1	s_0
s_1	s_2	s_1
s_2	s_3	s_2
s_3	s_3	s_3

24.

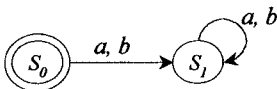
	a	b
s_0	s_0	s_1
s_1	s_0	s_2
s_2	s_2	s_2

 $T = \{s_0, s_1\}$

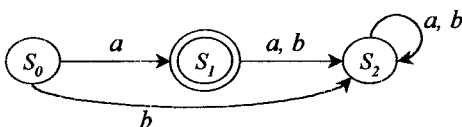
	a	b
s_0	s_0	s_1
s_1	s_0	s_2
s_2	s_2	s_2

25. R is reflexive because $f_w(x) = f_w(x)$. R is symmetric because if $f_w(s_i), f_w(s_j)$ are both (not) in T , then $f_w(s_j), f_w(s_i)$ are both (not) in T . R is transitive because $s_i R s_j, s_j R s_k$ if and only if $f_w(s_i), f_w(s_j), f_w(s_k)$ are all in (or not in) T .

26.



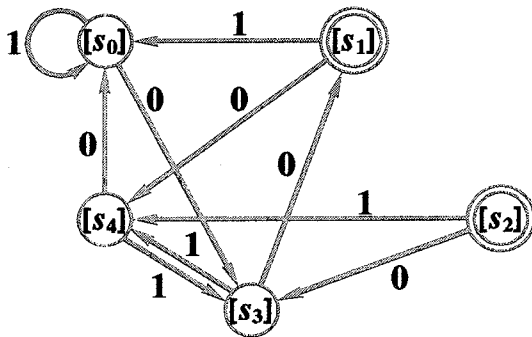
27.



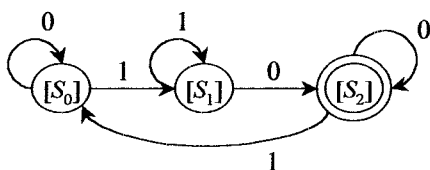
28. Remove the acceptance states from M_α (except s_0 if $\Lambda \in L(M_\alpha) \cap L(M_\beta)$); all edges that would have gone to an acceptance state should now go to the starting state of M_β .
29. The construction of Exercise 24 of Section 10.3 gives a machine that accepts $\alpha \vee \beta$.
30. Add a new starting state s'_0 which is also an acceptance state. Add edges from s'_0 that match edges from the original starting state. For each original acceptance state also redirect edges to match those from the original starting state.

Exercise Set 10.6, page 411

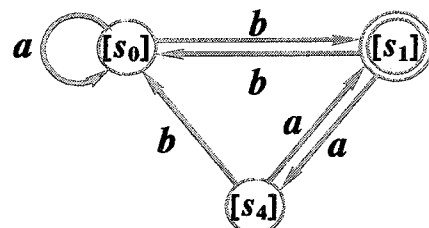
1. $R_0 = \{(s_0, s_0), (s_0, s_1), (s_1, s_0), (s_1, s_1), (s_2, s_2)\}$
2. $R_1 = \{(s_0, s_0), (s_1, s_1), (s_2, s_2)\}$
3. $R_1 = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_4, s_4), (s_0, s_3), (s_3, s_0), (s_1, s_2), (s_2, s_1)\}$
4. $R_2 = R_1$
5. $R_{127} = R_1$
6. $R_1 = \{(s_0, s_0), (s_0, s_1), (s_1, s_0), (s_1, s_1), (s_2, s_2), (s_2, s_3), (s_3, s_2), (s_3, s_3), (s_4, s_4), (s_4, s_5), (s_5, s_4), (s_5, s_5)\}$
7. $R_2 = R_1$
8. $R_1 = \{(s_0, s_0), (s_0, s_4), (s_4, s_0), (s_1, s_1), (s_2, s_2), (s_2, s_1), (s_1, s_2), (s_3, s_3), (s_4, s_4), (s_4, s_5), (s_5, s_4), (s_5, s_5), (s_6, s_6), (s_0, s_5), (s_5, s_0), (s_3, s_6), (s_6, s_3)\}$
9. $R = \{(s_0, s_0), (s_1, s_1), (s_2, s_2)\}$
10. $R = R_1$ as given in Exercise 3.
11. $R = R_1$ as given in Exercise 6.
12. $R = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3)\}$
13. $R = \{(s_0, s_0), (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_4, s_4), (s_4, s_5), (s_5, s_4), (s_5, s_5), (s_6, s_6), (s_3, s_6), (s_6, s_3)\}$



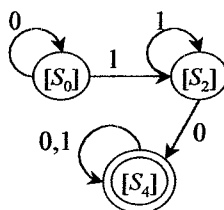
14.



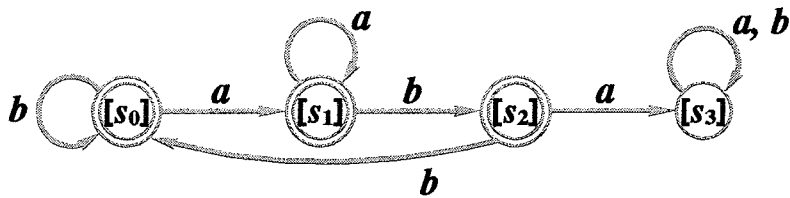
15.



16.



17.



18. $P = \{\{s_0, s_1, s_4\}, \{s_2\}, \{s_3\}, \{s_5, s_6\}\}$

	0	1
$[s_0]$	$[s_5]$	$[s_2]$
$[s_2]$	$[s_0]$	$[s_0]$
$[s_3]$	$[s_3]$	$[s_5]$
$[s_5]$	$[s_3]$	$[s_0]$

19. $P = \{\{a, g\}, \{f\}, \{b\}, \{c\}, \{d, e\}\}$

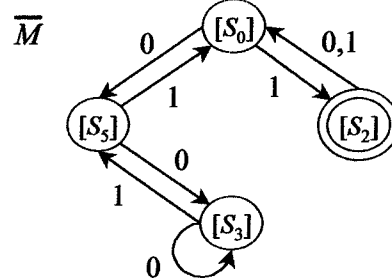
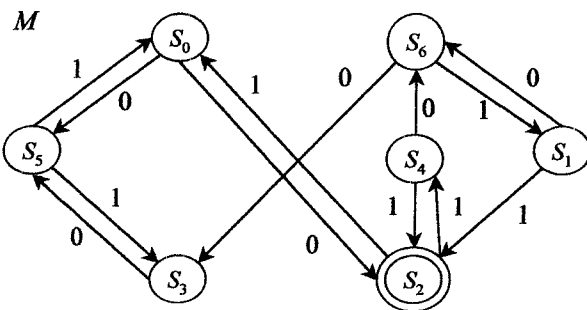
	0	1
$[a]$	$[a]$	$[c]$
$[b]$	$[a]$	$[d]$
$[c]$	$[f]$	$[d]$
$[d]$	$[a]$	$[d]$
$[f]$	$[a]$	$[f]$

20.

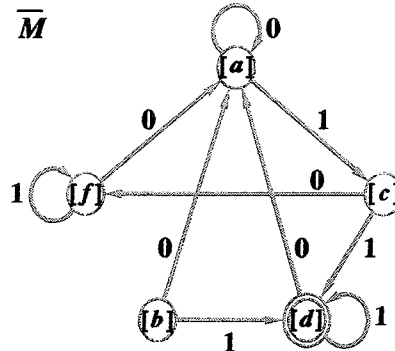
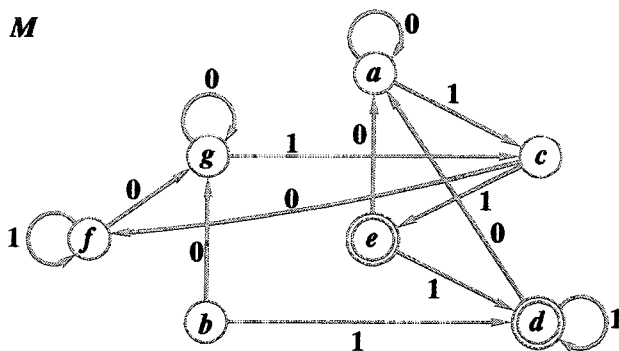
	0	1
$[s_0]$	$[s_1]$	$[s_1]$
$[s_1]$	$[s_1]$	$[s_3]$
$[s_3]$	$[s_5]$	$[s_3]$
$[s_5]$	$[s_5]$	$[s_5]$

21. The equivalence classes each contain a single state, so the table is essentially that given in Exercise 21.

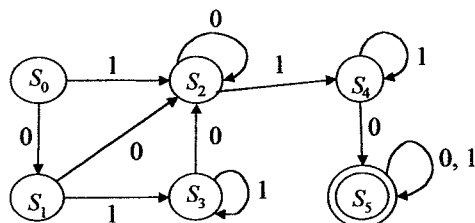
22.



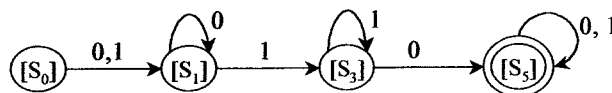
23.



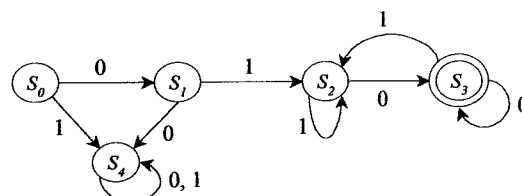
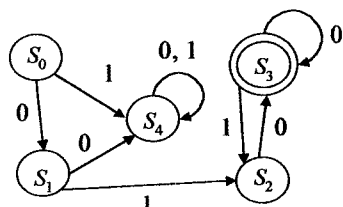
24. M



$\overline{M}$



25. M



26. strings that begin with 01 and end with 0

Review Questions, page 413

- No, type 0 and type 1 languages can not be given in BNF.
- Some type 2 and type 3 grammars can be represented by a syntax diagram.
- This language is type 2, context-sensitive, but importantly it is not regular.
- A machine accepts or recognizes a language if and only every string in the language leads from the starting state to an acceptance state; that is, $f_w(s_0)$ is an acceptance state.
- A language L is regular if and only if $L = L(M)$ for some Moore machine M .

Chapter 11 Answers

Exercise Set 11.1, page 426

- (a) 3 (b) 2 (c) 3
- (a) 4 (b) 5 (c) 3
- (a) Yes (b) No
- (a) Yes (b) No
- (a) No (b) No (c) Yes (d) No
- (a) Yes (b) Yes
- (a) Yes (b) Yes
- (a) Yes (b) Yes
- Let $a_1a_2\dots a_{10}$ be an ISBN. Suppose a_i and a_{i+1} are distinct and are transposed in transmission. Consider $a_1 + 2a_2 + 3a_3 + \dots + ia_i + (i+1)a_{i+1} + \dots + 10a_{10} - (a_1 + 2a_2 + 3a_3 + \dots + ia_{i+1} + (i+1)a_i + \dots + 10a_{10}) = a_{i+1} - a_i$. Because $0 \leq a_i, a_{i+1} \leq 9$, $a_{i+1} - a_i$ can not be congruent to 0 (mod 11) and neither can $a_1 + 2a_2 + 3a_3 + \dots + ia_{i+1} + (i+1)a_i + \dots + 10a_{10} = a_1 + 2a_2 + 3a_3 + \dots + ia_i + (i+1)a_{i+1} + \dots + 10a_{10} - (a_{i+1} - a_i)$. Hence an error would be detected.
- Let $a_1a_2\dots a_{12}$ be a valid bar code with $a = a_2 + a_4 + \dots + a_{12} + 3(a_1 + a_3 + \dots + a_{11}) \equiv 0 \pmod{10}$. Suppose that a_{2k} is changed to a'_{2k} . This error can change the sum a by at most 8, so the new sum can not be congruent to 0 (mod 10). 0-51000-01161-9 is a valid bar code, but errors changing the second and tenth positions to give 0-41000-01171-9 yields a valid bar code and the errors would not be detected.
- By definition, $x \oplus y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$ and $x_i + y_i = 1$ if and only if $x_i \neq y_i$.
- (a) 4 (b) 2

13. (a) 2 (b) 6
 14. (a) $\delta(x, y) = |x \oplus y| = |y \oplus x| = \delta(y, x)$ since $x \oplus y = y \oplus x$.
 (b) $\delta(x, y)$ is the number of positions in which x and y differ so $\delta(x, y) \geq 0$.
 15. If $x = y$, they differ in 0 positions and $\delta(x, y) = 0$. Conversely, if $\delta(x, y) = 0$, then x and y cannot differ in any position and $x = y$.
 16. 2
 17. 1
 18. (a) 2 (b) 1
 19. (a) 3 (b) 2 or fewer
 20. $\oplus$
- | | | | | |
|-------|-------|-------|-------|-------|
| | 00000 | 01110 | 10101 | 11011 |
| 00000 | 00000 | 01110 | 10101 | 11011 |
| 01110 | 01110 | 00000 | 11011 | 10101 |
| 10101 | 10101 | 11011 | 00000 | 01110 |
| 11011 | 11011 | 10101 | 01110 | 00000 |
21. Let $a = 0000000$, $b = 0010110$, $c = 0101000$, $d = 0111110$, $e = 1000101$, $f = 1010011$, $g = 1101101$, $h = 1111011$.

$\oplus$	a	b	c	d	e	f	g	h
a	a	b	c	d	e	f	g	h
b	b	a	d	c	f	e	h	g
c	c	d	a	b	g	h	e	f
d	d	c	b	a	h	g	f	e
e	e	f	g	h	a	b	c	d
f	f	e	h	g	b	a	d	c
g	g	h	e	f	c	d	a	b
h	h	g	f	e	d	c	b	a

22. 3
 23. 2

24. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ 25. $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

26. $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ 27. $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

28. (a) Yes (b) Yes (c) Yes (d) Yes
 29. (a) Yes (b) No (c) Yes (d) No
 30. $e_H(00) = 00000$ $e_H(10) = 10011$ $e_H(01) = 01011$ $e_H(11) = 11000$
 31. $e_H(000) = 000000$ $e_H(100) = 100100$ $e_H(001) = 001111$ $e_H(101) = 101011$
 $e_H(010) = 010011$ $e_H(110) = 110111$ $e_H(011) = 011100$ $e_H(111) = 111000$
 32. A case-by-case check shows $(d_{ij} + e_{ij}) \cdot f_{jk} = d_{ij} \cdot f_{jk} + e_{ij} \cdot f_{jk}$.

The ik -element of $(D \oplus E) * F$ is $\sum_{j=1}^p (d_{ij} + e_{ij}) \cdot f_{jk}$ and the ik -element of $(D * F) \oplus (E * F)$

is $\sum_{j=1}^p d_{ij} \cdot f_{jk} + \sum_{j=1}^p e_{ij} \cdot f_{jk}$. Hence $(D \oplus E) * F = (D * F) \oplus (E * F)$

33. The statement of Theorem 2 is of the form $p \Leftrightarrow q$. Part one of the proof shows that $q \Rightarrow p$. The second part establishes $p \Rightarrow q$ by showing $\sim q \Rightarrow \sim p$ (the contrapositive).